Part II — General Relativity

Based on lectures by Dr J.M. Evans

Notes taken by Xing Yu Yi

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These notes are not endorsed by the lecturer, and are intended only for personal use.
Additional notes and comments have been added.

Part IB Methods and Variational Principles are very useful.

Brief Review of Special Relativity
Notion of proper time. Equation of motion for free point particle derivable from a variational principle. Noether's theorem. [1]

Introduction and motivation for General Relativity
Curved and Riemannian spaces. The Pound–Rebka experiment. Introduction to general relativity: interpretation of the metric, clock hypothesis, geodesics, equivalence principles. Static spacetimes. Newtonian limit. [4]

Tensor calculus
Covariant and contravariant tensors, tensor manipulation, partial derivatives of tensors. Metric tensor, magnitudes, angles, duration of curve, geodesics. Connection, Christoffel symbols, covariant derivatives, parallel transport, autoparallels as geodesics. Curvature. Riemann and Ricci tensors, geodesic deviation. [5]

Vacuum field equations
Spherically symmetric spacetimes, the Schwarzschild solution. Birkhoff's theorem *with proof*. Rays and orbits, gravitational redshift, light deflection, perihelion advance. Shapiro time delay. [4]

Einstein's Equations coupled to matter
Concept of an energy momentum tensor. Maxwell stress tensor and perfect fluid as examples. Importance of Bianchi identities. The emergence of the cosmological term. Simple exact solutions: Friedmann–Lemaitre metrics, the Einstein Static Universe. Hubble expansion and redshift. De-Sitter spacetime, mention of Dark Energy and the problem of Dark matter. Notion of geodesic completeness and definition of a spacetime singularity. Schwarzschild and Friedmann-Lemaitre spacetimes as examples of spacetimes with singularities. [4]

Linearized theory
Linearized form of the vacuum equations. De-Donder gauge and reduction to wave equation. Comparison of linearized point mass solution with exact Schwarzschild solution and identification of the mass parameter. Gravitational waves in linearized theory. *The quadrupole formula for energy radiated.* Comparison of linearized gravitational waves with the exact pp-wave metric. [4]

Gravitational collapse and black holes
Non-singular nature of the surface $r = 2M$ in the Schwarzschild solution using Finkelstein and Kruskal coordinates. The idea of an event horizon and the one-way passage of timelike geodesics through it. Qualitative account of idealized spherically symmetric collapse. The final state: statement of Israel's Theorem. *Qualitative description of Hawking radiation.* [2]

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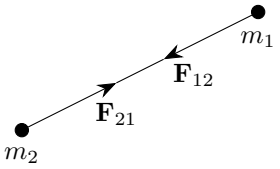
1 Introduction

1.1 Gravity & relativity

Newton's law of gravitation states that the force on mass m_1 at $\mathbf{x}_1$ due to mass m_2 at $\mathbf{x}_2$ is given by

$$\mathbf{F}_{12} = -\frac{Gm_1m_2\left(\mathbf{x}_1 - \mathbf{x}_2\right)}{\left|\mathbf{x}_1 - \mathbf{x}_2\right|^3} \tag{1.1}$$

where G is Newton's gravitational constant.



Note:

> We could distinguish between "active" and "passive" gravitational masses by writing $m_1^{(P)}m_2^{(A)}$ in (1.1), but Newton's 3rd law gives
>
> $$\begin{aligned} &\mathbf{F}_{12} = \mathbf{F}_{21} \\ \implies\ & m_1^{(A)}m_2^{(P)} = m_1^{(P)}m_2^{(A)} \\ \implies\ & \frac{m_1^{(A)}}{m_1^{(P)}} = \frac{m_2^{(A)}}{m_2^{(P)}} \end{aligned}$$
>
> so, without loss of generality, we can set this ratio to 1. [i.e. we have a single notion of gravitational mass.]

From (1.1), we can find the force on a mass m at $\mathbf{x}$ due to matter distribution with density $\rho(\mathbf{x})$; this is given by

$$\mathbf{F} = m\mathbf{g}(\mathbf{x})$$

where $\mathbf{g}(\mathbf{x}) = -\boldsymbol{\nabla}\Phi$ is the *gravitational field*, and $\Phi(\mathbf{x})$ is the *gravitational potential*. The gravitational potential satisfies

$$\boldsymbol{\nabla}^2\Phi = 4\pi G\rho.$$

Note that the gravitational field is the force per unit mass, whilst the gravitational potential is the potential energy per unit mass. Now, consider Newton's 2nd Law:

$$\mathbf{F} = m^{(I)}\ddot{\mathbf{x}} = m\mathbf{g}(\mathbf{x})$$

where $m^{(I)}$ is the *inertial mass*.

Remarkably, without loss of generality,

$$\frac{m^{(I)}}{m} = 1$$

is the same for all bodies (with any constant of proportionality absorbed in the definition for unit of force), and has been experimentally tested to $\mathcal{O}(10^{-12})$ by the Eötvös experiment.

Hence, we have that

$$\textit{inertial mass} = \textit{gravitational mass}$$

but we have no explanation of this in Newtonian theory.

Newtonian gravity (NG) and *Newtonian dynamics* (ND) are very successful (e.g. predicting planetary orbits), but ND only applies when speeds $v \ll c$. Otherwise, we must use *special relativity* (SR).

Recall:

Both ND and SR have a preferred class of reference frames, called *inertial frames* (IFs), in which objects move in straight lines at constant velocity, which is Newton's 1st law.

In ND: assume that absolute time is the same in all inertial frames, related by *Galilean transformations*.

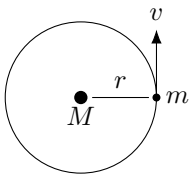
In SR: assume that the speed of light c is the same in all inertial frames, related by *Lorentz transformations*, which interlink space and time. For spacelike-separated events, there is no notion of simultaneity.

Now, note that positions $\mathbf{x}_1$, $\mathbf{x}_2$ in (1.1) are assumed to be measured at the same instant of absolute time, so this is not consistent with SR.

When is modification likely to be important?

Consider a small body of mass m in circular motion of radius r about mass M. By Newton's 2nd law, we have:

$$\Phi = -\frac{GM}{r}$$
$$\frac{v^2}{r} = \frac{GM}{r^2}$$
$$\frac{v^2}{c^2} = \frac{GM}{r}\frac{1}{c^2} = \frac{|\Phi|}{c^2}$$



so SR corrections are small for $\frac{v^2}{c^2} \ll 1$. This suggests that NG is adequate for $\frac{|\Phi|}{c^2} \ll 1$. In the solar system, this is $\frac{|\Phi|}{c^2} \sim 10^{-5}$, which is consistent with this regime.

1.2 Gravitational fields & accelerating frames

Einstein's "happiest thought" was the following:

The gravitational field has only a local existence because an observer falling freely detects no field (locally).

We discuss this idea with Newtonian theory. Consider coordinates $(\mathbf{x}, t)$, and call this choice of frame $\mathcal{S}$. The motion of a small body in $\mathcal{S}$ with gravitational field $\mathbf{g}(\mathbf{x}, t)$ is given by

$$\ddot{\mathbf{x}} = \mathbf{g}(\mathbf{x}, t).$$

Define new coordinates $(\tilde{\mathbf{x}}, t)$ for frame $\tilde{\mathcal{S}}$, with the same time t, given by

$$\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{b}(t)$$
$$\ddot{\tilde{\mathbf{x}}} = \ddot{\mathbf{x}} - \ddot{\mathbf{b}}(t),$$

so $\ddot{\tilde{\mathbf{x}}} = \tilde{\mathbf{g}}$, where $\tilde{\mathbf{g}}(\mathbf{x}, t) = \mathbf{g}(\mathbf{x}, t) - \ddot{\mathbf{b}}(t)$.

Note:

The origin of $\tilde{\mathcal{S}}$ follows the trajectory $\mathbf{x}(t) = \mathbf{b}(t)$ in frame $\mathcal{S}$.

Now, consider the case where $\mathbf{g}$ is uniform in $\mathcal{S}$ (so is independent of $\mathbf{x}$).

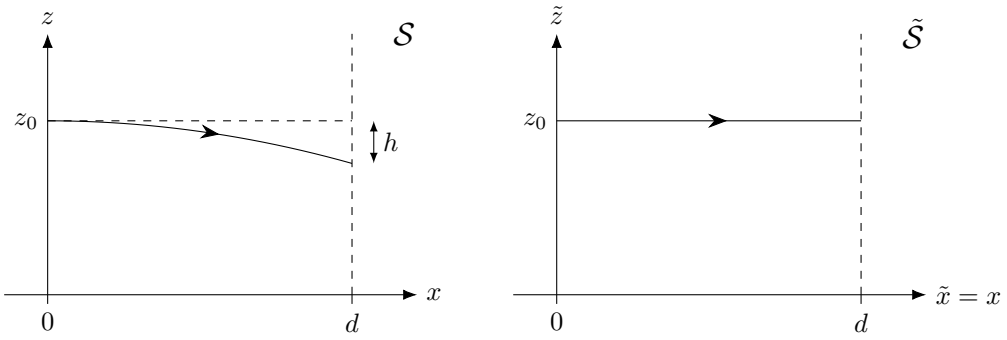
- If $\mathbf{g} = \mathbf{0}$, we have $\tilde{\mathbf{g}} = -\ddot{\mathbf{b}}(t)$ in $\mathcal{S}$, which is generally non-zero.
- If $\mathbf{g} \neq \mathbf{0}$, then we can choose $\mathbf{b}(t)$ such that $\tilde{\mathbf{g}} = \mathbf{0}$ in $\tilde{\mathcal{S}}$. In this case, we call $\tilde{\mathcal{S}}$ a *freely falling frame* (FFF).

1.2.1 Gravitational bending of light

Let $\mathcal{S}$ be the lab frame (e.g. on Earth's surface), where we have gravitational field (approximately)

$$\mathbf{g} = -g\hat{\mathbf{z}}$$

and let $\tilde{\mathcal{S}}$ be a FFF (and also an IF), where $\tilde{\mathbf{g}} = \mathbf{0}$ by choosing $\mathbf{b} = -\frac{1}{2}gt^2\hat{\mathbf{z}}$.



Consider light fired horizontally in the x-direction. In $\tilde{\mathcal{S}}$, light travels in a straight line

$$\tilde{z} = z_0,$$

whilst in $\mathcal{S}$, light travels on a parabola

$$z = z_0 - \frac{1}{2}gt^2.$$

In horizontal distance $d = ct$, light ray falls through a vertical distance

$$h = \frac{1}{2}gt^2 = \frac{gd^2}{2c^2},$$

so on Earth ($g \approx 10\,\mathrm{m\,s^{-2}}$), over a distance $d = 1\,\mathrm{km}$, $h \approx 5 \times 10^{-11}\,\mathrm{m}$.

In the above, we have used a simple version of *Einstein's Equivalence Principle*:

> *In an isolated laboratory, there is no experiment which can distinguish between a uniform gravitational field and uniform acceleration. Moreover, the results of an experiment in a freely falling frame are the same as those in an inertial frame.*

1.2.2 Gravitational redshift/blueshift

Consider two observers Alice (A) and Bob (B), where Bob is located on the surface of the Earth, and Alice is a distance h above Bob.

Alice sends a light signal at time t_A vertically downwards, and Bob receives this at time t_B, viewing this in the frame $\mathcal{S}$ where the Earth is at rest, as above, as well as the FFF $\tilde{\mathcal{S}}$.

For Alice, she is located at

$$z = h\,, \qquad \tilde{z} = h + \frac{1}{2}gt^2\,,$$

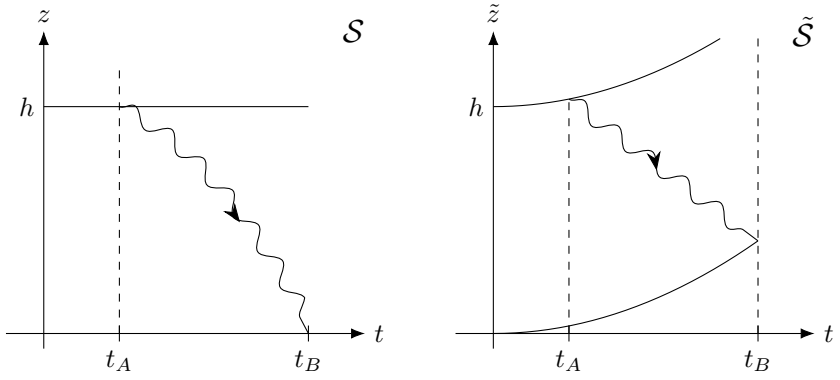
whilst Bob is located at

$$z = 0\,, \qquad \tilde{z} = \frac{1}{2}gt^2\,,$$

in the frames $\mathcal{S}$ and $\tilde{\mathcal{S}}$.

Note:

In the diagram, light does **not** follow a straight path in the frame $\mathcal{S}$ (due to gravitational bending), but does travel in a straight line in the freely falling frame $\tilde{\mathcal{S}}$.



$\tilde{\mathcal{S}}$ is a freely falling frame, so is a local inertial frame, so light signals move in straight lines with speed c in $\tilde{\mathcal{S}}$. Hence,

$$h + \frac{1}{2}gt_A^2 - \frac{1}{2}gt_B^2 = c(t_B - t_A)$$

Now, suppose light signals are sent by Alice at small time intervals Δt_A, and received by Bob at small time intervals Δt_B. Since the equation above is also true each time a light signal is sent, with $t_A \mapsto t_A + \delta t_A$ and $t_B \mapsto t_B + \delta t_B$, then, working to first order in Δt_A, Δt_B, we have

$$gt_A \Delta t_A - gt_B \Delta t_B = c(\Delta t_B - \Delta t_A)$$

$$\implies \frac{\Delta t_B}{\Delta t_A} = \frac{1 + gt_A/c}{1 + gt_B/c}.$$

All of this only uses non-relativistic motion, so $gt_A, gt_B \ll c$, so we can expand

$$\frac{\Delta t_B}{\Delta t_A} \simeq 1 - \frac{g}{c}(t_B - t_A) = 1 - \frac{gh}{c^2}.$$

Hence, to leading order,

$$\frac{\Delta t_B}{\Delta t_A} = 1 - \frac{1}{c^2}(\Phi_A - \Phi_B)$$

which is consistent with the assumption of weak gravity $\frac{|\Phi|}{c^2} \ll 1$.

Regarding each interval as ticks of a clock, we find that time runs at different rates depending on the gravitational potential.

We can write the same result in terms of frequencies $\nu = \frac{1}{\Delta t}$, so that

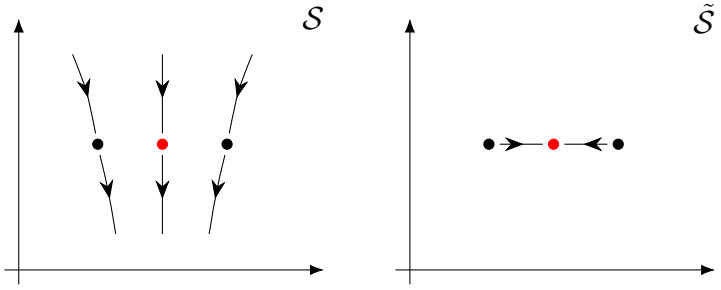
$$\frac{\nu_B}{\nu_A} = 1 + \frac{1}{c^2}(\Phi_A - \Phi_B)$$

Since Alice is above the Earth's surface, we have $\Phi_A > \Phi_B \implies \nu_B > \nu_A$. Hence, we expect there to be gravitational blueshift, and exchanging the positions of Alice and Bob would give rise to gravitational redshift.

This has been experimentally tested, and was first confirmed by the Pound–Rebka experiment (1960) with light falling inside a tower of $h = 22.5\,\mathrm{m}$, so $\frac{gh}{c^2} \sim 10^{-15}$, which has eventually been confirmed to $\pm 1\%$. This effect applies to GPS systems, which now take this into account.

Non-uniform fields

If $\mathbf{g}$ (in $\mathcal{S}$) is not uniform, the best we can do is to choose $\tilde{\mathcal{S}}$ to make $\tilde{\mathbf{g}} = 0$ at a point, by choosing a locally freely falling frame, which still sees the effects of gravity if we look close enough.



In $\mathcal{S}$, we can compare nearby trajectories, with spatial coordinate indices $i = 1, 2, 3$,

$$\ddot{x}_i = g_i(\mathbf{x}, t)$$
$$\ddot{(x_i + h_i)} = g_i(\mathbf{x} + \mathbf{h}, t)$$

By Taylor expanding,

$$g_i(\mathbf{x} + \mathbf{h}, t) = g_i(\mathbf{x}, t) + h_j \partial_j g_i(\mathbf{x}, t) + \mathcal{O}(h^2),$$

so to leading order in h,

$$\ddot{h}_i - h_j \partial_j g_i(\mathbf{x}, t) = 0$$
$$\ddot{h}_i + E_{ij} h_j = 0$$

where $E_{ij} = -\partial_j g_i$ is the *tidal tensor*.

In $\tilde{\mathcal{S}}$, the gravitational field and tidal tensor are related to those in $\mathcal{S}$ by

$$\tilde{g}_i = g_i - \ddot{b}_i(t) \qquad \& \qquad \tilde{E}_{ij} = E_{ij}$$

1.2.3 Special relativity & Minkowski space

An *inertial frame* in special relativity is described by *Minkowski space(time)* with *points* (or *events*) labelled by *coordinates*

$$x^\mu \quad (\mu = 0,\ 1,\ 2,\ 3) \quad \text{or} \quad (x^0, x^1, x^2, x^3) = (ct, \mathbf{x}),$$

equipped with an inner product defined by the *Minkowski metric*

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1)$$

and *line element*

$$\begin{aligned} ds^2 &= \eta_{\mu\nu}\, dx^\mu dx^\nu \\ &= -(dx^0)^2 + (dx^1)^2 + (dx^2)^2 + (dx^3)^2 \\ &= -c^2 dt^2 + d\mathbf{x}^2 \end{aligned}$$

For spacetime indices (labelled by Greek letters $\alpha, \beta, \mu, \nu, \ldots$), the position of indices is important, and the Einstein summation convention applies when up and down indices are the same, which are said to be *contracted*.

The metric gives the invariant separation between nearby points/events x^μ and $x^\mu + \delta x^\mu$:

$$\begin{aligned} \delta s^2 &= \eta_{\mu\nu}\, \delta x^\mu \delta x^\nu \\ &= -c^2 \delta t^2 + \delta \mathbf{x}^2 \end{aligned}$$

Definition 1.1

We say that two events in spacetime are

- *spacelike-separated* if $\delta s^2 > 0$.
  - δs is said to be the *proper distance*.
- *lightlike-separated* or *null-separated* if $\delta s^2 = 0$.
- *timelike-separated* if $\delta s^2 < 0$.
  - $\delta\tau$ is said to be the *proper time*, where $\delta s^2 = -c^2\delta\tau^2$.

Consider inertial frames related by the coordinate transformation

$$\tilde{x}^\mu = \Lambda^\mu{}_\nu x^\nu,$$

where Λ is constant, where we note that this family of transformations fixes the origin

$$x^\mu = 0 \iff \tilde{x}^\mu = 0,$$

but we could also consider translations in spacetime.

To ensure $ds^2 = \eta_{\mu\nu}dx^\mu dx^\nu = \eta_{\alpha\beta}d\tilde{x}^\alpha d\tilde{x}^\beta$ is invariant, and using

$$d\tilde{x}^\alpha = \Lambda^\alpha{}_\mu x^\mu \qquad \text{and} \qquad d\tilde{x}^\beta = \Lambda^\beta{}_\nu x^\nu,$$

the coordinate transformation satisfies

$$\eta_{\alpha\beta}\Lambda^\alpha{}_\mu\Lambda^\beta{}_\nu = \eta_{\mu\nu},$$

so preserves the Minkowski metric.

As matrices, this can be written as

$$(\Lambda^T)_\mu{}^\alpha\,\eta_{\alpha\beta}\,\Lambda^\beta{}_\nu = \eta_{\mu\nu} \qquad \text{or} \qquad \Lambda^T\eta\Lambda = \eta.$$

All such transformations Λ form the *Lorentz group*. If we also considered the group of translations, we would instead have the *Poincaré group*.

It is often helpful to consider the block forms of these transformations.

Example:

A *Lorentz boost* in the x^1 direction with speed $v = c\tanh\theta$ is given by

$$\Lambda = \begin{pmatrix} \cosh\theta & \sinh\theta & \multicolumn{2}{c}{\multirow{2}{*}{$\mathbf{0}$}} \\ -\sinh\theta & \cosh\theta & & \\ \multicolumn{2}{c}{\multirow{2}{*}{$\mathbf{0}$}} & 1 & 0 \\ & & 0 & 1 \end{pmatrix}.$$

Example:

Any spatial orthogonal transformation on $\mathbb{R}$ is a Lorentz transformation, given by

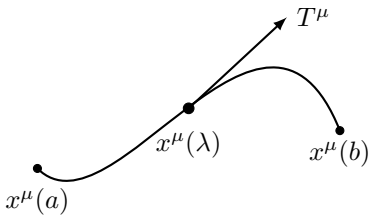
$$\Lambda = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R & \\ 0 & & & \end{pmatrix}.$$

Proper distance & time and geodesics

Consider the parameterised curve $x^\mu(\lambda)$ with parameter $\lambda \in [a, b]$, with tangent vector

$$T^\mu = \frac{dx^\mu}{d\lambda} = \dot{x}^\mu,$$

where $\delta x^\mu = T^\mu \, \delta\lambda$ to order $\delta\lambda$.



Definition 1.2

We say that a curve is

- *spacelike* if $\eta_{\mu\nu} T^\mu T^\nu > 0$.
- *lightlike* or *null* if $\eta_{\mu\nu} T^\mu T^\nu = 0$.
- *timelike* if $\eta_{\mu\nu} T^\mu T^\nu < 0$..

at all points on the curve.

Definition 1.3

The *proper distance* $s(\lambda)$ on spacelike curves is such that

$$\frac{ds}{d\lambda} = (\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{\frac{1}{2}} = \mathcal{L}.$$

The *proper time* $\tau(\lambda)$ on timelike curves is such that

$$c\,\frac{d\tau}{d\lambda} = (-\eta_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)^{\frac{1}{2}} = \mathcal{L}.$$

Note:

> For spacelike and timelike curves, we have different Lagrangians $\mathcal{L}$, since $\eta_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ takes different signs for each family of curves.

The proper distance between endpoints of a spacelike curve is

$$S = \int_a^b \mathcal{L}\, d\lambda = s(b) - s(a),$$

and the proper time between endpoints of a timelike curve is

$$cT = \int_a^b \mathcal{L}\, d\lambda = c\tau(b) - c\tau(a).$$

Under a change in parameterisation $\lambda \mapsto \hat{\lambda}$ (where this transformation is invertible),

$$\hat{T}^\mu = \frac{dx^\mu}{d\hat{\lambda}} = \frac{d\lambda}{d\hat{\lambda}}\frac{dx^\mu}{d\lambda} = \frac{d\lambda}{d\hat{\lambda}}\, T^\mu,$$

so the tangent vector scales in size, but maintains its direction, under reparameterisation.

Note:

> For an allowable reparameterisation of this curve, we must have
>
> $$\frac{d\lambda}{d\hat{\lambda}} \neq 0.$$
>
> A more rigorous treatment (for a fair amount of this course) can be found in Part II *Differential Geometry*.

S and T are unaffected, since (as seen for line integrals from Part IA *Vector Calculus*)

$$\int_a^b \mathcal{L}(T^\mu)\, d\lambda = \int_{\hat{a}}^{\hat{b}} \mathcal{L}(\hat{T}^\mu)\, d\hat{\lambda}$$

is parameterisation-invariant.

Definition 1.4

A *geodesic* is a curve obtained by extremising S or T for fixed endpoints.

Concentrating on T, we want the first variation $\delta T = 0$ to first-order changes in $\delta x^\mu(\lambda)$ with fixed endpoints $\delta x^\mu(a) = \delta x^\mu(b) = 0$.

Solution are given by the *Euler-Lagrange equations* (E-L) (see Part IB *Variational Principles*):

$$\frac{d}{d\lambda}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha}\right) - \frac{\partial \mathcal{L}}{\partial x^\alpha} = 0$$

where, noting $\mathcal{L}$ is independent of x^α,

$$\frac{\partial \mathcal{L}}{\partial x^\alpha} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} = -\frac{1}{\mathcal{L}}\eta_{\alpha\nu}\dot{x}^\nu.$$

Multiplying by $\eta^{\mu\alpha}$, where $\eta^{\mu\alpha}\eta_{\alpha\nu} = \delta^\mu{}_\nu$, we have

$$\frac{d}{d\lambda}\left(\frac{1}{\mathcal{L}}\dot{x}^\mu\right) = 0$$

This looks complicated, but we can choose the proper time $\lambda = \tau$ to be our parameter, such that $L = c$ is constant, and so, the Euler-Lagrange equations become

$$\frac{d^2x^\mu}{d\tau^2} = 0$$

$$\implies \quad U^\mu = \frac{\partial x^\mu}{\partial \tau} \quad \text{4-velocity is constant}$$

Hence, timelike geodesics follow straight lines in Minkowski space, given by

$$x^\mu(\tau) = U^\mu \tau + x^\mu(0).$$

Note also, if $x^\mu = (ct, \mathbf{x})$, choosing t to be our parameter, and $\mathbf{v} = \frac{\mathrm{d}\mathbf{x}}{dt}$, then the tangent vector is

$$\frac{dx^\mu}{dt} = V^\mu = (c, \mathbf{v}).$$

Since, for timelike curves parameterised by the proper time τ,

$$V^\mu = \frac{d\tau}{dt}U^\mu, \qquad \eta_{\mu\nu}U^\mu U^\nu = -c^2,$$

then, we have

$$\eta_{\mu\nu}V^\mu V^\nu = -c^2 + \mathbf{v}^2 = \left(\frac{d\tau}{dt}\right)^2(-c^2) = \left(\frac{d\tau}{dt}\right)^2 \eta_{\mu\nu}U^\mu U^\nu$$

$$\implies \quad U^\mu = \gamma(\mathbf{v})V^\mu$$

where $\gamma(\mathbf{v}) = \left(1 - \mathbf{v}^2/c^2\right)^{-\frac{1}{2}}$ is the *Lorentz factor*.

Remarks:

(i) Extremising T gives the trajectories of massive particles as timelike geodesics, and we can generalise this to curved spacetime.

(ii) Similarly, extremising S gives straight lines as shortest path between spacelike separated points.

(iii) How can we describe trajectories for massless particles? How can we make sense of a null geodesic? We will answer these in due course...

We can improve our treatment of the above!

Consider curves $x^\mu(\lambda)$ with fixed endpoints as before, but no further restriction to being spacelike or timelike, where we now extremise

$$I = \int_a^b L \, d\lambda, \qquad L = -\eta_{\mu\nu}\dot{x}^\mu \dot{x}^\nu.$$

The Euler-Lagrange equations are now simpler

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^\alpha}\right) = -2\eta_{\alpha\nu}\ddot{x}^\nu = 0$$

i.e. we have

$$\frac{d^2x^\mu}{d\lambda^2} = 0$$

$$\implies \qquad x^\mu(\lambda) = T^\mu \lambda + x^\mu(0),$$

with T^μ constant and this treatment allows T^μ to be timelike, spacelike or lightlike.

This gives trajectories of massive and massless particles, so automatically gives a convenient affine parameterisation, where the only freedom in reparameterisation is affine transformations $\lambda \mapsto A\lambda + B$, where A is non-zero. Using this, we can set $\lambda \mapsto \tau$ for timelike geodesics.

2 Metrics and geodesics on manifolds

2.1 Manifolds & metrics

2.1.1 Manifolds

A *manifold* is described locally by a set of coordinates $\{x^\mu\}$, and the number of coordinates is the *dimension* of the manifold.

More formally, (see Part II *Differential Geometry*)

- We say that $\mathcal{M} \subset \mathbb{R}^N$ is an *n-dimensional manifold* if for all points $p \in \mathcal{M}$, there exists an open neighbourhood $V \subset \mathcal{M}$ of p, which is diffeomorphic to an open subset $U \subset \mathbb{R}^n$.
- The smooth diffeomorphism $\varphi : U \to V$ is called a *parameterisation* of the neighbourhood V.
- The inverse diffeomorphism $\varphi^{-1} : V \to U$ (which is also smooth by definition of a smooth diffeomorphism φ) is called a *coordinate system* or a *chart* on V.
- If $\varphi^{-1} = (x^1, \dots, x^n)$, then the smooth functions $x^1, \dots, x^n$ are *coordinate functions*.
  - (i.e. equipped with a coordinate system, we can assign each point $p \in V \subset \mathcal{M}$ with an n-tuple $(x^1(p), x^2(p), \dots, x^n(p))$ from U.)

Example:

- Euclidean space $\mathbb{R}^3$ has dimension 3.
- Minkowski space has dimension 4

In general, a manifold is **not** a vector space, and coordinates are not components of position vectors.

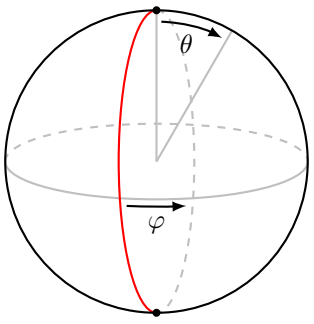
Example:

A 2-dimensional unit sphere S^2 can be described by spherical polar coordinates

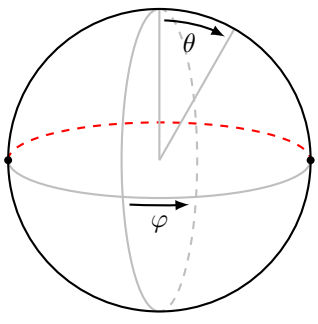
$$x^1 = \theta, \quad x^2 = \varphi$$

with $\theta \in (0, \pi)$, $\varphi \in (0, 2\pi)$, which is well-defined away from the meridian ($\varphi = 0,\ 2\pi$), as shown in the left figure.

To describe all points, we can use another set of coordinates $\{\tilde{x}^\mu\}$ with respect to different, non-intersecting meridian, an example of which is shown in the right figure.



Meridian:
$\varphi = 0, \quad \theta \in [0, \pi]$



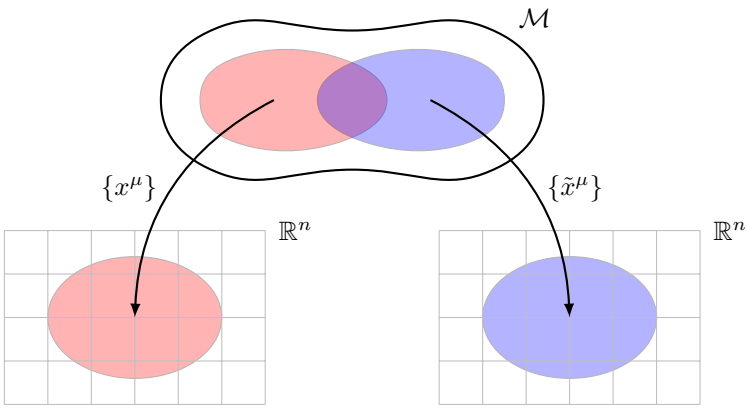
Meridian:
$\theta = \frac{\pi}{2}, \quad \varphi \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

Most notably, S^2 does **not** form a vector space under usual vector addition and scalar multiplication.

Where these two sets of coordinates overlap, $\{x^\mu\}$ and $\{\tilde{x}^\mu\}$ are smoothly related.

In general, a *smooth manifold* $\mathcal{M}$ (of dimension n) is defined by a set of *coordinate patches* (open sets and domains) and *charts*. On $\mathcal{M}$, for a sufficiently small neighbourhood of any point on $\mathcal{M}$, it looks "locally like $\mathbb{R}^n$".

This allows for sensible definitions of smooth functions, vectors, tensors, etc. on $\mathcal{M}$.



Small changes in coordinates are related by (to first order)

$$\delta\tilde{x}^\alpha = \frac{\partial\tilde{x}^\alpha}{\partial x^\mu}\,\delta x^\mu\,, \qquad \delta x^\mu = \frac{\partial x^\mu}{\partial\tilde{x}^\alpha}\delta\,\tilde{x}^\alpha$$

Matrix of partial derivatives must satisfy

$$\det\left(\frac{\partial\tilde{x}^\alpha}{\partial x^\mu}\right) \neq 0.$$

Chain rule implies that

$$\frac{\partial x^\mu}{\partial\tilde{x}^\alpha}\,\frac{\partial\tilde{x}^\alpha}{\partial x^\nu} = \delta^\mu{}_\nu\,, \qquad \frac{\partial\tilde{x}^\alpha}{\partial x^\nu}\,\frac{\partial x^\nu}{\partial\tilde{x}^\beta} = \delta^\alpha{}_\beta$$

Example:

$\mathbb{R}^2$ (which we will revisit many times in different examples) can be equipped with either Cartesian or plane polar coordinates, given by

$$x^1 = x, \qquad x^2 = y,$$
$$\tilde{x}^1 = r, \qquad \tilde{x}^2 = \theta,$$

with $r \in (0,\infty)$, $\theta \in (0, 2\pi)$.

These coordinates systems are related by

$$x^1 = r\cos\theta, \qquad x^2 = r\sin\theta$$

with Jacobian matrix

$$J = \left(\frac{\partial x^\mu}{\partial\tilde{x}^\alpha}\right) = \begin{pmatrix}\cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi\end{pmatrix}$$

with $\det J = r \neq 0$.

These sets of coordinates have overlap on $\mathbb{R}^2 \setminus \{x^1 \geqslant 0,\, x^2 = 0\}$.

Note:

We may denote the components of a tensor as a matrix by writing round brackets around the tensor, with the first index indicating rows, and second index indicating columns, e.g. $(T^\mu{}_\nu)$ or $(g_{\mu\nu})$ (as above).

2.1.2 Metrics

A *metric* on a manifold $\mathcal{M}$ (of dimension n) is an additional structure specified by a symmetric tensor with components

$$g_{\mu\nu}(x) = g_{\nu\mu}(x)$$

with respect to some (or any) set of coordinates $\{x^\mu\}$, which can be described by an $n \times n$ mnon-singular matrix. (We will meet a more general definition of a tensor later.)

Along with the metric, we can define a *line element*

$$ds^2 = g_{\mu\nu}(x)dx^\mu dx^\nu,$$

which defines the invariant separation between nearby points on $\mathcal{M}$.

Now, in a different coordinate system $\{\tilde{x}^\alpha\}$,

$$ds^2 = g_{\alpha\beta}d\tilde{x}^\alpha d\tilde{x}^\beta,$$

so for ds^2 to be invariant, by applying the chain rule, we find that

$$\tilde{g}_{\alpha\beta} = g_{\mu\nu}\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial x^\nu}{\partial \tilde{x}^\beta}$$

As matrices,

$$(g_{\alpha\beta}) = J^T(g_{\mu\nu})J,$$

where J is given as seen in the previous example.

Example:

$\mathbb{R}^2$ with Cartesian coordinates has metric with components

$$(g_{\mu\nu}) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

whilst with respect to polar coordinates, we instead have

$$(\tilde{g}_{\alpha\beta}) = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} = M^T(g_{\mu\nu})M = M^T M$$

where the line element is given by

$$\begin{aligned} ds^2 &= (dx^2)^2 + (dx^2)^2 \\ &= (d\tilde{x}^1)^2 + r^2(d\tilde{x}^2)^2. \end{aligned}$$

In general, $(g_{\mu\nu})$ is diagonalisable (as a quadratic form), since it is a non-singular symmetric matrix, so at any given point, we can choose $\{\tilde{x}^\alpha(x^\mu)\}$ and hence J such that $(\tilde{g}_{\alpha\beta})$. By Sylvester's law of inertia (see Part IB *Linear Algebra*), we can set these diagonal entries to be ± 1, which defines the *signature* of the metric.

Here, we consider

- Euclidean signature with entries $(+1, +1, \ldots, +1)$, which is positive-definite. We mainly look at this in the cases $n = 2$, 3 for background and comparison.
- Lorentzian signature $(-1, +1, \ldots, +1)$, which is indefinite. We mainly look at this in the case $n = 4$ for curved spacetime.

In general, from $g_{\mu\nu}$, we can define the *inverse metric* $g^{\mu\nu}$, such that

$$g^{\mu\alpha}g_{\alpha\nu} = \delta^\mu{}_\nu.$$

In the two cases above, the pair $(\mathcal{M}, g)$ is referred to as a *Riemannian/pseudo-Riemannian manifold*, respectively.

Examples:

- n-dimensional Euclidean space with Cartesian coordinates $\{\tilde{x}^i\}$ has metric

$$g_{ij} = \delta_{ij} = \text{diag}(+1, +1, \ldots, +1)$$

  and we use *Latin indices* for positive-definite signature.

- n-dimensional Minkowski space has metric

$$g_{\mu\nu} = \eta_{\mu\nu} = \text{diag}(-1, +1, \ldots, +1)$$

For the examples above, we see that the inverse metric has $(g^{ij}) = (g_{ij})$, $(g^{\mu\nu}) = (g_{\mu\nu})$.

A more specific example:

- S^2 with spherical polar coordinates

$$x^1 = \theta, \qquad x^2 = \varphi$$

  has line element

$$ds^2 = g_{ij}\, dx^i dx^j = d\theta^2 + \sin^2\theta\, d\varphi^2,$$

  with metric

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\sin^2\theta} \end{pmatrix}.$$

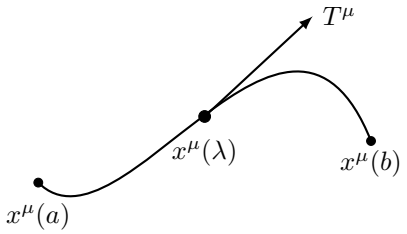
2.2 Curves, trajectories & geodesics

2.2.1 Definitions & key principles in general relativity

Consider a parameterised curve $x^\mu(\lambda)$ on a manifold $\mathcal{M}$, with tangent vector

$$T^\mu = \frac{dx^\mu}{d\lambda} = \dot{x}^\mu$$

in all coordinates $\{x^\mu\}$.



In a different set of coordinates $\tilde{x}^\mu$, we have

$$\tilde{T} = \frac{\mathrm{d}\tilde{x}^\alpha}{d\lambda}.$$

By applying the chain rule,

$$\tilde{T}^\alpha = \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} T^\mu$$

Combining with the transformation rule for metric components in Section 2.1, we have

$$g_{\mu\nu} T^\mu T^\nu = \tilde{g}_{\alpha\beta} \tilde{T}^\alpha \tilde{T}^\beta,$$

so this inner product is independent of coordinates.

Definition 2.1

For $L = -g_{\mu\nu}\dot{x^\mu}\dot{x}^\nu$, then

- The curve is *spacelike* if $L < 0$.
- The curve is *lightlike* or *null* if $L = 0$.
- The curve is *timelike* if $L > 0$.

Furthermore, the *proper distance* $s(\lambda)$ on spacelike curves is such that

$$L = -\left(\frac{ds}{d\lambda}\right)^2,$$

and the *proper time* $\tau(\lambda)$ on timelike curves is such that

$$L = c^2\left(\frac{d\tau}{d\lambda}\right)^2.$$

Definition 2.2

An (affinely parameterised) *geodesic* is a curve $x^\mu(\lambda)$, $\lambda \in [a, b]$, which extremises

$$I = \int_a^b L(x^\mu, \dot{x}^\mu)\, d\lambda$$

where the endpoints of the curve $x^\mu(a)$, $x^\mu(b)$ are fixed.

It follows that L is constant $\left(\frac{dL}{d\lambda} = 0\right)$ along geodesics (see later, since L is quadratic in $\dot{x}$ and invariant under translations in λ), and timelike/spacelike geodesics extremise

$$cT = \int_a^b \left(-g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu\right)^{\frac{1}{2}} d\lambda = c(\tau(b) - \tau(a)),$$

$$S = \int_a^b \left(g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu\right)^{\frac{1}{2}} d\lambda = s(b) - s(a),$$

respectively.

We can choose the affine parameter λ to be

$$\begin{aligned} \lambda = \tau \quad &\longrightarrow \quad L = c^2, \\ \lambda = s \quad &\longrightarrow \quad L = -1. \end{aligned}$$

In general relativity, spacetime is a 4D (pesudo-)Riemannian manifold with metric signature $(-1, +1, +1, +1)$.

- The *trajectory* or *world line* of a *massive* test particle is a *timelike curve*, and the *proper time* along the curve is the time measured by a clock moving with the particle. This is known as the **clock postulate**.
- The trajectory of a passive particle which is *free* moves along a *timelike geodesic*.
  - This optimises the proper time between two points.
- The trajectory of a *massless particle* moves along a *lightlike* or *null geodesic*.

2.2.2 Euler-Lagrange equations

We extremise

$$I = \int_a^b L(q^r, \dot{q}^r, \lambda)\, d\lambda$$

for $q^r(\lambda)$ $(r = 1, 2, \ldots, n)$, $\lambda \in [a, b]$ and $q^r(a)$, $q^r(b)$ are fixed.

Requiring the first variation $\delta I = 0$ to first order for any $\delta q^r(\lambda)$, with $\delta q(a) = \delta q(b) = 0$, we get the *Euler-Lagrange equations*

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{q}^r}\right) - \frac{\partial L}{\partial q^r} = 0$$

First integrals and conservation laws

- L is independent of $q^r \implies \frac{\partial L}{\partial \dot{q}^r}$ constant
- L has no explicit dependence on $\lambda \implies \dot{q}^r \frac{\partial L}{\partial \dot{q}^r} - L$ constant

We can regard these conditions as symmetries of L, and are examples of *Noether's theorem*.

Example:

We look at geodesics in $\mathbb{R}^2$ in polar coordinates, with metric

$$ds^2 = dr^2 + r^2 d\varphi^2,$$

with Lagrangian

$$L = \dot{r}^2 + r^2\dot{\varphi}^2.$$

The Euler-Lagrange equations are

$$r: \qquad \frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0 \qquad \implies \qquad \ddot{r} - r\dot{\varphi}^2 = 0$$

$$\varphi: \qquad \frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{\varphi}}\right) = 0 \qquad \implies \qquad r^2\dot{\varphi} = l \quad \text{constant}$$

Also, we have the first integral

$$\dot{r}\frac{\partial L}{\partial \dot{r}} + \dot{\varphi}\frac{\partial L}{\partial \dot{\varphi}} - L = L \ \text{ constant}.$$

By choosing the affine parameter $\lambda = s$, we can choose $L = 1$.
Hence,

$$\dot{r}^2 + r^2\dot{\varphi}^2 = \dot{r}^2 + \frac{l^2}{r^2} = 1$$

$$\implies \qquad \frac{r}{\sqrt{r^2 - l^2}}\frac{dr}{ds} = \pm 1$$

$$\implies \qquad \sqrt{r^2 - l^2} = \pm s + C$$

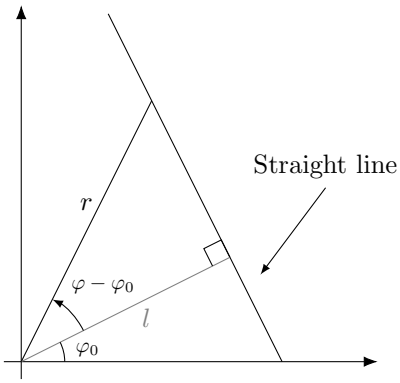
$$\implies \qquad r^2 = l^2 + s^2$$

where we make the choice (without loss of generality) that $C = 0$, since this simply shifts where we measure proper distance s from.

Substituting this back, from the first integral for φ,

$$\dot{\varphi} = \frac{l}{l^2 + s^2} \qquad \implies \qquad \varphi - \varphi_0 = \arctan\left(\frac{s}{l}\right)$$

and we find that a geodesic in $\mathbb{R}^2$ is given by a straight line, as expected.



2.2.3 Geodesic equations & the connection

We extremise

$$I = \int_a^b L(x^\mu, \dot{x}^\mu)\, d\lambda$$

where $x^\mu(a)$, $x^\mu(b)$ are fixed, and the Lagrangian is

$$L = -g_{\alpha\beta}(x)\, \dot{x}^\alpha \dot{x}^\beta .$$

The Euler-Lagrange equations are

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^\mu}\right) - \frac{\partial L}{\partial x^\mu} = 0$$

We have

$$\frac{\partial L}{\partial \dot{x}^\mu} = -2 g_{\mu\beta}\dot{x}^\beta$$

so using the chain rule on the metric $g_{\alpha\beta}(x)$,

$$\begin{aligned} \implies \quad \frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^\mu}\right) &= -2 g_{\mu\beta}\ddot{x}^\beta - 2(\partial_\gamma g_{\mu\beta})\dot{x}^\gamma \dot{x}^\beta , \\ &= -\big(2 g_{\mu\beta}\ddot{x}^\beta + (\partial_\gamma g_{\mu\beta} + \partial_\beta g_{\mu\gamma})\dot{x}^\gamma \dot{x}^\beta\big), \end{aligned}$$

where we symmetrise the final term.

Also,

$$\frac{\partial L}{\partial x^\mu} = -\partial_\mu g_{\alpha\beta}\dot{x}^\alpha \dot{x}^\beta .$$

Hence, the Euler-Lagrange equations are

$$2 g_{\mu\beta}\ddot{x}^\beta + (\partial_\gamma g_{\mu\beta} + \partial_\beta g_{\mu\gamma} - \partial_\mu g_{\beta\gamma})\dot{x}^\beta \dot{x}^\gamma = 0.$$

Multiplying by $\frac{1}{2}g^{\alpha\mu}$, we get the *geodesic equations*

$$\ddot{x}^\alpha + \Gamma_\beta{}^\alpha{}_\gamma\, \dot{x}^\beta \dot{x}^\gamma = 0\,,$$

where the *Levi-Civita* (or *metric*) *connection* is

$$\Gamma_\beta{}^\alpha{}_\gamma = \frac{1}{2} g^{\alpha\mu}(\partial_\beta g_{\mu\gamma} + \partial_\gamma g_{\mu\beta} - \partial_\mu g_{\beta\gamma})\,.$$

Example:

We revisit geodesics in $\mathbb{R}^2$ with polar coordinates, where

$$x^1 = \theta, \qquad x^2 = \varphi.$$

From the Euler-Lagrange equations, we have the geodesic equations

$$\ddot{r} - r\dot{\varphi}^2 = 0, \qquad \ddot{\varphi} + \frac{2}{r}\dot{r}\dot{\varphi} = 0.$$

Hence, we can read off the non-zero connection components, which are

$$\Gamma_2{}^1{}_2 = -r,$$
$$\Gamma_1{}^2{}_2 = \Gamma_2{}^2{}_1 = \frac{1}{r},$$

or alternatively, these can be computed directly from the expression for the Levi-Civita connection.

Remarks:

(i) In terms of tangent vectors, the geodesic equation can also be written as

$$\dot{T}^\alpha + \Gamma_\beta{}^\alpha{}_\gamma \, T^\beta \, T^\gamma = 0 \, .$$

(ii) The alternative notation for the Levi-Civita connection are the *Christoffel symbols*, which are notated by braces

$$\left\{ \begin{array}{cc} \alpha \\ \beta \quad \gamma \end{array} \right\}.$$

(iii) For Minkowski space,

$$g_{\mu\nu} = \eta_{\mu\nu} \qquad \Longrightarrow \qquad \Gamma_\beta{}^\alpha{}_\gamma = 0 \, .$$

This will be useful when we talk about a specific choice of coordinates at any given point on our spacetime manifold: *normal coordinates*. (See Sec 4.2.5)

(iv) As seen from the example above, we can find $\Gamma_\beta{}^\alpha{}_\gamma$ directly from the expression with the metric $g_{\alpha\beta}$, or via the Euler-Lagrange equations.

(v) The choice of sign for the Langragian

$$L = -g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$$

is not important, but this choice of sign is convenient for $L \geqslant 0$ for lightlike geodesics.

However, if the metric has Euclidean signature, it is more natural to take the choice of sign

$$L = g_{ij}\dot{x}^i\dot{x}^j \, .$$

Example:

Recall that we can describe S^2 with spherical polar coordinates

$$x^1 = \theta, \qquad x^2 = \varphi,$$

with line element

$$ds^2 = d\theta^2 + \sin^2\theta \, d\varphi^2,$$

and metric components

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\sin^2\theta} \end{pmatrix}.$$

The Lagrangian on S^2 is given by

$$L = \dot{\theta}^2 + \sin^2\theta \, \dot{\varphi}^2.$$

Since this metric has Euclidean signature, then we can choose proper distance s as our affine parameter so that $L = 1$ for geodesics on S^2.

The Euler-Lagrange equations are

$$\theta : \qquad \ddot{\theta} - \sin\theta \, \cos\theta \, \dot{\varphi}^2 = 0$$

$$\varphi : \qquad \ddot{\varphi} + 2\cot\theta \, \dot{\theta} \, \dot{\varphi} = 0.$$

so the only non-zero components of the connection are

$$\Gamma_2{}^1{}_2 = -\sin\theta \, \cos\theta$$
$$\Gamma_1{}^2{}_2 = \Gamma_2{}^2{}_1 = \cot\theta$$

We have the first integral (since L has no explicit dependence on φ)

$$\sin^2\theta \, \dot{\varphi} = \frac{1}{l} \quad \text{constant}, \qquad l^2 \geqslant 1.$$

The simplest solution is $\theta = \frac{\pi}{2}$, $\varphi = s$ when $l = 1$, and more generally,

$$\operatorname{cosec}^2\theta \, \frac{\mathrm{d}\theta}{\mathrm{d}\varphi} = (\alpha^2 - \cot^2\theta)^{\frac{1}{2}} \quad \text{with} \quad \alpha^2 = l^2 - 1 \geqslant 0 \implies \quad \cot\theta = \alpha \cos(\varphi - \varphi_0)$$

which are equations of great circles on S^2.

2.3 Static spacetimes & Newtonian limits

Consider a spacetime manifold with coordinates $\{x^0 = ct, x^i\}$ such that

$$\begin{aligned} ds^2 &= g_{\mu\nu} \, dx^\mu dx^\nu \\ &= g_{00} \, c^2 dt^2 + g_{ij} dx^i dx^j. \end{aligned}$$

Definition 2.3

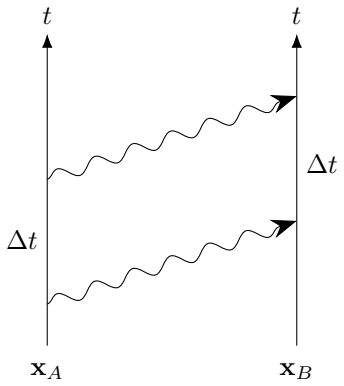
We say that the above metric is

- *stationary* if $\partial_t g_{\mu\nu} = 0$.
- *static* if it is stationary, and furthermore, $g_{0i} = g_{i0} = 0$.

For a static metric, the only non-zero components of the metric are $g_{00}(\mathbf{x})$, $g_{ij}(\mathbf{x})$ with $\mathbf{x} = (x^1, x^2, x^3)$.

2.3.1 Gravitational redshift

Consider Alice (A) and Bob (B) at fixed locations $\mathbf{x}_A$, $\mathbf{x}_B$. Alice sends and Bob receives light signals, which travel along null geodesics.



Since we work with a metric $g_{\mu\nu}$ which is stationary, therefore independent of time, the coordinate time interval Δt between light signals is the same at $\mathbf{x}_A$ and $\mathbf{x}_B$.

Changes in proper time at $\mathbf{x}_A$ and $\mathbf{x}_B$ are given by

$$\begin{aligned}(\Delta\tau_A)^2 &= -g_{00}(\mathbf{x}_A)(\Delta t)^2\\ (\Delta\tau_B)^2 &= -g_{00}(\mathbf{x}_B)(\Delta t)^2\end{aligned}$$

$$\implies \quad \frac{\Delta\tau_B}{\Delta\tau_A} = \left[\frac{-g_{00}(\mathbf{x}_B)}{-g_{00}(\mathbf{x}_A)}\right]^{\frac{1}{2}}$$

For weak gravity, set $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, with $h_{\mu\nu}$ small.

In particular,

$$\begin{aligned}g_{00} &= \eta_{00} + h_{00}\\ &= -1 + h_{00}\end{aligned}$$

$$\begin{aligned}\implies \quad \frac{\Delta\tau_B}{\Delta\tau_A} &= \left[\frac{1 - h_{00}(\mathbf{x}_B)}{1 - h_{00}(\mathbf{x}_A)}\right]^{\frac{1}{2}}\\ &\approx 1 - \frac{1}{2}\big(h_{00}(\mathbf{x}_B) - h_{00}(\mathbf{x}_A)\big)\end{aligned}$$

From Section 1.2,

$$\frac{\Delta\tau_B}{\Delta\tau_A} = 1 + \frac{1}{c^2}\big(\Phi(\mathbf{x}_B) - \Phi(\mathbf{x}_A)\big)\,,$$

where Φ is the Newtonian potential.

We deduce that

$$h_{00}(\mathbf{x}) = -\frac{2}{c^2}\,\Phi(\mathbf{x})\,,$$

which is, indeed, small from the assumption $\frac{|\Phi|}{c^2} \ll 1$.

Hence,

$$g_{00}(\mathbf{x}) = -\left(1 + \frac{2\Phi}{c^2}\right)\,.$$

2.3.2 Newtonian limit

Now consider non-relativistic motion in the weak field metric (as seen above in Section 2.3.1):

$$g_{00} = -1 + h_{00}, \qquad g_{ij} = \delta_{ij} + h_{ij}\,,$$

where h_{00}, h_{ij} are small.

Consider a timelike geodesic, choosing proper time τ to be the choice of affine parameter, so we have the first integral

$$L = -g_{00}\,c^2\left(\frac{dt}{d\tau}\right)^2 - g_{ij}\,\frac{dx^i}{d\tau}\frac{dx^j}{d\tau} = c^2\,. \tag{2.1}$$

From the geodesic equations,

$$t: \qquad -g_{00}c^2\frac{dt}{d\tau} = E \quad \text{constant} \tag{2.2}$$

$$x^i: \qquad \frac{d^2x^i}{d\tau^2} = -\Gamma_{\alpha\ \beta}^{\ i}\,\frac{dx^\alpha}{d\tau}\,\frac{dx^\beta}{d\tau} \tag{2.3}$$

We re-express these in terms of $v^i = \frac{\mathrm{d}x^i}{\mathrm{d}t} = \frac{dx^i}{d\tau}\frac{d\tau}{dt} \ll c$.

(2.1):
$$\left(\frac{d\tau}{dt}\right)^2 = -g_{00} - \frac{1}{c^2}g_{ij}v^iv^j$$
$$= \left(1 + \frac{2\Phi}{c^2}\right) - \frac{1}{c^2}\delta_{ij}v^iv^j$$

where we work to $\mathcal{O}(h, \Phi, v^2)$.

For $\Phi = 0$, we have

$$\left(\frac{d\tau}{dt}\right)^2 = \frac{1}{\gamma(\mathbf{v})^2}\,,$$

which we have seen in special relativity.

By using a Taylor expansion in (2.1),

(2.2):
$$E = \left(1 + \frac{2\Phi}{c^2}\right)c^2\left(1 - \frac{\Phi}{c^2} + \frac{1}{2}\frac{1}{c^2}\delta_{ij}v^iv^j\right)$$
$$= c^2 + \frac{1}{2}\delta_{ij}v^iv^j + \Phi + \ldots\,,$$

so the t equation gives the statement for *conservation of energy* (per unit mass), where we see that the terms written above correspond to rest energy, kinetic energy, and potential energy, respectively.

(2.3):
$$\frac{\mathrm{d}^2x^i}{d\tau^2} = -\Gamma_{\alpha\ \beta}^{\ i}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}$$

Note that, by looking at our expression for the connection in terms of the metric, $\Gamma_{\alpha\ \beta}^{\ \gamma} = \mathcal{O}(h)$, so the only leading order contribution in the summation of this equation is when $\alpha = \beta = 0$, and we take $\frac{dt}{d\tau} = 1 + \mathcal{O}(h, v^2)$ by Taylor expansion.

Working to this order, (2.3) simplifies to

$$\frac{d^2x^i}{dt^2} = \frac{dv^i}{dt} = -\Gamma_{0\ 0}^{\ i}\,c^2\,.$$

Now, since the only relevant connection component, to this order, is

$$\begin{aligned}\Gamma_0{}^i{}_0 &= \frac{1}{2}g^{i\alpha}(2\,\partial_0 g_{\alpha 0} - \partial_\alpha g_{00}) \\ &= \frac{1}{2}g^{ij}(-\partial_j g_{00}) \\ &= \frac{1}{2}\delta^{ij}(-\partial_j h_{00}) \\ &= \frac{1}{2}\delta^{ij}\,\partial_j\left(\frac{2\Phi}{c^2}\right) \\ &= \frac{1}{c^2}\delta^{ij}\,\partial_j\Phi\,,\end{aligned}$$

where we have used the fact that the metric is stationary in the second equality.

(2.3) then gives us *Newton's 2nd law* with a gravitational potential Φ:

$$\frac{dv^i}{dt} = -\delta^{ij}\,\partial_j\Phi\,.$$

Note:

h_{ij} was nowhere used in the above argument.

2.4 Changing coordinates & the equivalence principle

Consider a metric in two sets of coordinates $\{x^\mu\}$, $\{\tilde{x}^\alpha\}$, with components related by

$$\tilde{g}_{\alpha\beta} = g_{\mu\nu}\frac{\partial x^\mu}{\partial\tilde{x}^\alpha}\frac{\partial x^\nu}{\partial\tilde{x}^\beta}\,.$$

At a given point, (without loss of generality, take this to be at the origin O)

$$\begin{aligned} g_{\mu\nu} &= \eta_{\mu\nu}, & x^\mu &= 0\,, \\ \tilde{g}_{\mu\nu} &= \eta_{\mu\nu}, & \tilde{x}^\mu &= 0\,. \end{aligned}$$

Now, consider the behaviour as we move away from O in some small neighbourhood of O, so to first order,

$$g_{\mu\nu} = \eta_{\mu\nu} + C_{\mu\nu\rho}x^\rho + \dots\,,$$

with $C_{\mu\nu\rho} = C_{\nu\mu\rho}$ by symmetry of the metric. Again, note that this all holds only at a point.

Consider also a change of coordinates, to second order,

$$\tilde{x}^\mu = x^\mu + \frac{1}{2}A^\mu{}_{\nu\rho}x^\nu x^\rho + \dots\,, \qquad x^\mu = \tilde{x}^\mu - \frac{1}{2}A^\mu{}_{\nu\rho}\tilde{x}^\nu\tilde{x}^\rho + \dots\,,$$

with $A^\mu{}_{\nu\rho} = A^\mu{}_{\rho\nu}$.

To find the new metric components, we use

$$\frac{\partial x^\mu}{\partial\tilde{x}^\alpha} = \delta^\mu{}_\alpha - A^\mu{}_{\alpha\rho}\tilde{x}^\rho + \dots\,,$$

so that

$$\begin{aligned}\tilde{g}_{\alpha\beta} &= g_{\mu\nu}\frac{\partial x^\mu}{\partial\tilde{x}^\alpha}\frac{\partial x^\nu}{\partial\tilde{x}^\beta} \\ &= (\eta_{\mu\nu} + C_{\mu\nu\rho}x^\rho + \dots)(\delta^\mu{}_\alpha - A^\mu{}_{\alpha\sigma}\tilde{x}^\sigma + \dots)(\delta^\nu{}_\beta - A^\nu{}_{\beta\gamma}\tilde{x}^\gamma + \dots) \\ &= \eta_{\alpha\beta} + C_{\alpha\beta\rho}\tilde{x}^\rho - A_{\beta\alpha\sigma}\tilde{x}^\sigma - A_{\alpha\beta\gamma}\tilde{x}^\gamma + \dots\,,\end{aligned}$$

where we note that we have used $x^\mu = \tilde{x}^\mu$ to leading order, and we lower/raise indices with the Minkowski metric $A_{\alpha\beta\gamma} = \eta_{\alpha\mu}A^\mu{}_{\beta\gamma}$.

Hence, we find that

$$\tilde{g}_{\alpha\beta} = \eta_{\alpha\beta} + \tilde{C}_{\alpha\beta\rho}\tilde{x}^{\rho} + \dots ,$$

where

$$\tilde{C}_{\alpha\beta\gamma} = C_{\alpha\beta\gamma} - A_{\alpha\beta\gamma} - A_{\beta\alpha\gamma} .$$

We can set $\tilde{C}_{\alpha\beta\gamma} = 0$ by choosing $A_{\alpha\beta\gamma} = \frac{1}{2}(A_{\alpha\beta\gamma} + A_{\alpha\gamma\beta} - A_{\beta\gamma\alpha})$.

To see this, by cycling through indices,

$$C_{\alpha\beta\gamma} = A_{\alpha\beta\gamma} + A_{\beta\alpha\gamma} , \tag{1}$$

$$C_{\beta\gamma\alpha} = A_{\beta\gamma\alpha} + A_{\gamma\beta\alpha} , \tag{2}$$

$$C_{\gamma\alpha\beta} = A_{\gamma\alpha\beta} + A_{\alpha\gamma\beta} . \tag{3}$$

Notice that taking (1) − (2) + (3) gives us the required cancellation (recalling by symmetry, $A_{\alpha\beta\gamma} = A_{\alpha\gamma\beta}$, so the highlighted terms are the same).

In conclusion:

*For any metric, we can choose coordinates $\{x^{\mu}\}$ such that **at any, but only that, point** in spacetime (without loss of generality, use the origin O with $x^{\mu} = 0$),*

$$g_{\mu\nu} = \eta_{\mu\nu} \quad \& \quad \partial_{\rho} g_{\mu\nu} = 0 \quad \Longrightarrow \quad \Gamma_{\mu}{}^{\rho}{}_{\nu} = 0 .$$

This conclusion defines a *local inertial frame* (LIF), *local inertial coordinates*, or *freely falling frame*. This holds at any single point, but also for small neighbourhoods around it in spacetime, provided changes in the metric can be neglected.

We can restate *Einstein's Equivalence Principle*:

In a local inertial frame, the results of all non-gravitational experiments are indistinguishable from results of the same experiments performed in an inertial frame in Minkowski space.

This will be the key to generalising from special relativity to general relativity.

3 Motion in Schwarzschild metric

This is a very important case of Newtonian gravity, where $\Phi = -\frac{GM}{r}$ for a point mass M at $r = 0$, and no other point masses in $r > 0$.

The generalisation of this in GR is given by the *Schwarzschild metric*:

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)$$

It is an exact solution of Einstein's equations (see Section 5) with sources in $r > 0$.

Properties

(i) The Schwarzschild metric has *symmetries*:
- It is *static* (invariant under time translation)
- It is *spherically symmetric* (for r, t constant, we have the standard metric on S^2)

(ii) For $r \gg \frac{2GM}{c^2}$, we recover the previous expression for the weak field

$$g_{00} = -\left(1 + \frac{2GM}{c^2 r}\right)$$
$$g_{ij} = \delta_{ij} + h_{ij}$$

with $\Phi = -\frac{GM}{r}$, but now in exact form.

(iii) We have interesting behaviour as $r \to 0$, but also at the *Schwarschild radius*

$$r = r_s = \frac{2GM}{c^2}$$

For a realistic model of a gravitational source e.g. planets or most stars, we expect a change in solution as we cross the surface (since in reality, they are not point-like and have a surface). From this point of view, singularities in components at $r = r_s$ is not a problem, since the Schwarzschild radius is relatively small. For example,

Sun	$r_s \sim 3\,\text{km}$
Earth	$r_s \sim 9\,\text{mm}$
Moon	$r_s \sim 0.1\,\text{mm}$

3.1 Timelike & null geodesics

Take units where $c = G = 1$, so the Schwarzschild radius is $r_s = 2M$, and the Schwarzschild metric in these units is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)$$

The geodesic equations for massive/massless particles, with an affine parameter λ, and $\lambda = \tau$ is proper time for massive particles, are

t : $$E = \left(1 - \frac{2M}{r}\right)\frac{dt}{d\lambda} \qquad \text{constant}$$

φ : $$h = r^2 \sin^2\theta \left(\frac{d\varphi}{d\lambda}\right) \qquad \text{constant}$$

θ : $$0 = \frac{d}{d\lambda}\left(r^2 \frac{d\theta}{d\lambda}\right) - r^2 \sin\theta\cos\theta \left(\frac{d\varphi}{d\lambda}\right)^2$$

Generally, the geodesic equation for r is more complicated, so we instead use the fact that $\mathcal{L}$ is a first integral:

$$\begin{aligned}\kappa &= \left(1-\frac{2M}{r}\right)\dot{t}^2 - \left(1-\frac{2M}{r}\right)^{-1}\dot{r}^2 - r^2\dot{\theta}^2 - r^2\sin^2\theta\,\dot{\varphi}^2 \\ &= \begin{cases} 0 & \text{for massless particle on null geodesic} \\ 1 & \text{for massive particle on timelike geodesic} \end{cases}\end{aligned}$$

where dots denote derivatives with respect to λ.

We can interpret the constants of motion in the regime for large r and small velocities $v \ll c$ (so the metric looks like the Minkowski metric) as

E: energy per unit mass $\qquad$ h: angular momentum per unit mass

Note:

θ equation has solution $\theta = \frac{\pi}{2}$, and we can assume this, without loss of generality by rotating the sphere and choose (θ, φ) coordinates accordingly

So, we make the choice (at $\lambda = 0$)

$$\theta = \frac{\pi}{2}, \quad \frac{d\theta}{d\lambda} = 0$$

Combining these first integrals with this change of coordinates, we have

$$E^2 - \dot{r}^2 - \left(1-\frac{2M}{r}\right)\frac{h^2}{r^2} = \kappa\left(1-\frac{2M}{r}\right)$$

and after some rearranging,

$$\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = \frac{1}{2}(E^2 - \kappa), \tag{3.1a}$$

where the *effective potential* is given by

$$V_{\text{eff}}(r) = -\frac{\kappa M}{r} + \frac{h^2}{2r^2} - \frac{Mh^2}{r^3}. \tag{3.1b}$$

In this form, we can make a non-relativistic, mechanical analogy, as the radial motion $r(\lambda)$ is like that of a particle moving in the effective potential $V_{\text{eff}}(r)$ according to Newton's 2nd law. (See Part IA *Dynamics and Relativity*)

- $V_{\text{eff}}(r)$ tells us about the motion of the particle.
- $V'_{\text{eff}}(r)$ is the effective inward acceleration.
- $V'_{\text{eff}}(r) = 0$ for constant r are *circular orbits*.
- $V''_{\text{eff}}(r)$ determines the stability of orbits.

To find the shape of orbits, it is convenient to change variables

$$\begin{aligned} u(\varphi) &= \frac{1}{r} \\ \frac{\mathrm{d}u}{\mathrm{d}\varphi} &= -\frac{1}{r^2}\frac{dr}{d\lambda}\frac{d\lambda}{d\varphi} \\ \dot{r} &= -hu' \end{aligned}$$

where primes denote derivatives with respect to φ.

Using this, the equation of motion (3.1a) becomes

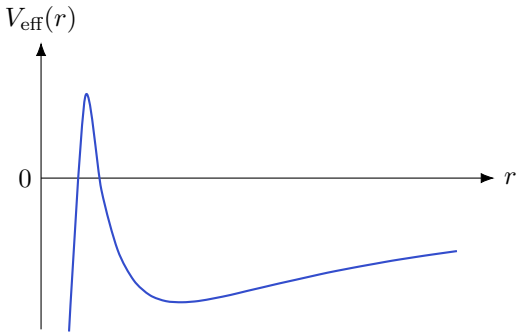
$$\frac{1}{2}(u')^2 + \frac{1}{h^2}V_{\text{eff}}\left(\frac{1}{u}\right) = \frac{1}{2h^2}(E^2 - \kappa)\,. \tag{3.2a}$$

Differentiating, we have

$$\frac{\mathrm{d}^2 u}{\mathrm{d}\varphi^2} + u = \frac{\kappa M}{h^2} + 3Mu^2\,. \tag{3.2b}$$

Example:

Consider the case where $\kappa = 1$ for a massive particle, and $h > 4M$, where the latter condition ensures $V_{\text{eff}}(r)$ crosses the axis.



3.2 Massive particles: circular & near-circular orbits

3.2.1 Newtonian gravitation

We look at Newtonian gravity, for the purposes of comparison. With the standard treatment, we have the gravitational potential

$$\Phi = -\frac{M}{r}\,,$$

where we have set $G = 1$ for convenience.

For motion in a plane with coordinates r, φ, which depend on t, we have

$$\frac{1}{2}\dot{r}^2 + V_{\text{eff}}(r) = \text{ constant}\,,$$

with effective potential

$$V_{\text{eff}}(r) = \frac{-M}{r} + \frac{h^2}{2r^2}\,,$$

and we still have conservation of angular momentum, so

$$r^2\dot{\varphi} = h \quad \text{constant.}$$

For $h > 0$, orbits exist, with stable circular orbit $r = l = \frac{h^2}{M}$, where

$$V'_{\text{eff}}(l) = 0\,, \qquad V''_{\text{eff}}(l) > 0\,.$$

Taking $u(\varphi) = \frac{1}{r}$, we have

$$u'' + u = \frac{1}{l},$$

which has general solution

$$u = \frac{1}{r} = \frac{1}{l}\left(1 + e\cos(\varphi - \varphi_0)\right)\,,$$

which are conic sections with eccentricity e, with bound, elliptical orbits for $e < 1$.

3.2.2 The effective potential & circular orbits in general relativity

In general relativity, we now have the effective potential

$$V_{\text{eff}}(r) = -\frac{M}{r} + \frac{h^2}{2r^2} - \frac{Mh^2}{r^3}\,,$$
$$V'_{\text{eff}}(r) = \frac{M}{r^2} - \frac{h^2}{r^3} + \frac{3Mh^2}{r^4}\,,$$

where the highlighted term is the general relativistic correction.

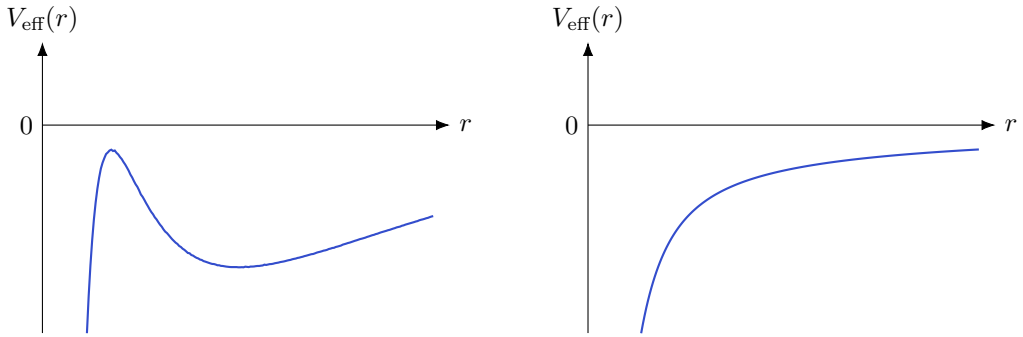
$V'_{\text{eff}}(r) = 0$ general has solutions, given by

$$r = r_{\pm} = \frac{h^2}{2M}\left[1 \pm \left(1 - \frac{12M^2}{h^2}\right)^{\frac{1}{2}}\right].$$

For solutions to exist, we need that $h > 2\sqrt{3}M$. However, we also have that

$$h^2 = \frac{Mr^2}{r - 3M}\,,$$

so solutions must only exist for $r > 3M$ (which we can already see directly from the condition $V'_{\text{eff}}(r) = 0$).



$2\sqrt{3}M < h < 4M$ $\qquad$ $h < 2\sqrt{3}M$

For the circular orbits to be stable, consider

$$V''_{\text{eff}}(r) = -\frac{2M}{r^3} + \frac{3h^2}{r^4} - \frac{12Mh^2}{r^5} = \frac{M}{r^3}\frac{r - 6M}{r - 3M}\,,$$

where we have substituted our expression for h^2.

From this, we can deduce that we have circular orbits that are

- stable for $r > 6M$.
- unstable for $3M < r < 6M$.

Note:

- Generally, the Euler-Lagrange equation for r is complicated, but it is simple for the case of circular orbits $r = R$:

$$-\frac{2M}{R^2}\left(\frac{dt}{d\tau}\right)^2 + 2R\left(\frac{d\varphi}{d\tau}\right)^2 = 0$$
$$\Longrightarrow \quad \left(\frac{d\varphi}{dt}\right)^2 = \frac{M}{R^3}\,.$$

Hence, using $\dot{\varphi} = 2\pi/T$, then

$$\frac{T^2}{R^3} = \frac{4\pi^2}{M},$$

where T is the orbital period (in coordinate time), where we note that this is alike to *Kepler's 3rd law of planetary motion.*

- The time experienced by an observer on this geodesic is proper time τ, and is **not**, in general, the same as coordinate time t.

3.2.3 Near-circular orbits in general relativity

From (3.2b) equation for the shape of orbits with $\kappa = 1$ for massive particles on timelike geodesics is

$$u'' + u = \frac{M}{h^2} + 3Mu^2,$$

where the highlighted term is the general relativistic correction to Newtonian gravity.

For corrections to be small,

$$3Mu^2 \ll u \implies Mu \ll 1 \implies r \gg r_s = 2M \quad (= \frac{2GM}{c^2})$$
$$3Mu^2 \ll \frac{M}{h^2} \implies u^2 \ll \frac{1}{h^2} \implies (r\dot{\varphi})^2 \ll 1 \quad (= c^2)$$

Note:

To fully restore constants, take $t \to ct$, $M \to \frac{GM}{c^2}$, and $\kappa \to \kappa c^2$.

Define the lengthscale

$$l = \frac{h^2}{M},$$

which is the exact radius for the solution of the circular orbit with Newtonian gravity without general relativistic corrections.

We look for an approximation to the general relativistic solution of the form

$$u = \frac{1}{l}(1 + v(\varphi)),$$

where $v(\varphi)$ is small.

Substituting this into the equation, and cancelling out leading order terms and constants, we find, to $\mathcal{O}(v)$,

$$\begin{aligned} v'' + v &= \frac{3M}{l}(1+v)^2 \\ &= \frac{3M}{l}(1+2v) \end{aligned}$$
$$\implies \quad v'' + \omega^2 v = \frac{3M}{l}$$

where

$$\omega^2 = 1 - \frac{6M}{l}.$$

Hence, we can solve this to find

$$v(\varphi) = \frac{3M}{l}\frac{1}{\omega^2} + A\cos[\omega(\varphi - \varphi_0)],$$

where A is a small constant.

Now note, for $u \approx \frac{1}{l}$, our previous conditions for small general relativistic corrections are

$$\frac{M}{l} \ll 1\,, \qquad \left(\frac{h}{l}\right)^2 \ll 1\,,$$

noting that the first condition is for weak gravity.

However, by definition of l,

$$\frac{M}{h} = \frac{h}{l} \qquad \Longrightarrow \qquad \left(\frac{M}{h^2}\right) \ll 1\,.$$

Since $\frac{M}{l}$ is small, then $\omega \approx 1 - \frac{3M}{l}$.

Hence, to leading order approximation,

$$\begin{aligned} v &= \frac{3M}{l} + A\cos[\omega(\varphi - \varphi_0)] \\ \Longrightarrow \quad u &= \frac{1}{l}\left(1 + \frac{3M}{l} + A\cos[\omega(\varphi - \varphi_0)]\right) \\ &= \frac{1}{l'}\left(1 + e\cos[\omega(\varphi - \varphi_0)]\right) \end{aligned}$$

where

$$l' = \frac{l}{1 + 3M/l} \approx l\left(1 - \frac{3M}{l}\right)\,, \qquad e = A\left(1 - \frac{3M}{l}\right)\,.$$

Note:

- Taking the formula for r_+ from before when looking at exact circular orbits, we have

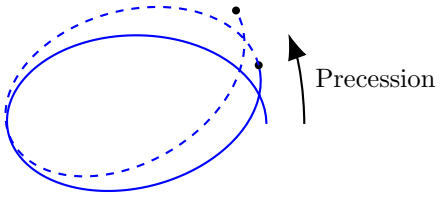
$$\begin{aligned} r_+ &= \frac{h^2}{2M}\left[1 + \left(1 - \frac{12M^2}{h^2}\right)^{\frac{1}{2}}\right] \\ &= \frac{h^2}{2M}\left[2 - \frac{6M^2}{h^2}\right] \\ &= l\left(1 - \frac{3M^2}{h^2}\right) = l' \end{aligned}$$

  where we have expanded to $\mathcal{O}(\frac{M^2}{h^2})$ in the second line. We see that this agrees with the result above.

- The formula for $u = \frac{1}{r}$ is nearly an ellipse, but with corrections to this ellipse involving the angular frequency $\omega = 1 - \frac{3M}{l}$. We see that the elliptical orbit precesses through an angle $\Delta\varphi$ per revolution, where

$$\omega(2\pi + \Delta\varphi) = 2\pi \qquad \Longrightarrow \qquad \Delta\varphi = \frac{6\pi M}{l}$$

  The predicted precession for Mercury is $\Delta\varphi = 43''$ (arcseconds) per century. However, we would also need to take into account Earth's precession about its axis, and the effects of other planets which are more than 100 times larger.



3.2.4 Massless particles & bending of light

Setting $\kappa = 0$ in (3.1b), as light travels on null geodesics, the effective potential is

$$V_{\text{eff}}(r) = \frac{h^2}{2r^2} - \frac{Mh^2}{r^3}\,,$$
$$V'_{\text{eff}}(r) = -\frac{h^2}{r^3} + \frac{3Mh^2}{r^4}\,,$$

The most extreme example of light bending occurs when light travels along circular orbits at radius $r = 3M = \frac{3}{2}r_s$, but these are unstable.

Instead, consider the shape of orbits in the more familiar regime $r \gg r_s$.

The equation of motion, with $\kappa = 0$ in (3.2b) is

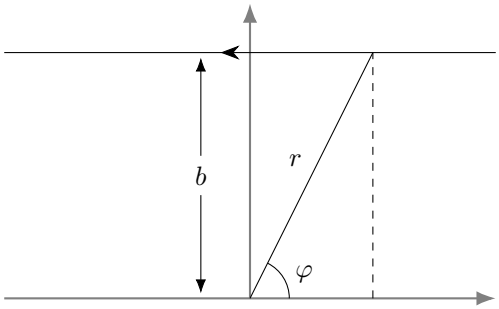
$$u'' + u = 3Mu^2\,,$$

with $Mu \ll 1$ for small general relativistic corrections.

To leading order in small terms,

$$u'' + u = 0 \qquad \Longrightarrow \qquad u = \frac{1}{b}\sin\varphi$$

where we impose the symmetry $u(\pi - \varphi) = u(\varphi)$. We see that this solution is a straight line, and the constant b is the *impact parameter*, which is the closest distance that the light ray is to the origin.



Note that to this order, we have no deflection, where $u \to 0$ (or $r \to \infty$) as $\varphi \to 0\,, \pi$.

We take $b \gg M$ (to ensure that $r \gg M \iff Mu \ll 1$), and we seek a solution to the equation of motion of the form

$$u(\varphi) = \frac{1}{b}\left(\sin\varphi - v(\varphi)\right),$$

with $v(\varphi)$ small, and we impose symmetry to have that $v(\pi - \varphi) = v(\varphi)$.

Substituting this into the equation of motion, to first order in $\frac{M}{b}$,

$$\begin{aligned} v'' + v &= \frac{3M}{b}(\sin\varphi + v(\varphi))^2 \\ &= \frac{3M}{b}\sin^2\varphi \\ &= \frac{3M}{2b}(1 - \cos 2\varphi). \end{aligned}$$

Hence, the general solution, with the imposed symmetry and α a small constant, is

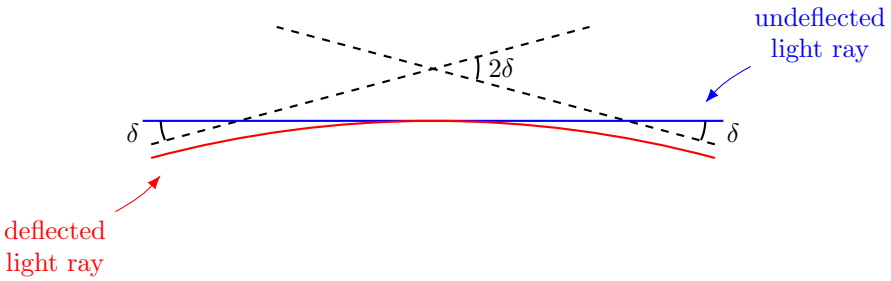
$$v(\varphi) = \alpha\frac{M}{b}\sin\varphi + \frac{M}{b}\left(\frac{3}{2} + \frac{1}{2}\cos 2\varphi\right).$$

We can eliminate the first, homogeneous term for v, by redefining

$$b \to b' = b\left(1 - \frac{\alpha M}{b}\right) ,$$

so without loss of generality, we can set $\alpha = 0$.

Given the symmetry, and that light deflects, we expect $u \to 0$ as $\varphi \to -\delta$ and $\varphi \to \pi + \delta$, where δ is a small angle of deflection.



Finding the angle of deflection when $u = 0$, and using the small angle approximations $\sin\delta \approx \delta$ and $\cos\delta \approx 1$ for small δ,

$$\begin{aligned} \sin(-\delta) + v(-\delta) &= 0 \\ -\sin(\delta) + \frac{M}{b}\left(\frac{3}{2} + \frac{1}{2}\cos(2\delta)\right) &= 0 \\ -\delta + \frac{2M}{b} &= 0 \end{aligned}$$

Hence,

$$\delta = \frac{2M}{b}$$

and the total angle of deflection is $2\delta = \frac{4M}{b}$ $\left(= \frac{4GM}{bc^2}\right)$.

For light grazing the sun, we have that $\delta = 1.75''$, which was measured by the Eddington experiment in 1919 during a total solar eclipse, and $\frac{M}{b} \approx 2 \times 10^{-6}$ for these light rays.

Note:

In this section, we have seen two classic tests of general relativity, which are

- perihelion precession of Mercury's orbit;
- deflection of light by the Sun;
- gravitational redshift of light.

Another classic test of general relativity using null geodesics is *ranging*, which involves *Shapiro time delay* in null geodesics passing close to a gravitational source. [This is illustrated in Example 2, Question 10]

4 Tensors, connections and curvature

We will work on general manifolds $\mathcal{M}$ as in Section 2.1, but we will be primarily concerned with spacetime.

4.1 Vectors & tensors

4.1.1 Definitions & transformations

Vectors and *tensors* are defined as objects with *components* in each coordinate system, related by a transformation rule under a change of coordinates $\{x^\mu\} \to \{\tilde{x}^\alpha\}$.

Definition 4.1

A *vector* has components $V^\mu, \tilde{V}^\alpha$ which obey the *vector transformation rule*

$$\tilde{V}^\alpha = \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} V^\mu \,.$$

Definition 4.2

A *covector* has components $U_\mu, \tilde{U}_\alpha$ which obey the *covector transformation rule*

$$\tilde{U}_\alpha = \frac{\partial x^\mu}{\partial \tilde{x}^\alpha} U_\mu \,.$$

Note:

- Vectors/covectors are sometimes called contravariant/covariant vectors, respectively.
- These definitions only apply at a point.

Definition 4.3

A smooth (co)vector-valued function is a called a *(co)vector field*.

Note:

$\tilde{V}^\alpha \tilde{U}_\alpha = V^\mu U_\mu$ is invariant (i.e. a *scalar*).

Examples:

- For a parameterised curve $x^\mu(\lambda), \tilde{x}^\alpha$, then the tangent vectors

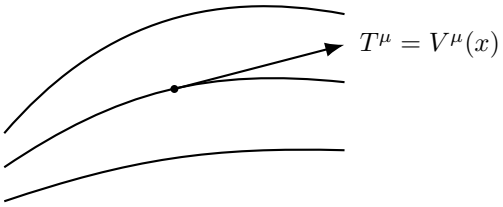
$$T^\mu = \frac{dx^\mu}{d\lambda}, \qquad \tilde{T}^\alpha = \frac{d\tilde{x}^\alpha}{d\lambda}$$

are related by a vector transformation

$$\tilde{T}^\alpha = \frac{\partial \tilde{x}^\alpha}{\partial x^\mu} T^\mu \,.$$

Conversely, given any vector field $V^\mu(x)$, we can find *integral curves* $x^\mu(\lambda)$ by solving the set of ODEs

$$\frac{dx^\mu}{d\lambda} = V^\mu(x(\lambda)).$$



- For functions $f(x)$ on a manifold $\mathcal{M}$, which is specified by $f(x^\mu)$ or $f(\tilde{x}^\alpha)$, consider the analogous gradient

$$U_\mu = \frac{\partial f}{\partial x^\mu}\,, \qquad \tilde{U}_\alpha = \frac{\partial f}{\partial \tilde{x}^\alpha}\,,$$

  which transform as covectors:

$$\tilde{U}_\alpha = \frac{\partial f}{\partial \tilde{x}^\alpha} = \frac{\partial f}{\partial x^\mu}\frac{\partial x^\mu}{\partial \tilde{x}^\alpha} = \frac{\partial x^\mu}{\partial \tilde{x}^\alpha} U_\mu\,.$$

- Along a curve,

$$\frac{d}{d\lambda}[f(x^\mu(\lambda))] = \frac{\partial f}{\partial x^\mu}\frac{\partial x^\mu}{\partial \lambda} = U_\mu T^\mu$$

  is independent of coordinates.

We can generalise the above to objects with multiple indices.

Definition 4.4

- A *tensor of type* (p, q) has components

$$T^{\mu_1\ldots\mu_p}{}_{\nu_1\ldots\nu_q} \quad \& \quad \tilde{T}^{\alpha_1\ldots\alpha_p}{}_{\beta_1\ldots\beta_q}$$

  which obey the *tensor transformation rule*

$$\tilde{T}^{\alpha_1\ldots\alpha_p}{}_{\beta_1\ldots\beta_q} = \left(\frac{\partial \tilde{x}^{\alpha_1}}{\partial x^{\mu_1}}\cdots\frac{\partial \tilde{x}^{\alpha_p}}{\partial x^{\mu_p}}\right)\left(\frac{\partial x^{\nu_1}}{\partial \tilde{x}^{\beta_1}}\cdots\frac{\partial x^{\nu_q}}{\partial \tilde{x}^{\beta_q}}\right) T^{\mu_1\ldots\mu_p}{}_{\nu_1\ldots\nu_q}\,.$$

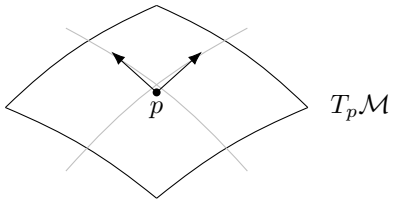
- A *tensor field* is a tensor-valued function on a manifold $\mathcal{M}$.
- For a tensor of rank (p, q), $(p + q)$ is the *rank* of the tensor.

Examples:

- Vectors are type $(1, 0)$ tensors.
- Covectors are type $(0, 1)$ tensors.
- Scalars are type $(0, 0)$ tensors.
- A metric $g_{\mu\nu}$ is a tensor of type $(0, 2)$, and the inverse metric $g^{\mu\nu}$ is a tensor of type $(2, 0)$.
- $\delta^\mu{}_\nu$ is a tensor of type $(1, 1)$, and is *invariant* (i.e. it has the same components in all coordinates).

4.1.2 Nature of vectors and tensors

Vectors at a point p on a manifold $\mathcal{M}$ (of dimension n) live in the *tangent space at* p, $T_p\mathcal{M}$, which is a vector space of dimension n. Covectors at a point p live in the *cotangent space at* p, $T_p^*\mathcal{M}$, which is dual to $T_p\mathcal{M}$. Any general tensor lives in a tensor space which is constructed from tensor products of these spaces.



We can picture $T_p\mathcal{M}$ as the "tangent plane" to $\mathcal{M}$ at p.
Using coordinates $\{x^\mu\}$ around p allow us to

(i) label points on $\mathcal{M}$ and gives a consistent notion of any function varying smoothly.

(ii) provide a basis for $T_p\mathcal{M}$, $T_p^*\mathcal{M}$, and tensor spaces.

We can compare values of a scalar function at nearby points and differentiate to get a covector.

However, we shouldn't expect to do the same for components of a tensor and get a 'nice' (tensor-valued) result, since a change of coordinates produces a change in basis that can vary from point to point.

Example:

Indeed, $\partial_\mu V^\nu$ is **not** a tensor:

$$\begin{aligned}\tilde{\partial}_\alpha \tilde{V}^\beta &= \frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\,\partial_\mu\left(\frac{\partial \tilde{x}^\beta}{\partial x^\nu}V^\nu\right)\\ &= \frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial \tilde{x}^\beta}{\partial x^\nu}\,\partial_\mu V^\nu + \frac{\partial^2 \tilde{x}^\beta}{\partial x^\mu \partial x^\nu}\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}V^\nu\\ &\neq \frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial \tilde{x}^\beta}{\partial x^\nu}\,\partial_\mu V^\nu\end{aligned}$$

We will need to generalise this to the *covariant derivative* which produces tensors from tensors.

Multilinear maps & quotient rule

A tensor of type (p,q) is equivalent to a multilinear map from p covectors $U,\ldots,W$ and q vectors $V,\ldots,Y$ to a scalar:

$$T^{\alpha_1\ldots\alpha_p}{}_{\beta_1\ldots\beta_q}\,U_{\alpha_1}\ldots W_{\alpha_p}V^{\beta_1}\ldots Y^{\beta_q}\,.$$

This expression is invariant for any vectors and covectors if and only if T obeys the transformation rule, so is a tensor of type (p,q).

The use of this result to deduce that T is a tensor is called the *quotient rule*.

Example:

$$\begin{aligned}T_\beta V^\beta &= \tilde{T}_\alpha \tilde{V}^\alpha\\ &= \tilde{T}^\alpha \frac{\partial \tilde{x}^\alpha}{\partial x^\beta}V^\beta\end{aligned}$$

If this is true for all vectors V^α, then we deduce that

$$T_\beta = \frac{\partial \tilde{x}^\alpha}{\partial x^\beta}\tilde{T}^\alpha\,,\quad \tilde{T}_\mu = \frac{\partial x^\beta}{\partial \tilde{x}^\mu}T_\beta.$$

Hence, T is a tensor of type $(0,1)$, i.e. a covector.

The result for general tensors of any rank (p, q) follows similarly, by treating every vector/covector index similarly, as in the above example. There are also many possible variations of the quotient rule.

Examples:

- If $T^{\alpha}{}_{\beta\mu\nu} V^{\beta}$ is a tensor of type $(1, 2)$ for any vector V^{β}, then $T^{\alpha}{}_{\beta\mu\nu}$ is a tensor of type $(1, 3)$.
- If $T^{\alpha\beta\mu\nu}{}_{\gamma\delta\rho} S^{\rho}{}_{\mu\nu}$ is a tensor of type $(2, 2)$ for any tensor $S^{\rho}{}_{\mu\nu}$ of type $(1, 2)$, then $T^{\alpha\beta\mu\nu}{}_{\gamma\delta\rho}$ is a tensor of type $(4, 3)$.

4.1.3 Tensor algebra

Operations on tensors can be defined as operations on components, provided they obey the tensor transformation rule.

(i) **Addition:**

For tensors S, T of the same type (p, q), then $T + S$ has components

$$(T + S)^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q} = T^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q} + S^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q} \,.$$

(ii) **Scalar multiplication:**

For any scalar λ, then λT has components

$$(\lambda T)^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q} = \lambda \left(T^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q} \right) .$$

(iii) **Tensor product:**

For tensors T of type (p, q), and S of type (m, n), then $T \otimes S$ is a tensor of type $(p + m, q + n)$, defined by

$$(T \otimes S)^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q}{}^{\mu_1 \ldots \mu_m}{}_{\nu_1 \ldots \nu_n} = T^{\alpha_1 \ldots \alpha_p}{}_{\beta_1 \ldots \beta_q} S^{\mu_1 \ldots \mu_m}{}_{\nu_1 \ldots \nu_n}$$

Example:

Given a vector V, and covectors U, W, the tensor product $T = V \otimes U \otimes W$ is a tensor of type $(1, 2)$, with

$$T^{\mu}{}_{\alpha\beta} = V^{\mu} U_{\alpha} W_{\beta} \,.$$

Note:

- It is **not** the case that all tensors can be written as tensor products.
- However, a general tensor can always be written as a linear combination of tensor products of vectors and covectors.

(iv) **Index contraction:**

For a tensor T of type (p, q), define S to be a tensor of type $(p - 1, q - 1)$ by

$$T^{\alpha\mu_1 \ldots \mu_{p-1}}{}_{\alpha\nu_1 \ldots \nu_{q-1}} = S^{\mu_1 \ldots \mu_{p-1}}{}_{\nu_1 \ldots \nu_{q-1}} \,.$$

where we contract any upper index with any lower index.

(v) **Symmetrisation & antisymmetrisation:**

We can have symmetry/antisymmetry in any pair of indices of the same type.

e.g. $T_{\alpha\beta} = \pm T_{\beta\alpha}$ is a tensor property (and respects the transformation rule).

Notation:

We denote the *symmetrisation* of a tensor by

$$T_{(\alpha\beta)} = \frac{1}{2}\left(T_{\alpha\beta} + T_{\beta\alpha}\right),$$

and the *antisymmetrisation* of a tensor by

$$T_{[\alpha\beta]} = \frac{1}{2}\left(T_{\alpha\beta} - T_{\beta\alpha}\right),$$

which can also be thought of as the symmetric and antisymmetric parts of $T_{\alpha\beta}$:

$$T_{\alpha\beta} = T_{(\alpha\beta)} + T_{[\alpha\beta]}\,.$$

4.1.4 Raising & lowering indices with metrics

A metric provides an additional structure on a manifold, which allows us to change between vectors and covectors by raising and lowering indices.

Definition 4.5

Given a metric $g_{\mu\nu}$, a vector V^μ, and a covector U_μ, we can *lower* an index by

$$V_\mu = g_{\mu\nu}V^\nu\,,$$

and *raise* an index by

$$U^\mu = g^{\mu\nu}U_\nu\,.$$

This is consistent, since, for example,

$$g^{\alpha\mu}(g_{\mu\nu}V^\nu) = \delta^\alpha{}_\nu V^\nu = V^\alpha.$$

With these definitions, the inner product can be written as

$$V^\mu U_\mu = g_{\mu\nu}V^\mu U^\nu = g^{\mu\nu}V_\mu\, U_\nu = V_\mu\, U^\mu.$$

Similarly, for a general tensor T of type (p, q), we can lower an index.

$$g_{\gamma\alpha}T^{\alpha\mu_1\ldots\mu_{p-1}}{}_{\nu_1\ldots\nu_q} = T_\gamma{}^{\mu_1\ldots\mu_{p-1}}{}_{\nu_1\ldots\nu_q}$$

which is a tensor of type $(p-1, q+1)$.

4.2 Connections & covariant derivatives

4.2.1 Motivation & definitions

As noted in the previous section, for a vector field V^μ, $\partial_\alpha V^\mu$ is **not** a tensor.
Similarly, the derivative of $V^\mu(x(\lambda))$ along a curve $x^\alpha(\lambda)$ with tangent vector $T^\alpha = \frac{dx^\alpha}{d\lambda}$ is

$$\frac{dV^\mu}{d\lambda} = T^\alpha\partial_\alpha V^\mu \left(= \frac{dx^\alpha}{d\lambda}\frac{dV^\mu}{dx^\alpha}\right),$$

and is **not** a vector.

However, if we have a metric $g_{\mu\nu}$ and the (Levi-Civita) connection $\Gamma_\beta{}^\alpha{}_\gamma$, then the geodesic equation can be written as

$$\frac{DT^\mu}{D\lambda} = \frac{dT^\mu}{d\lambda} + \Gamma_\alpha{}^\mu{}_\beta T^\alpha T^\beta = 0,$$

This holds in all coordinate systems (since it was derived from a coordinate-independent Lagrangian).

We therefore expect that $\frac{DT^\mu}{D\lambda}$ transforms as a vector, which can be checked to be true, and follows from the *connection transformation rule*.

Definition 4.6

A *connection* has components $\Gamma_\beta{}^\alpha{}_\gamma$, $\tilde{\Gamma}_\beta{}^\alpha{}_\gamma$ which obey the *connection transformation rule*

$$\tilde{\Gamma}_\beta{}^\alpha{}_\gamma = \frac{\partial x^\mu}{\partial \tilde{x}^\beta}\frac{\partial x^\nu}{\partial \tilde{x}^\gamma}\frac{\partial \tilde{x}^\alpha}{\partial x^\rho}\Gamma_\mu{}^\rho{}_\nu + \frac{\partial \tilde{x}^\beta}{\partial x^\sigma}\frac{\partial^2 x^\sigma}{\partial \tilde{x}^\alpha \partial \tilde{x}^\gamma}\,.$$

We will take this as the general definition for a connection on a manifold, irrespective of whether it arises from a metric.

Given such a connection, we can define the *covariant derivative*.

Definition 4.7

The *covariant derivative along a curve*, with tangent vector T^α is given by

$$\frac{DV^\mu}{D\lambda} = \frac{dV^\mu}{d\lambda} + \Gamma_\alpha{}^\mu{}_\nu T^\alpha V^\nu\,.$$

Furthermore,

$$\frac{DV^\mu}{D\lambda} = T^\alpha\left(\nabla_\alpha V^\mu\right),$$

where the *covariant derivative of a vector field* is given by

$$\nabla_\alpha V^\mu = \partial_\alpha V^\mu + \Gamma_\alpha{}^\mu{}_\nu V^\nu.$$

We will later check that the covariant derivative of a vector field is a tensor of type $(1,1)$. by explicit calculation.

To extend the covariant derivative to any tensor, we define its action on scalars and covector fields as well.

Definition 4.8

The *covariant derivative of a scalar* is

$$\nabla_\alpha f = \partial_\alpha f\,.$$

The *covariant derivative of a covector field* is

$$\nabla_\alpha U_\mu = \partial_\alpha U_\mu - \Gamma_\alpha{}^\rho{}_\mu U_\rho\,.$$

This ensures that the following Leibniz property holds:

$$\nabla_\alpha(V^\mu U_\mu) = (\nabla_\alpha V^\mu)U_\mu + V^\mu(\nabla_\alpha U_\mu)$$

We see that this is true, since, using the fact that $V^\mu U_\mu$ is a scalar,

$$\begin{aligned}\text{LHS} &= (\partial_\alpha V^\mu)U_\mu + V^\mu(\partial_\alpha U_\mu)\\ \text{RHS} &= (\partial_\alpha V^\mu + \Gamma_\alpha{}^\mu{}_\nu V^\nu)U_\mu + V^\mu(\partial_\alpha U_\mu - \Gamma_\alpha{}^\rho{}_\mu U_\rho),\end{aligned}$$

where the highlighted terms cancel.

Note:

It follows that $\nabla_\alpha U_\mu$ is a tensor since, for any covector V_μ,

$$V^\mu(\nabla_\alpha U_\mu) = \nabla_\alpha(V^\mu U_\mu) - (\nabla_\alpha V^\mu)U_\mu\,,$$

so it follows from the quotient rule that $\nabla_\alpha U_\mu$ is a tensor.

For a general tensor of type (p, q), we take its covariant derivative by treating each upper index as a vector, and each lower index as a covector.

Examples:

- If W is a tensor of type $(1, 1)$, then

$$\nabla_\alpha W^\mu{}_\nu = \partial_\alpha W^\mu{}_\nu + \Gamma_\alpha{}^\mu{}_\rho W^\rho{}_\nu - \Gamma_\alpha{}^\sigma{}_\nu W^\mu{}_\sigma\,.$$

- If S is a tensor of type $(0, 2)$, then

$$\nabla_\alpha S_{\mu\nu} = \partial_\alpha S_{\mu\nu} - \Gamma_\alpha{}^\rho{}_\mu S_{\rho\nu} - \Gamma_\alpha{}^\rho{}_\nu S_{\mu\rho}\,.$$

The results are tensors by following a similar argument:

- Use the Leibniz property, and check connection terms involving Γ cancel.
- Use the quotient rule to deduce that it is a tensor.

Example:

Looking at W from before, we have that

$$\nabla_\alpha(W^\mu{}_\nu U_\mu V^\nu) = (\nabla_\alpha W^\mu{}_\nu)U_\mu V^\nu + W^\mu{}_\nu(\nabla_\alpha U_\mu)V^\nu + W^\mu{}_\nu U_\mu(\nabla_\alpha V^\nu).$$

Every other term is a tensor.

$\implies$ $(\nabla_\alpha W^\mu{}_\nu)\,U_\mu V^\nu$ is a tensor for all U_μ, V^ν

$\implies$ $(\nabla_\alpha W^\mu{}_\nu)$ is a tensor (by the quotient rule).

Explicit check for $\nabla_\mu V^\nu$ is a tensor

In a different set of coordinates $\{\tilde{x}^\alpha\}$,

$$\begin{aligned}\tilde{\nabla}_\alpha \tilde{V}^\beta &= \tilde{\partial}_\alpha \tilde{V}^\beta + \tilde{\Gamma}^\beta{}_{\alpha\gamma}\tilde{V}^\gamma \\ &= \frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial \tilde{x}^\beta}{\partial x^\nu}(\partial_\mu V^\nu + \Gamma_\mu{}^\nu{}_\rho V^\rho) + \left[\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial^2 \tilde{x}^\beta}{\partial x^\mu \partial x^\rho} + \frac{\partial \tilde{x}^\rho}{\partial x^\sigma}\frac{\partial^2 x^\sigma}{\partial \tilde{x}^\alpha \partial \tilde{x}^\gamma}\frac{\partial \tilde{x}^\gamma}{\partial x^\rho}\right] V^\rho\end{aligned}$$

To show that the term in square brackets vanishes, notice that the first of these terms can be written as

$$\begin{aligned}\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial}{\partial x^\rho}\left(\frac{\partial \tilde{x}^\beta}{\partial x^\mu}\right) &= \frac{\partial}{\partial x^\rho}\left(\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial \tilde{x}^\beta}{\partial x^\mu}\right) - \frac{\partial \tilde{x}^\beta}{\partial x^\mu}\frac{\partial}{\partial x^\rho}\left(\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\right) \\ &= -\frac{\partial \tilde{x}^\beta}{\partial x^\mu}\frac{\partial \tilde{x}^\gamma}{\partial x^\rho}\frac{\partial^2 x^\mu}{\partial \tilde{x}^\alpha \partial \tilde{x}^\gamma}\end{aligned}$$

where the first term vanishes, since

$$\frac{\partial}{\partial x^\rho}\left(\frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial \tilde{x}^\beta}{\partial x^\mu}\right) = \frac{\partial}{\partial x^\rho}\left(\delta_\alpha{}^\beta\right) = 0\,,$$

and we see that the term in square brackets does vanish.

Hence,

$$\tilde{\nabla}_\alpha \tilde{V}^\beta = \frac{\partial x^\mu}{\partial \tilde{x}^\alpha}\frac{\partial \tilde{x}^\beta}{\partial x^\nu}\nabla_\mu V^\nu\,,$$

is a tensor of type $(1, 1)$. □

Warning:

Note that, in general, we can not mix partials $\partial/\partial x^\mu$ and $\partial/\partial \tilde{x}^\alpha$, since we implicitly fix all other coordinates in some coordinate system, but $\{x^\mu\}$ and $\{\tilde{x}^\alpha\}$ can still depend on each other. Instead, we have to use the chain rule to simplify.

4.2.2 Metric compatibility & torsion

For a general connection Γ, and for any scalar f,

$$\begin{aligned}\nabla_\alpha\nabla_\beta f - \nabla_\beta\nabla_\alpha f &= \nabla_\alpha(\partial_\beta f) - \nabla_\beta(\partial_\alpha f)\\ &= \Gamma_\alpha{}^\gamma{}_\beta\,\partial_\gamma f - \Gamma_\beta{}^\gamma{}_\alpha\,\partial_\gamma f\\ &= T_\alpha{}^\gamma{}_\beta\,\partial_\gamma f\end{aligned}$$

where $T_\alpha{}^\gamma{}_\beta = \Gamma_\beta{}^\gamma{}_\alpha - \Gamma_\alpha{}^\gamma{}_\beta$ is the *torsion tensor*.

Note:

The fact that torsion is a tensor is consistent with the connection transformation rule.

In this course, we only deal with *symmetric* or *torsion-free* connections:

$$\Gamma_\alpha{}^\gamma{}_\beta = \Gamma_\beta{}^\gamma{}_\alpha \quad \text{or} \quad T_\alpha{}^\gamma{}_\beta = 0 \quad \text{or} \quad \nabla_\alpha\nabla_\beta f - \nabla_\beta\nabla_\alpha f = 0$$

Now, we have the following useful theorem:

Theorem 4.1: Fundamental Theorem of Riemannian Geometry

For a Riemannian manifold $\mathcal{M}$ with metric $g_{\mu\nu}$, there exists a *unique*, *torsion-free* connection Γ for which it is covariantly constant

$$\nabla_\alpha g_{\mu\nu} = 0$$

and this choice of connection is the *Levi-Civita connection*.

Proof. Consider the cyclic permutations of indices

$$\nabla_\alpha g_{\mu\nu} = \partial_\alpha g_{\mu\nu} - \Gamma_\alpha{}^\rho{}_\mu g_{\rho\nu} - \Gamma_\alpha{}^\rho{}_\nu g_{\mu\rho} = 0\,, \tag{1}$$

$$\nabla_\mu g_{\nu\alpha} = \partial_\mu g_{\nu\alpha} - \Gamma_\mu{}^\rho{}_\nu g_{\rho\alpha} - \Gamma_\mu{}^\rho{}_\alpha g_{\nu\rho} = 0\,, \tag{2}$$

$$\nabla_\nu g_{\alpha\mu} = \partial_\nu g_{\alpha\mu} - \Gamma_\nu{}^\rho{}_\alpha g_{\rho\mu} - \Gamma_\nu{}^\rho{}_\mu g_{\alpha\rho} = 0\,. \tag{3}$$

Taking (2) + (3) − (1), and using the fact that Γ is symmetric, we have

$$\partial_\mu g_{\nu\alpha} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu} = 2g_{\alpha\rho}\Gamma_\mu{}^\rho{}_\nu$$

Rearranging, and using the inverse metric, we have the *Levi-Civita connection*

$$\Gamma_\mu{}^\rho{}_\nu = \frac{1}{2}g^{\rho\alpha}(\partial_\mu g_{\nu\alpha} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu})\,.$$

□

4.2.3 Summary of definitions & properties

Definition 4.9

For a tensor T of type (p, q), and connection Γ, the *covariant derivative of a tensor* ∇T is a tensor of type $(p, q+1)$, with

$$\begin{aligned}\nabla_\alpha T^{\mu_1\dots\mu_p}{}_{\nu_1\dots\nu_q} &= \partial_\alpha T^{\mu_1\dots\mu_p}{}_{\nu_1\dots\nu_q} \\ &+ \Gamma_\alpha{}^{\mu_1}{}_\rho\, T^{\rho\mu_2\dots\mu_p}{}_{\nu_1\dots\nu_q} + \dots + \Gamma_\alpha{}^{\mu_p}{}_\rho\, T^{\mu_1\dots\mu_{p-1}\rho}{}_{\nu_1\dots\nu_q} \\ &- \Gamma_\alpha{}^{\rho}{}_{\nu_1}\, T^{\mu_1\dots\mu_p}{}_{\rho\nu_2\dots\nu_q} - \dots - \Gamma_\alpha{}^{\rho}{}_{\nu_q}\, T^{\mu_1\dots\mu_p}{}_{\nu_1\dots\nu_{q-1}\rho}\end{aligned}$$

Properties

(i) For all α, β constants, and T, S are tensors of the same type, then

$$\nabla(\alpha T + \beta S) = \alpha\nabla T + \beta\nabla S\,.$$

(ii) For any scalar f, then

$$\nabla f = \partial f\,.$$

(iii) For any tensors T, S, then we have the *Leibniz property for tensor products*

$$\nabla(T\otimes S) = (\nabla T)\otimes S + T\otimes(\nabla S).$$

(iv) ∇ commutes with index contraction.

(v) For the Levi-Civita connection,

$$\nabla_\alpha g_{\mu\nu} = 0\,,$$

so ∇ commutes with the raising/lowering of indices.

For our purposes, we restrict to a *symmetric, torsion-free* connection

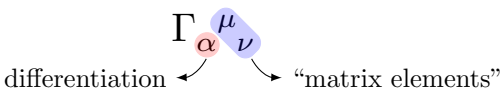
$$\Gamma_\alpha{}^\gamma{}_\beta = \Gamma_\beta{}^\gamma{}_\alpha \quad\Longleftrightarrow\quad \nabla_\alpha\nabla_\beta f - \nabla_\beta\nabla_\alpha f = 0$$

and choose Γ to be the *Levi-Civita connection*, if and only if

$$\nabla_\alpha g_{\mu\nu} = 0\,.$$

Note:

- In formulas, we have, with the following convention



  although α and ν above could also change roles in a different convention, but these two conventions are equivalent for the case of a symmetric, torsion-free connection.

- It is **not** true in general that, for example,

$$\nabla_3 S^1{}_2 = \partial_3 S^1{}_2 + \Gamma_3{}^1{}_\alpha\, S^\alpha{}_2 - \Gamma_3{}^\alpha{}_2\, S^1{}_\alpha$$

  depends only on $S^1{}_2$.

Punctuation notation

It can be convenient to introduce the following notation

$$\partial_\alpha T \equiv T_{,\alpha} \quad \text{and} \quad \nabla_\alpha T \equiv T_{;\alpha}$$

so that we can write

$$\begin{aligned} \nabla_\alpha f = \partial_\alpha f \quad \text{as} \quad f_{;\alpha} &= f_{,\alpha} \,, \\ \nabla_\alpha V^\beta &= V^\beta_{\ ;\alpha} \,, \\ \partial_\alpha V^\beta &= V^\beta_{\ ,\alpha} \,, \\ V^\beta_{\ ;\alpha} &= V^\beta_{\ ,\alpha} + \Gamma_\alpha{}^\beta{}_\gamma V^\gamma \,. \end{aligned}$$

For double derivatives, we use only one comma or semicolon (, or ;), and use the ordering convention

$$\nabla_\alpha \nabla_\beta U_\gamma = U_{\gamma;\beta\alpha} \,.$$

Later in this section, when we introduce *normal coordinates*, we will denote the use of normal coordinates by "=" , for example,

$$\nabla_\alpha V^\mu \text{ “=” } \partial_\alpha V^\mu .$$

4.2.4 Parallel transport

Consider a manifold $\mathcal{M}$ with metric $g_{\mu\nu}$, and a curve $x^\mu(\lambda)$ with tangent vector $T^\mu = \dot{x}^\mu(\lambda)$.

Recall that the covariant derivative along a curve

- for a scalar f,
$$\frac{Df}{D\lambda} = \frac{df}{d\lambda} \,.$$
- for a vector V^μ,
$$\frac{D}{D\lambda} V^\mu = \frac{dV^\mu}{d\lambda} + \Gamma_\alpha{}^\mu{}_\beta T^\alpha V^\beta \,.$$

If a tensor S is defined in some neighbourhood of the curve, then

$$\frac{dS}{d\lambda} = T^\alpha \partial_\alpha S$$

$$\frac{DS}{D\lambda} = T^\alpha \nabla_\alpha S = \nabla_T S$$

where $\nabla_V = V^\alpha \nabla_\alpha$ is the *directional covariant derivative along a vector field* V^μ.

If S is defined only on the curve, we can choose ∇_T as a convenient shorthand, and replace

$$T^\alpha \partial_\alpha \longrightarrow \frac{d}{d\lambda}$$

where necessary.

Definition 4.10

We say that S is *parallel transported* along the curve if

$$\frac{DS}{D\lambda} = \nabla_T S = 0 \,.$$

Note:

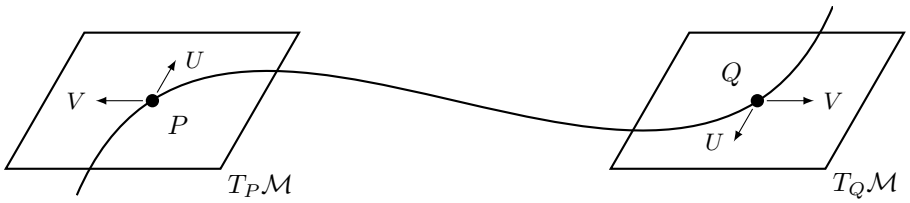
- The geodesic equation (with an affine parameter λ) is

$$\frac{DT^\alpha}{D\lambda} = \nabla_T T^\alpha = 0\,,$$

  hence, the tangent vector is parallel transported along a geodesic.

- Geodesics are sometimes called *autoparallels*.

Consider vectors U^α, V^α which are parallel transported from point P to point Q on $\mathcal{M}$ along a curve.



Then, along the curve, the covariant derivative of the inner product of U and V is

$$\begin{aligned}\frac{d}{d\lambda}(g_{\alpha\beta}U^\alpha V^\beta) &= \frac{D}{D\lambda}(g_{\alpha\beta}U^\alpha V^\beta)\\ &= g_{\alpha\beta}\frac{D}{D\lambda}(U^\alpha V^\beta)\\ &= g_{\alpha\beta}\left(\frac{DU^\alpha}{D\lambda}V^\beta + U^\alpha\frac{DV^\beta}{D\lambda}\right)\\ &= 0\,,\end{aligned}$$

where we note that the metric is always parallel transported ($\nabla_\alpha g_{\mu\nu} = 0$ so commutes with covariant derivative), and that U, V are both parallel transported along the curve.

Therefore, inner products, and hence lengths, and hence angles are preserved under parallel transport.

4.2.5 Normal coordinates

Recall, from Section 2.4, in a small neighbourhood of a given point P, we can choose coordinates $\{x^\mu\}$, or a *local inertial frame*, where, at the point P,

$$g_{\mu\nu} = \eta_{\mu\nu}\,, \quad \partial_\alpha g_{\mu\nu} = 0 \qquad \Longrightarrow \qquad \Gamma_\alpha{}^\mu{}_\beta = 0\,.$$

We can now understand this directly from the transformation of a connection.

Definition 4.11

The choice of coordinates at a point P for which, at the point P,

$$g_{\mu\nu} = \eta_{\mu\nu}\,, \quad \partial_\alpha g_{\mu\nu} = 0 \qquad \Longrightarrow \qquad \Gamma_\alpha{}^\mu{}_\beta = 0\,.$$

is called *normal coordinates*.

Since a tensor equation holds for any coordinate system, if it holds in one coordinate system, then we can adopt normal coordinates to simplify calculations.

Example:

We have

$$\nabla_\alpha T^{\mu\nu}{}_\rho \text{ “=” } \partial_\alpha T^{\mu\nu}{}_\rho\,, \quad \Longleftrightarrow \quad T^{\mu\nu}{}_{\rho;\alpha} \text{ “=” } T^{\mu\nu}{}_{\rho,\alpha}\,,$$

where “=” denotes equality in normal coordinates.

4.3 Curvature & Riemann tensor

4.3.1 Definitions

Claim

If V^μ is any vector field, then the *Ricci identity* is

$$(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) V^\mu = R^\mu{}_{\nu\alpha\beta} V^\nu \,.$$

Here, $R^\mu{}_{\nu\alpha\beta}$ is the *Riemann curvature tensor*, which is a tensor of type $(1,3)$. The Ricci identity provides a definition of the Riemann tensor in terms of ∇.

Note:

The LHS is a tensor for all vectors V^μ, so by the quotient rule, R is a tensor.

Proof.

$$\begin{aligned}
\nabla_\alpha \nabla_\beta V^\mu - \nabla_\beta \nabla_\alpha V^\mu &= \partial_\alpha (\nabla_\beta V^\mu) - \Gamma_\alpha{}^\sigma{}_\beta \nabla_\sigma V^\mu + \Gamma_\alpha{}^\mu{}_\sigma \nabla_\beta V^\sigma - (\alpha \leftrightarrow \beta) \\
&= \partial_\alpha (\partial_\beta V^\mu + \Gamma_\beta{}^\mu{}_\nu V^\nu) + \Gamma_\alpha{}^\mu{}_\sigma (\partial_\beta V^\sigma + \Gamma_\beta{}^\sigma{}_\nu V^\nu) - (\alpha \leftrightarrow \beta) \\
&= V^\nu \partial_\alpha \Gamma_\beta{}^\mu{}_\nu + \Gamma_\alpha{}^\mu{}_\sigma \Gamma_\beta{}^\sigma{}_\nu V^\nu \\
&\quad + \Gamma_\beta{}^\mu{}_\sigma \partial_\alpha V^\sigma + \Gamma_\alpha{}^\mu{}_\sigma \partial_\beta V^\sigma - (\alpha \leftrightarrow \beta) \\
&= R^\mu{}_{\nu\alpha\beta} V^\nu \,.
\end{aligned}$$

Hence, the Riemann tensor, in terms of the connection Γ, is given by

$$R^\mu{}_{\nu\alpha\beta} = \partial_\alpha \Gamma_\beta{}^\mu{}_\nu - \partial_\beta \Gamma_\alpha{}^\mu{}_\nu + \Gamma_\alpha{}^\mu{}_\sigma \Gamma_\beta{}^\sigma{}_\nu - \Gamma_\beta{}^\mu{}_\sigma \Gamma_\alpha{}^\sigma{}_\nu$$

□

There is an equivalent form of the Ricci identity for covectors:

$$(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) U_\mu = -R^\nu{}_{\mu\alpha\beta} U_\nu$$

Similarly, for a general tensor, we sum terms on the RHS for each vector and covector index

Example:

For a tensor $T^\mu{}_\nu$ of type $(1,1)$, it satisfies the Ricci identity

$$(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) T^\mu{}_\nu = R^\mu{}_{\rho\alpha\beta} T^\rho{}_\nu - R^\rho{}_{\nu\alpha\beta} T^\mu{}_\rho \,.$$

From the Riemann tensor, it is useful to define the following:

Definition 4.12

The *Ricci tensor* and *Ricci scalar* are

$$R_{\alpha\beta} = R^\mu{}_{\alpha\mu\beta} \,, \qquad R = g_{\alpha\beta} R_{\alpha\beta} \,,$$

respectively.

Example:

Consider the 2-sphere S^2 with coordinates $\{\theta, \varphi\}$, and metric

$$(g_{\mu\nu}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}.$$

The only non-zero connection components are

$$\Gamma_{\theta\ \varphi}^{\ \varphi} = \Gamma_{\varphi\ \theta}^{\ \varphi} = \cot\theta\,,$$
$$\Gamma_{\varphi\ \varphi}^{\ \theta} = -\sin\theta\cos\theta\,.$$

We calculate the components of $\nabla_\alpha V^\beta$ for $V^\theta = 0$, $V^\varphi = 1$:

$$\begin{aligned} \nabla_\theta V^\theta &= \partial_\theta V^\theta + \Gamma_{\theta\ \alpha}^{\ \theta} V^\alpha & \nabla_\theta V^\varphi &= \partial_\theta V^\varphi + \Gamma_{\theta\ \alpha}^{\ \varphi} V^\alpha \\ &= 0 & &= \Gamma_{\theta\ \varphi}^{\ \varphi} V^\varphi \\ & & &= \cot\theta \\ \nabla_\varphi V^\theta &= \partial_\varphi V^\theta + \Gamma_{\varphi\ \alpha}^{\ \theta} V^\alpha & \nabla_\varphi V^\varphi &= \partial_\varphi V^\varphi + \Gamma_{\varphi\ \alpha}^{\ \varphi} V^\alpha \\ &= \Gamma_{\varphi\ \varphi}^{\ \theta} V^\varphi & &= \Gamma_{\varphi\ \theta}^{\ \varphi} V^\theta \\ &= -\sin\theta\cos\theta & &= 0 \end{aligned}$$

where the terms highlighted cancel.

Then,

$$\begin{aligned} \nabla_\theta(\nabla_\varphi V^\theta) &= \partial_\theta(\nabla_\varphi V^\theta) + \Gamma_{\theta\ \alpha}^{\ \theta}(\nabla_\varphi V^\alpha) - \Gamma_{\theta\ \varphi}^{\ \alpha}(\nabla_\alpha V^\theta) \\ &= \partial_\theta(-\sin\theta\cos\theta) - \cot\theta(-\sin\theta\cos\theta) \\ &= \sin^2\theta \end{aligned}$$

$$\begin{aligned} \nabla_\varphi(\nabla_\theta V^\theta) &= \partial_\varphi(\nabla_\theta V^\theta) + \Gamma_{\varphi\ \alpha}^{\ \theta}(\nabla_\theta V^\alpha) - \Gamma_{\varphi\ \theta}^{\ \alpha}(\nabla_\alpha V^\theta) \\ &= (-\sin\theta\cos\theta)\cot\theta - \cot\theta(-\sin\theta\cos\theta) \\ &= 0 \end{aligned}$$

Hence,

$$(\nabla_\theta\nabla_\varphi - \nabla_\varphi\nabla_\theta) = \sin^2\theta = R^\theta{}_{\alpha\theta\varphi}V^\alpha = R^\theta{}_{\varphi\theta\varphi}V^\varphi\,.$$

In this case, due to symmetry (which will see in the next section), the only independent component is

$$R^\theta{}_{\varphi\theta\varphi} = \sin^2\theta \qquad \Longrightarrow \qquad R_{\alpha\beta} = g_{\alpha\beta}\,, \quad R = 2\,,$$

where we note the only non-zero components of the inverse metric are

$$g^{\theta\theta} = 1\,, \qquad g^{\varphi\varphi} = \frac{1}{\sin^2\theta}\,.$$

4.3.2 Symmetries of the Riemann tensor

Notation:

We denote the *symmetrisation* and *antisymmetrisation* on any n indices of the same type by $(\dots)$, $[\dots]$ as before, respectively, summing over all permutations of indices, and accounting for the sign of each permutation and a factor of $\frac{1}{n!}$.

Example:

$$T_{(\alpha\beta\gamma)} = \frac{1}{3!}(T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} + T_{\beta\alpha\gamma} + T_{\alpha\gamma\beta} + T_{\gamma\beta\alpha})$$
$$T_{[\alpha\beta\gamma]} = \frac{1}{3!}(T_{\alpha\beta\gamma} + T_{\beta\gamma\alpha} + T_{\gamma\alpha\beta} - T_{\beta\alpha\gamma} - T_{\alpha\gamma\beta} - T_{\gamma\beta\alpha})$$

Symmetries

(i) $R^{\mu}{}_{\nu\alpha\beta} = -R^{\mu}{}_{\nu\beta\alpha}$ (by definition)

(ii) $R^{\mu}{}_{[\nu\alpha\beta]} = 0 \overset{(i)}{\implies} R^{\mu}{}_{\nu\alpha\beta} + R^{\mu}{}_{\alpha\beta\nu} + R^{\mu}{}_{\beta\nu\alpha} = 0\,.$

(iii) $R_{\mu\nu\alpha\beta} = -R_{\nu\mu\alpha\beta}$

(iv) $R_{\mu\nu\alpha\beta} = R_{\alpha\beta\mu\nu}$

(v) $R_{\alpha\beta} = R_{\beta\alpha}$

We note that (v) can be deduced from (iv).

Taking these symmetries into account, the number of independent components in dimension n is

$$\frac{1}{12}n^2(n^2-1) = \begin{cases} 20, & n = 4 \\ 6, & n = 3 \\ 1, & n = 2 \end{cases}$$

Note:

- There are $N = \frac{1}{2}n(n-1)$ values for each pair (α, β) & (μ, ν) from (i) and (iii).
- From (iv), we have $\frac{1}{2}N(N+1)$ total possible indices given the symmetries (i) and (iii).
- However, by (ii), there are $\binom{n}{4}$ constraints, since (ii) is only a constraint when there are four distinct index values.

To establish these symmetries,

(i) We can see this directly from the definition of the Riemann tensor.

(ii) In normal coordinates at a point,

$$R^{\mu}{}_{\nu\alpha\beta} = \partial_\alpha \Gamma_\beta{}^{\mu}{}_{\nu} - \partial_\beta \Gamma_\alpha{}^{\mu}{}_{\nu}\,,$$
$$R^{\mu}{}_{\alpha\beta\nu} = \partial_\beta \Gamma_\nu{}^{\mu}{}_{\alpha} - \partial_\nu \Gamma_\beta{}^{\mu}{}_{\alpha}\,,$$
$$R^{\mu}{}_{\beta\nu\alpha} = \partial_\nu \Gamma_\alpha{}^{\mu}{}_{\beta} - \partial_\alpha \Gamma_\nu{}^{\mu}{}_{\beta}\,.$$

Adding these together, we see that the RHS cancels.

(iii) This follows from the Ricci identity for $g_{\mu\nu}$ (where $\nabla_\alpha g_{\mu\nu} = 0$), but we can also use normal coordinates at a point.

(iv) Recall that, from the definition of the Levi-Civita connection,

$$g_{\mu\sigma}\Gamma_\beta{}^{\sigma}{}_{\nu} = \frac{1}{2}(g_{\mu\beta,\nu} + g_{\mu\nu,\beta} - g_{\beta\nu,\mu}).$$

Taking a partial derivative, in normal coordinates at a point, where $\Gamma = 0$ and $\partial g = 0$,

$$g_{\mu\sigma}\partial_\alpha\Gamma_\beta{}^{\sigma}{}_{\nu} \text{ “=” } \frac{1}{2}(g_{\mu\beta,\nu\alpha} + g_{\mu\nu,\alpha\beta} - g_{\nu\beta,\mu\alpha})$$
$$\implies \quad g_{\mu\sigma}R^{\sigma}{}_{\nu\alpha\beta} \text{ “=” } \frac{1}{2}(g_{\mu\beta,\nu\alpha} + g_{\nu\alpha,\mu\beta} - g_{\nu\beta,\mu\alpha} - g_{\mu\alpha,\nu\beta}).$$

All symmetries manifest from this expression.

4.3.3 Parallel transport & curvature

Parallel transport of a vector field $V^\mu(\lambda) = V^\mu(x^\alpha(\lambda))$ along a curve $x^\alpha(\lambda)$ starting at $x^\mu(0)$ is given by

$$\frac{D}{D\lambda}V^\mu = 0 \qquad \Longrightarrow \qquad \frac{d}{d\lambda}V^\mu = -\left(\Gamma_{\alpha}{}^{\mu}{}_{\nu}\,\dot{x}^\alpha V^\nu\right)(\lambda)$$
$$\Longrightarrow \qquad V^\mu(\lambda) = V^\mu(0) - \int_0^\lambda \left(\Gamma_{\alpha}{}^{\mu}{}_{\nu}\,\dot{x}^\alpha V^\nu\right)(\tau)\,d\tau\,.$$

We use this to generate a series expansion, arranged in increasing order in terms of the size of the curve, as given by the size of x^α, $\dot{x}^\beta$, etc:

$$V^\mu(\lambda) = V^\mu(0) + [\Delta V^\mu]_1(\lambda) + [\Delta V^\mu]_2(\lambda) + \dots\,.$$

To leading order, we have

$$V^\mu(\lambda) = V^\mu(0).$$

Substituting this into the above, we get the first order correction

$$[\Delta V^\mu]_1(\lambda) = -\int_0^1 \left(\Gamma_{\alpha}{}^{\mu}{}_{\nu}V^\nu\right)(0)\,\dot{x}^\alpha(\tau)\,d\tau$$
$$= -\left(\Gamma_{\alpha}{}^{\mu}{}_{\nu}V^\nu\right)(0)\,x^\alpha(\tau)\,.$$

Using this, we can get the second order correction

$$[\Delta V^\mu]_2(\lambda) = S^{\mu}{}_{\nu\alpha\beta}V^\nu(0)\int_0^1 (x^\beta\dot{x}^\alpha)(\tau)\,d\tau$$

where $S^{\mu}{}_{\nu\alpha\beta} = -\Gamma_{\alpha}{}^{\mu}{}_{\nu,\beta} + \Gamma_{\alpha}{}^{\mu}{}_{\sigma}\Gamma_{\beta}{}^{\sigma}{}_{\nu}$, and the first term here comes from the correction to the connection

$$\Gamma_{\alpha}{}^{\mu}{}_{\nu}(\lambda) = \Gamma_{\alpha}{}^{\mu}{}_{\nu}(0) + \Gamma_{\alpha}{}^{\mu}{}_{\nu,\beta}x^\beta(\lambda)\,.$$

Consider integrating over a closed loop, so for $\lambda \in [0,1]$, and $x^\mu(0) = x^\mu(1) = 0$ so that

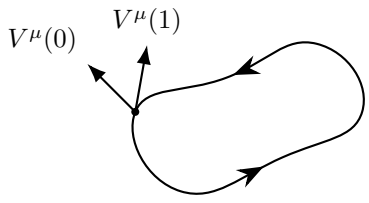
$$[\Delta V^\mu]_1(1) = 0\,,$$

and, using integration by parts, the second order corrections gives

$$W^{\alpha\beta} = -\int_0^1 (x^\beta\dot{x}^\alpha)(\tau)\,d\tau$$
$$= \frac{1}{2}\int_0^1 (x^\alpha\dot{x}^\beta - x^\beta\dot{x}^\alpha)\,d\tau$$
$$= -W^{\beta\alpha}\,, \qquad (\text{“coordinate area”})$$

to find

$$[\Delta V^\mu]_2 = -\frac{1}{2}R^{\mu}{}_{\nu\alpha\beta}W^{\alpha\beta}V^\nu(0)\,.$$



4.3.4 Bianchi identity

Claim

The Riemann tensor obeys the *Bianchi identity*:

$$R^{\mu}{}_{\nu[\alpha\beta;\gamma]} = 0\,,$$

or, alternatively,

$$\nabla_\gamma R^{\mu}{}_{\nu\alpha\beta} + \nabla_\alpha R^{\mu}{}_{\nu\beta\gamma} + \nabla_\beta R^{\mu}{}_{\nu\gamma\alpha} = 0\,.$$

Proof. Note that

$$\left(\Gamma_\beta{}^{\mu}{}_{\nu,\alpha} + \Gamma_\alpha{}^{\mu}{}_{\sigma}\,\Gamma_\beta{}^{\sigma}{}_{\nu} \;-\; (\alpha \leftrightarrow \beta)\right)_{,\gamma} \;\text{“=”}\; \left(\Gamma_\beta{}^{\mu}{}_{\nu,\alpha} - \Gamma_\beta{}^{\mu}{}_{\nu,\alpha}\right)_{,\gamma}$$

$$\implies \begin{cases} R^{\mu}{}_{\nu\alpha\beta;\gamma} \;\text{“=”}\; \Gamma_\nu{}^{\mu}{}_{\beta,\alpha\gamma} - \Gamma_\nu{}^{\mu}{}_{\alpha,\beta\gamma}\,, \\ R^{\mu}{}_{\nu\beta\gamma;\alpha} \;\text{“=”}\; \Gamma_\nu{}^{\mu}{}_{\gamma,\beta\alpha} - \Gamma_\nu{}^{\mu}{}_{\beta,\gamma\alpha}\,, \\ R^{\mu}{}_{\nu\gamma\alpha;\beta} \;\text{“=”}\; \Gamma_\nu{}^{\mu}{}_{\alpha,\gamma\beta} - \Gamma_\nu{}^{\mu}{}_{\gamma,\alpha\beta}\,. \end{cases}$$

Adding these together gives us the result in normal coordinates, hence true in all coordinates, since this is a tensor identity. □

Contracted Bianchi identity

Claim

The Riemann tensor also obeys the *contracted Bianchi identity*:

$$\nabla^\beta R_{\alpha\beta} - \frac{1}{2}\nabla_\alpha R = 0\,,$$

or, alternatively,

$$\nabla^\beta G_{\alpha\beta} = 0\,.$$

Definition 4.13

The *Einstein tensor* is given by

$$G_{\alpha\beta} = R_{\alpha\beta} - \frac{1}{2}R g_{\alpha\beta}\,.$$

Proof. Contract indices μ and γ in the Bianchi identity, and use the Ricci tensor $R_{\alpha\beta} = R^{\mu}{}_{\alpha\mu\beta}$ to get

$$\nabla_\gamma R^{\gamma}{}_{\nu\alpha\beta} - \nabla_\alpha R_{\nu\beta} - \nabla_\beta R_{\nu\alpha} = 0\,.$$

Contract again on indices ν and β to get

$$\nabla^\gamma R_{\gamma\alpha} - \nabla_\alpha R + \nabla^\nu R_{\nu\alpha} = 0\,,$$

and the result follows by rearranging. □

4.3.5 Geodesic deviation

In flat Minkowski space, geodesics are straight lines, and separation doesn't change as you move along them. On the other hand, in curved space, the separation of geodesics varies, governed by the curvature of space.

To explore this, first, consider vector fields T^α, S^α and their directional derivatives

$$\nabla_T = T^\alpha \nabla_\alpha \,, \qquad \nabla_T = T^\alpha \nabla_\alpha$$

along a vector field V^μ.

We have

$$\begin{aligned}(\nabla_T \nabla_S - \nabla_S \nabla_T) V^\mu &= T^\alpha \nabla_\alpha \left(S^\beta \nabla_\beta V^\mu\right) - S^\beta \nabla_\beta \left(T^\alpha \nabla_\alpha V^\mu\right) \\ &= T^\alpha S^\beta \left(\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha\right) V^\mu + \left(T^\alpha \nabla_\alpha S^\beta - S^\alpha \nabla_\alpha T^\beta\right) \nabla_\beta V^\mu\end{aligned}$$

We note that the *commutator* is a vector field:

Definition 4.14

For vector fields T^α, S^α, the *commutator* of T and S is

$$\begin{aligned}[T, S]^\beta &= T^\alpha \nabla_\alpha S^\beta - S^\alpha \nabla_\alpha T^\beta \\ &= T^\alpha \partial_\alpha S^\beta - S^\alpha \partial_\alpha T^\beta\end{aligned}$$

Hence,

$$(\nabla_T \nabla_S - \nabla_S \nabla_T) V^\mu = T^\alpha S^\beta R^\mu{}_{\nu\alpha\beta} V^\nu + \nabla_{[T,S]} V^\mu \,, \tag{4.1}$$

which is another version of the Ricci identity.

We now apply this to a family of geodesics $x^\mu(\tau, \sigma)$, where τ is an affine parameter, and σ labels each member of the family of geodesics, so we have a geodesic for fixed σ.

We also have the *tangent vector*

$$T^\mu = \frac{\partial x^\mu}{\partial \tau} \,,$$

where $\nabla_T T^\alpha = 0$ satisfies the geodesic equation, and the *separation vector* between geodesics

$$S^\mu = \frac{\partial x^\mu}{\partial \sigma} \,.$$

In these circumstances,

$$\begin{aligned}[T, S]^\beta &= T^\alpha \partial_\alpha S^\beta - S^\alpha \partial_\alpha T^\beta \\ &= \frac{\partial S^\beta}{\partial \tau} - \frac{\partial T^\beta}{\partial \sigma} \\ &= 0\end{aligned}$$

where we use the chain rule, and note that

$$\frac{\partial S^\beta}{\partial \tau} = \frac{\partial^2 x^\mu}{\partial \tau \partial \sigma} = \frac{\partial T^\beta}{\partial \sigma} \,.$$

From (4.1), with $V^\mu = T^\mu$, and using the geodesic equation,

$$\nabla_T \nabla_S - \nabla_S \nabla_T T^\mu = R^\mu{}_{\nu\alpha\beta} T^\alpha S^\beta T^\nu \,.$$

Using the fact that $\nabla_S T^\mu = \nabla_T S^\mu$ (since $[T,S]^\beta = 0$), we have the *equation of geodesic deviation*

$$\nabla_T^2 S^\mu + E^\mu{}_\beta S^\beta = 0$$

where

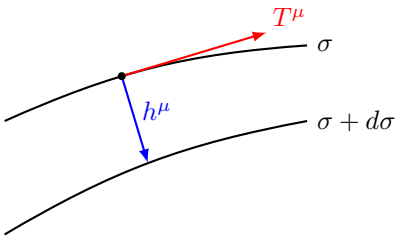
$$E^\mu{}_\beta = -R^\mu{}_{\nu\alpha\beta} T^\nu T^\alpha \,.$$

For closely-separated geodesics $X^\mu(\tau,\sigma)$, and

$$x^\mu(\tau,\sigma+\delta\sigma) \approx x^\mu(\tau,\sigma) + h^\mu(\tau)\,,$$

where $h^\mu(\tau) = S^\mu\, d\sigma$, so in terms of h^μ, the equation of geodesic deviation can also be expressed as

$$\nabla_T^2 h^\mu + E^\mu{}_\beta h^\beta = 0\,.$$



5 Einstein's equations

5.1 Introduction & motivation

We compare Newtonian gravity and general relativity:

Property	NG	GR
Basic field	potential $\Phi(x,t)$	metric $g_{\alpha\beta}(x)$
Equation of motion	$\frac{d^2x_i}{dt^2} = g_i = -\frac{\partial\Phi}{\partial x_i}$	$\frac{d^2x^\alpha}{d\tau^2} + \Gamma_{\beta}{}^{\alpha}{}_{\gamma}\frac{dx^\beta}{d\tau}\frac{dx^\gamma}{d\tau} = 0$ *geodesic equation*
LIF at a point	$g = 0$	$\Gamma = 0$
Nearby trajectories	$\frac{d^2h_i}{dt^2} + E_{ij}h_j = 0$	$\nabla_T^2 h^\alpha + E^\alpha{}_\beta h^\beta = 0$ *geodesic deviation*
	$E_{ij} = -\partial_j g_i = \frac{\partial^2\Phi}{\partial x_i \partial x_j}$	$E^\alpha{}_\beta = R^\alpha{}_{\mu\beta\nu}T^\mu T^\nu$
Field equations	$\nabla^2\Phi = 4\pi G\rho = E_{ii}$	*Einstein's equations*

We see that in Newtonian gravity, the source to the field equations is mass density ρ, but for Einstein's field equations,

→ What is our source?

→ How does it determine the metric $g_{\mu\nu}$?

→ The comparison above suggests it involves $R_{\mu\nu}$.

5.1.1 Statement of Einstein's equations

(i) **No course**

In Newtonian gravity,

$$\nabla^2\Phi = 0\,,$$

which still has interesting solutions, e.g. $\Phi = -\frac{GM}{r}$ for $r > 0$.

In general relativity, *Einstein's equations with no sources* (in a vacuum) are

$$R_{\mu\nu} = 0\,.$$

(ii) **With sources**

We replace mass from Newtonian gravity with energy and momentum in special relativity, which is described by the *energy-momentum* (or *stress-energy*) *tensor*

$$T^{\mu\nu} = T^{\nu\mu}\,,$$

which is conserved: $\partial_\mu T^{\mu\nu} = 0$.

The natural generalisation of this to curved spacetime is

$$\nabla_\mu T^{\mu\nu} = 0\,.$$

It suggests we take $T^{\mu\nu}$ as a source, and we look for a tensor related to $R_{\mu\nu}$ which is consistent with conservation of sources.

→ This is the *Einstein tensor*, $G_{\mu\nu}$.

This gives us *Einstein's field equations with sources*

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu} \ ,$$

where $\kappa = \frac{8\pi G}{c^4}$ by comparison with Newtonian gravity $\nabla^2\Phi = 4\pi G\rho$ (see later).

We have, from the contracted Bianchi identity, $\nabla_\mu G^{\mu\nu} = 0$, which is consistent with source conservation.

We see that (ii) is consistent with (i), since if $T_{\mu\nu} = 0$, then

$$\begin{aligned} & G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0 \\ \implies \quad & G^\mu{}_\mu = R^\mu{}_\mu - \frac{1}{2}g^\mu{}_\mu R = 0 \\ \implies \quad & R = 0 \\ \implies \quad & R_{\mu\nu} = 0 \qquad \text{as in (i),} \end{aligned}$$

where we note in $1+3$ spacetime, $g^\mu{}_\mu = 4$.

(iii) **Cosmological constant**

An additional term can be added to $G_{\mu\nu}$ which is consistent with source conservation:

$$\nabla_\mu \left(G^{\mu\nu} + \Lambda g^{\mu\nu}\right) = 0 \, ,$$

where Λ is a constant *cosmological constant*.

This results in *Einstein's equations with a cosmological constant* Λ

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \ .$$

Other possibilities are ruled out by *Lovelock's theorem*:

Theorem 5.1

If $H_{\alpha\beta} = H_{\beta\alpha}$ is a tensor with $\nabla_\alpha H^{\alpha\beta} = 0$ (or any tensor which is symmetric and divergence-free) and $H_{\alpha\beta}$ depends on $g_{\mu\nu}$, $\partial_\alpha g_{\mu\nu}$, $\partial_\alpha\partial_\beta g_{\mu\nu}$ at each point, then

$$H_{\alpha\beta} = a\, G_{\alpha\beta} + b\, g_{\alpha\beta}$$

for some constants a, b.
This is true in $n = 4$, and $n > 4$ if the dependence on $\partial^2 g$ is linear.

Observations suggest that Λ is a very small, positive constant:

$$|\Lambda|^{-\frac{1}{2}} \sim 10^9 \text{ light years (comparable to size of the universe).}$$

We can set $\Lambda = 0$ (so has negligible effects), unless we work on cosmological scales.

We can also reduce (iii) to (ii) by taking Λ term to the RHS as an extra contribution to $T_{\mu\nu}$:

$$G_{\mu\nu} = \kappa T_{\mu\nu} - \Lambda g_{\mu\nu} \, .$$

This contribution to $T_{\mu\nu}$ is called *vacuum energy* or *dark energy*.

Note:

> In (i), (ii), (iii), we have second order, non-linear PDEs for $g_{\mu\nu}$, which are generally hard to solve. Also, note that $g_{\mu\nu}$ appears in $T_{\mu\nu}$.

5.1.2 Newtonian limit

Consider a metric

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,,$$

with $h_{\mu\nu}$ small for weak gravity, and coordinate system

$$\{x^0 = t,\, x^i\} \qquad \text{(in units } c = 1\text{)}$$

in which the metric static/almost static, so we can neglect t derivatives.

Assume that, in addition, the only non-negligible components of $T_{\alpha\beta}$ is

$$T_{00} = \rho \quad \text{mass density} \qquad (\rho c^2 \text{ with units restored})$$

where $\kappa\rho$ is small.

In general,

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu}$$

can be rewritten by contracting with $g^{\mu\nu}$

$$\begin{aligned} R - \frac{1}{2}Rg^{\mu}{}_{\mu} &= \kappa T \qquad (\text{where } T = T^{\mu}{}_{\mu}) \\ -R &= \kappa T\,, \end{aligned}$$

and substituting back to get the *trace-reversed form* of Einstein's equations

$$R_{\mu\nu} = \kappa\left(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T\right).$$

Now, we now look at the 00 component with the above assumptions. To find the Ricci tensor, the Riemann curvature tensor, to first order in $h_{\mu\nu}$ (similar to that in normal coordinates), is

$$R_{\mu\nu\alpha\beta} = \frac{1}{2}\left[h_{\mu\beta,\nu\alpha} + h_{\nu\alpha,\mu\beta} - h_{\nu\beta,\mu\alpha} - h_{\mu\alpha,\nu\beta}\right].$$

So,

$$\begin{aligned} R_{00} &= g^{\mu\alpha}R_{\mu0\alpha0} = \eta^{\mu\alpha}R_{\mu0\alpha0} \\ &= \delta^{ij}R_{i0j0} \\ &= -\frac{1}{2}\delta^{ij}h_{00,ij} \\ &= -\frac{1}{2}\nabla^2 h_{00} \end{aligned}$$

where $\nabla^2 = \delta^{ij}\partial_i\partial_j$ is the usual 3D Laplacian.

For sources,

$$\begin{aligned} \kappa T = g^{\mu\nu}\kappa T_{\mu\nu} &= \eta^{\mu\nu}\kappa T_{\mu\nu} = -\kappa\rho \\ \implies \qquad \kappa\left(T_{00} - \frac{1}{2}g_{00}T\right) &= \frac{1}{2}\kappa\rho \qquad (\text{“trace-reversed”}) \\ \implies \qquad -\nabla^2 h_{00} &= \kappa\rho \end{aligned} \tag{1}$$

Now, recall from Section 2.3, motion in such a metric which is non-relativistic is given by

$$\frac{d^2x^i}{dt^2} = -\Gamma_0{}^i{}_0 = \frac{1}{2}\delta^{ij}h_{00,j}\,, \tag{2}$$

noting that $t \approx \tau$.

We noted that $h_{00} = -2\Phi$, which means that (2) is Newton's 2nd law for motion in gravitational potential Φ.

Then, (1) becomes $\nabla^2\Phi = 4\pi G\rho$ as required, with the choice of

$$\kappa = 8\pi G \qquad \left(\text{or } \frac{8\pi G}{c^4} \text{ with units restored}\right).$$

5.2 Spherically symmetric vacuum solutions: the Schwarzschild metric

Theorem 5.2: Birkhoff's theorem

The most general spherically symmetric solution of the vacuum Einstein equations is given by the Schwarzschild metric for some mass M, in units $c = 1$,

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2\,,$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\, d\varphi^2$.

Hence, every spherically metric symmetric solution of the vacuum Einstein equations is necessarily static and asymptotically flat.

In this case, assume that rotational symmetry acts on 2-spheres defined by constant r, t (parameterised by θ, φ), which are orbits of the symmetry group.

Outline of the proof: ***[Non-examinable]***

A general Lorentzian metric with spherical symmetry takes the form

$$ds^2 = -A\, dt^2 + 2B\, dr\, dt + C\, dr^2 + R^2\, d\Omega^2$$

where A, B, C, R are functions of r, t.

Spherical symmetry is maintained under the coordinate transformations of the form

$$\begin{aligned} r &\longrightarrow \tilde{r}(r,t)\,, \\ t &\longrightarrow \tilde{t}(r,t)\,. \end{aligned}$$

We use this repeatedly (and drop $\tilde{(\cdot)}$ to simplify notation).

First, choose $\tilde{r} = R(r,t)$, then $\tilde{t} = t + f(r,t)$, where we choose $f(r,t)$ to eliminate $dr\, dt$ term from the metric.

In these new coordinates,

$$ds^2 = -e^{\nu} dt^2 + e^{\lambda} dr^2 + r^2\, d\Omega^2$$

for some $\nu(r,t)$, $\lambda(r,t)$.

Compute the Ricci/Einstein tensor and set it to zero:

$$\begin{aligned} G_{tt} = 0 \quad &\Longrightarrow \quad -1 + e^{\lambda} + r\lambda' = 0 \qquad (1) \\ G_{rr} = 0 \quad &\Longrightarrow \quad 1 - e^{\lambda} + r\nu' = 0 \qquad (2) \\ G_{tr} = 0 \quad &\Longrightarrow \quad \dot{\lambda} = 0 \qquad (3) \\ G_{\varphi\varphi} = \sin^2\theta\, G_{\theta\theta} \quad &\longrightarrow \quad \text{more complicated} \qquad (4) \end{aligned}$$

From (3), $\lambda(r)$ is independent of t, and (1) + (2) gives

$$\begin{aligned} &(\lambda + \nu)' = 0 \\ \Longrightarrow \qquad &\nu(r,t) = -\lambda(r) + h(t) \end{aligned}$$

for some $h(t)$.

Then, eliminate $h(t)$ by further redefinition of $t \to \tilde{t}$, such that

$$d\tilde{t} = e^{\frac{1}{2}h(t)} dt\,.$$

Hence,

$$ds^2 = -e^{-\lambda} d\lambda + e^{\lambda} dr^2 + r^2 d\Omega^2\,.$$

Now, (1) and (2) coincide:

$$e^{-\lambda}(1 - r\lambda') = \frac{d}{dr}(re^{-\lambda}) = 1\,. \qquad (*)$$

(3) is already satisfied. (4) becomes

$$\lambda'' - (\lambda')^2 + \frac{2\lambda'}{r} = 0$$

which is satisfied if $(*)$ is satisfied.

Solution of $(*)$ is

$$e^{-\lambda(r)} = 1 + \frac{K}{r}\,,$$

for some constant K.

Comparing this to the weak field metric,

$$g_{tt} = -(1 + 2\Phi) = -\left(1 - \frac{2M}{r}\right)$$

and $K = -2M$. □

5.3 Matter-energy sources & $T^{\mu\nu}$

5.3.1 General theory

In special relativity, the conservation equation for 4-momentum J^α in Minkowski space is

$$\nabla_\mu J^\mu = 0\,,$$

where, with the choice $c = 1$,

$$x^\alpha = (t, x^i), \qquad \partial_\alpha = \left(\frac{\partial}{\partial t}, \frac{\partial}{\partial x^i}\right), \qquad J^\alpha = (J^0, J^i)\,.$$

$$\implies \qquad \frac{\partial J^0}{\partial t} = -\frac{\partial J^i}{\partial x^i}$$

$$\implies \qquad \frac{d}{dt}\int_V J^0\, d^3\mathbf{x} = -\int_V \frac{\partial J^i}{\partial x^i}\, d^3\mathbf{x} = -\int_{\partial V} J^i n_i\, dS$$

for any time-independent volume V in coordinates $\{x^\alpha\}$.

If $J^i = 0$ outside V, or $J^i \to 0$ as $V \to$ all space, then

$$Q = \int_V J^0\, d^3\mathbf{x}$$

is a *conserved scalar charge*.

To generalise this to curved space, by the equivalence principle,

$$\eta_{\alpha\beta} \longrightarrow g_{\alpha\beta}, \qquad \partial_\alpha \longrightarrow \nabla_\alpha$$

The conservation equation is now

$$\nabla_\alpha J^\alpha = 0\,,$$

which reduces to $\nabla_\alpha J^\alpha = 0$ in a local inertial frame, but we must be careful.

The full content of the conservation equation is

$$\nabla_\alpha J^\alpha \ = \partial_\alpha J^\alpha + \Gamma_{\alpha\ \beta}^{\ \alpha} J^\beta = 0\,,$$

however,

$$\Gamma_{\alpha\ \beta}^{\ \alpha} = \frac{1}{\sqrt{-g}} \nabla_\beta(\sqrt{-g}),$$

where $g = \det(g_{\alpha\beta})$. (See Example Sheet 3.)

So,

$$\nabla_\alpha J^\alpha = 0 \quad \Longrightarrow \quad \partial_\alpha (J^\alpha \sqrt{-g}) = 0 \,,$$

and we have the conservation of charge $\int_V J^\alpha \sqrt{-g}\, d^3\mathbf{x}$.

In special relativity, the energy-momentum tensor $T^{\alpha\beta}$ is conserved:

$$\partial_\alpha T^{\alpha\beta} = 0 \qquad \text{(in Minkowski space)}.$$

So, we have a 4-vector of conserved quantities.

We can give a physical interpretation of each component of the energy-momentum tensor:

T^{00}:	density of energy;
T^{i0}:	flux of energy in i^{th} direction;
T^{0j}:	density of jth momentum component;
T^{ij}:	flux of jth momentum component in i^{th} direction.

The symmetry of the energy-momentum tensor $T^{\alpha\beta} = T^{\beta\alpha}$ is implied by Lorentz invariance.

In general relativity, in curved space, we have a covariant, or "local", conservation equation:

$$\nabla_\alpha T^{\alpha\beta} = 0 \,.$$

Warning:

To get a conserved quantity, we need some symmetry of the metric, given by a *Killing vector*, ξ^α, which satisfies *Killing's equation*

$$\nabla_{(\alpha}\xi_{\beta)} = 0 \,.$$

With this assumption,

$$J^\alpha = T^{\alpha\beta}\xi_\beta$$

satisfies

$$\begin{aligned} \nabla_\alpha J^\alpha &= \nabla_\alpha (T^{\alpha\beta}\xi_\beta) \\ &= (\nabla_\alpha T^{\alpha\beta})\xi_\beta + T^{\alpha\beta}(\nabla_\alpha \xi_\beta) \\ &= 0 \,, \end{aligned}$$

where we use $\nabla_\alpha T^\alpha = 0$, and the fact that $T^{\alpha\beta}$ is symmetric, whilst $\nabla_\alpha \xi_\beta$ is antisymmetric (from Killing's equation).

We can then proceed as before.

5.3.2 Electromagnetism – Maxwell theory

In special relativity, the electric $\mathbf{E}$ and magnetic $\mathbf{B}$ fields are combined in the antisymmetric *electromagnetic tensor* (or *field strength tensor*)

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -F_{\nu\mu} \,,$$

where A_μ is the 4-vector/covector field containing potentials φ $\mathbf{A}$.

The generalisation to general relativity in curved space is immediate, since

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$$

is covariant. (See Example Sheet 2.)

Furthermore, we have the *Bianchi identity*

$$\partial_{[\rho} F_{\mu\nu]} = 0 = \nabla_{[\rho} F_{\mu\nu]} \,,$$

which contains half of Maxwell's equations.

The remaining Maxwell's equations are contained in

$$\begin{aligned} &\text{SR:} \qquad \partial_\mu F^{\mu\nu} = -J^\nu \\ &\text{GR:} \qquad \nabla_\mu F^{\mu\nu} = -J^\nu \end{aligned}$$

where J^μ is the charge/current vector.

The *energy momentum tensor* in curved space is

$$T_{\alpha\beta} = F_{\alpha\mu} F_\beta{}^\mu - \frac{1}{4} g_{\alpha\beta} F_{\gamma\mu} F^{\gamma\mu}.$$

(In Minkowski space, $g_{\mu\nu} = \eta_{\mu\nu}$, in units $\epsilon_0 = \mu_0 = 1$.)

If $J^\mu = 0$, then $\nabla_\mu T^{\mu\nu} = 0$. (See Example Sheet 3.)

An example of a solution with an electromagnetic source is the *Reissner-Nordström metric*, which the modifies the Schwarzschild metric to include an electric charge Q at $r = 0$, and $T^{\mu\nu}$ incorporates the electric field due to Q, which modifies the geometry:

$$\left(1 - \frac{r_s}{r}\right) \longrightarrow \left(1 - \frac{r_s}{r} + \frac{r_q^2}{r^2}\right)$$

where $r_s = 2GM$ is the usual Schwarzschild radius, and $r_Q^2 = \frac{GQ^2}{4\pi}$.

5.3.3 Perfect fluids

In special relativity, fluid motion is given by a *velocity field* U^α corresponding to behaviour of "fluid elements" (which may contain constituent particles).

At each point in spacetime, we can go to the *rest frame* of the fluid, so that the velocity field is

$$U^\alpha = (1, 0, 0, 0) \qquad (\text{with } c = 1).$$

Let n be the *scalar* number density of particles measured in the rest frame, and define the *number-flux 4-vector*

$$N^\alpha = nU^\alpha = (n, 0, 0, 0),$$

where we have conservation of particle number $\partial_\alpha N^\alpha = 0$.

The simplest case for a perfect fluid is if there is no relative motion within each fluid element, so the velocity field tells us exactly how all fluid particles are moving.

If each particle is of rest mass m, then in the rest frame of the fluid,

$$T^{00} = \rho = mn, \qquad T^{\alpha\beta} = 0 \quad \text{otherwise.}$$

In any frame of reference,

$$T^{\alpha\beta} = \rho U^\alpha U^\beta,$$

where ρ is the *density*.

More generally, in the rest frame, we may have an additional term

$$T^{ij} = p\delta^{ij},$$

where p is the *pressure*, or

$$T^{\alpha\beta} = \text{diag}(\rho, p, p, p).$$

Then, in any frame of reference,

$$T^{\alpha\beta} = (\rho + p) U^\alpha U^\beta + p\eta_{\alpha\beta},$$

and the conservation equation is

$$\partial_\alpha T^{\alpha\beta} = 0.$$

We can now generalise this to curved spacetimes in general relativity:

Definition 5.1

For a velocity field U^μ with $g_{\mu\nu}U^\mu U^\nu = -1$ (for $c = 1$), a *perfect fluid* is defined by the scalar functions ρ *density* and p *pressure*, with

$$T^{\alpha\beta} = (\rho + p)U^\alpha U^\beta + pg^{\alpha\beta}$$

and

$$\nabla_\alpha T^{\alpha\beta} = 0\,.$$

We can reduce this to a rest frame (of a co-moving observer), as before.

Examples:

(i) **Dust:**

$$p = 0$$

This is a good model for matter in the universe on the largest scales.

(ii) **Radiation/photon gas in thermal equilibrium (blackbody radiation):**

$$p = \frac{1}{3}\rho$$

(iii) **Vacuum/dark energy:**

$$\rho = -p = \frac{\Lambda}{8\pi G}$$

In general, $p = w\rho$ is an equation of state, where $w = 0$, $\frac{1}{3}$, -1 for the above examples.

5.4 Friedmann-Lemaître-Robertson-Walker spacetimes

5.4.1 Spatially homogeneous & isotropic metrics

Models of the Universe at large scales should be

- *spatially homogeneous*

  → "The Universe looks the same at all points in space." (Copernican principle)

- *isotropic*

  → "The Universe looks the same in all spatial directions."

These form the *cosmological principle*, of which we have good evidence to support it, e.g. maps of galaxies, cosmic microwave background (CMB), ...

Mathematically, we have a spacetime metric

$$ds^2 = -dt^2 + a(t)^2 d\Sigma^2$$

in coordinates $\{t, x^i\}$, where

$$d\Sigma^2 = h_{ij}(x)dx^i dx^j\,,$$

is homogeneous and isotropic, and $a(t)$ is called the *scale factor*.

Spatial manifolds at constant t must have constant curvature

$$R_{ijkl} = \kappa(h_{ik}h_{jl} - h_{il}h_{jk})\,,$$

with κ constant.

The metric can then be re-expressed as

$$d\Sigma_k^2 = \frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)$$

in spatial coordinates $\{x^i\} = \{r, \theta, \varphi\}$, where

$$k = \begin{cases} 1, & \text{closed universe;} \\ 0, & \text{flat universe;} \\ -1, & \text{open universe.} \end{cases}$$

These are the *Friedmann-Lemaître-Robertson-Walker metrics/spacetimes/universes*, ... (abbreviated as FLRW, FL, FRW, RW, ...).

The geodesic equations have solutions

$$t = \tau, \qquad x^i = \text{constant},$$

which we can get directly from the Euler-Lagrange equations without using the detailed form of h_{ij}.

These are trajectories of co-moving observers, particles, galaxies, ..., which are perpendicular to chosen spatial slices at fixed t.

Note:

For an observer who is not co-moving, they do **not** see the spacetime as isotropic.

Alternative coordinates can be introduced for

(i) $k = 1$:

$$r = \sin\chi \quad \Longrightarrow \quad d\Sigma^2_{+1} = d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta\, d\varphi^2)$$

is the metric on a 3-dimensional sphere with positive curvature.

(ii) $k = 0$:

$$d\Sigma^2_0 = dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\varphi^2)$$

is the flat metric on $\mathbb{R}^3$.

(iii) $k = -1$:

$$r = \sinh\chi \quad \Longrightarrow \quad d\Sigma^2_{-1} = d\chi^2 + \sinh^2\chi(d\theta^2 + \sin^2\theta\, d\varphi^2)$$

is the metric on 3-dimensional hyperbolic space with negative curvature.

5.4.2 General properties

(i) **Hubble's Law**

The distance between two co-moving galaxies is

$$d(t) = a(t)R\,,$$

where R is the distance calculated using the time-independent metric h_{ij}, e.g. $R^2 = \Delta x^i \Delta x^i$ in flat space ($k = 0$).

Then,

$$\begin{aligned} v(t) = \dot{d}(t) = \dot{a}(t)R = \left(\frac{\dot{a}}{a}\right) d \\ = H(t)d(t) \end{aligned}$$

Thus, $v \propto d$ at any fixed time, which is *Hubble's law*, and $H_0 = H(t_0) = \frac{\dot{a}(t_0)}{a(t_0)}$ at the present time t_0 is called *Hubble's constant*.

(ii) **Cosmological redshift**

For a radial, null geodesic in the FLRW metric,

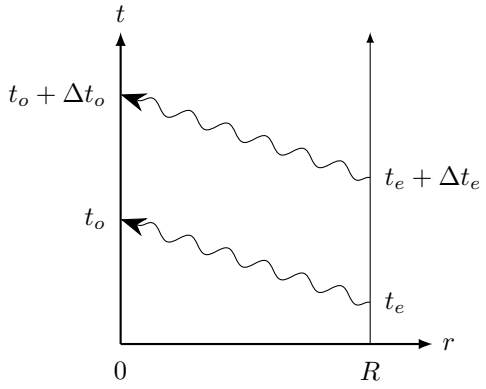
$$ds^2 = -dt^2 + \frac{a^2}{1-kr^2}dr^2 = 0$$

$$\Longrightarrow \qquad \frac{dt}{a(t)} = \pm\frac{dr}{\sqrt{1-kr^2}}\,.$$

Suppose an observer at $r = 0$ sees pulses of light at coordinate times t_o and $t_o + \Delta t_o$, which are emitted from a galaxy at $r = R$ at coordinate times t_e and $t_e + \Delta t_e$.

Then, integrating the above over the time taken for light to be sent from $r = R$ to $r = 0$.

$$\int_{t_e}^{t_o} \frac{dt}{a(t)} = \int_{t_e+\Delta t_e}^{t_o+\Delta t_o} \frac{dt}{a(t)} \qquad \left(= -\int_R^0 \frac{dr}{\sqrt{1-kr^2}} \right) .$$



If Δt_e, Δt_o is small compared to the scale of variation of $a(t)$, then

$$\begin{aligned} 0 &= \int_{t_e+\Delta t_e}^{t_o+\Delta t_o} \frac{dt}{a(t)} - \int_{t_e}^{t_o} \frac{dt}{a(t)} \\ &= \int_{t_o}^{t_o+\Delta t_o} \frac{dt}{a(t)} - \int_{t_e}^{t_e+\Delta t_e} \frac{dt}{a(t)} \\ &\approx \frac{\Delta t_o}{a(t_o)} - \frac{\Delta t_e}{a(t_e)} , \end{aligned}$$

where we use a Taylor expansion to approximate the integrals.

We can re-express this in terms of the frequency/wavelength of light:

$$\frac{\Delta t_o}{\Delta t_e} = \frac{\Delta \tau_o}{\Delta \tau_e} = \frac{\nu_e}{\nu_o} = \frac{\lambda_o}{\lambda_e} = \frac{a(t_o)}{a(t_e)} \quad \begin{cases} > 1 & \text{redshift} \\ < 1 & \text{blueshift} \end{cases}$$

For nearby galaxies,

$$\begin{aligned} & a(t_e) \approx a(t_o) + (t_e - t_o)\dot{a}(t_o) + \dots \\ \implies \qquad & \frac{\Delta t_o}{\Delta t_e} \approx 1 - (t_e - t_o)H_0 . \end{aligned}$$

5.5 Cosmological models

5.5.1 FLRW spacetimes

Note that FLRW metrics have

$$\begin{aligned} g_{tt} &= -1 , & g_{ij} &= a^2 h_{ij} , \\ g^{tt} &= -1 , & g^{ij} &= \frac{1}{a^2} h^{ij} , \end{aligned}$$

where $h^{ij} h^{jk} = \delta^i{}_k$.

Finding the non-zero, independent connection components,

$$\Gamma_i{}^t{}_j = a\dot{a}h_{ij}, \qquad \Gamma_t{}^i{}_j = \frac{\dot{a}}{a}\delta^i{}_j$$

and $\Gamma_j{}^i{}_k$ is the connection coming from h^{ij}, which is independent of t.

Direct computation gives the Riemann/Ricci tensor, and the only non-zero components of the Einstein tensor are

$$G_{tt} = \frac{3}{a^2}(\dot{a}^2 + k), \qquad G_{ij} = -(2a\ddot{a} + \dot{a}^2 + k)h_{ij}\,.$$

As a source, take a comoving perfect fluid, defined by the density $\rho(t)$ and pressure $p(t)$ (which are independent of x^i) with

$$T_{tt} = \rho(t), \qquad T_{ij} = p(t)a(t)^2 h_{ij}\,.$$

Einstein's equations (with units $c = G = 1$) give

$$tt: \qquad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi\rho}{3} - \frac{k}{a^2}\,, \tag{5.1}$$

$$ij: \qquad \frac{\ddot{a}}{a} = -\frac{4\pi}{3}(\rho + 3p)\,, \tag{5.2}$$

where we have simplified (5.2) using (5.1) to eliminate terms involving $\dot{a}$. These are known as the *Friedmann equations* (where the first equation can sometimes be referred to as the *Friedmann equation*).

Conservation of the energy-momentum tensor gives

$$\dot{\rho} + \frac{3\dot{a}}{a}(\rho + p) = 0\,,$$

which can be expressed as

$$\frac{d}{dt}(a^3\rho) = -p\frac{d}{dt}(a^3)\,, \tag{5.3}$$

where the LHS is the rate of increase of energy in some comoving volume, and the RHS is the rate of working against pressure as the fluid expands.

Note:

(5.1), (5.2), and (5.3) are **not** independent.

Suppose the fluid has the equation of state

$$p = w\rho$$

for a constant w, where

$$w = \begin{cases} 0 & \text{dust} \\ \frac{1}{3} & \text{radiation} \\ -1 & \text{vacuum energy} \end{cases}$$

Substituting this into (5.3), and integrate (for $w \geqslant -1$) to get

$$\rho(t) = \rho_0\left(\frac{a_0}{a(t)}\right)^{3(1+w)}\,,$$

with the convention that ρ_0, a_0 are taken to be the values at the present time (but still true for any general reference value).

The energy density dilutes at different rates as $a(t)$ increases (in an expanding universe):

$\rho(t) \propto a(t)^{-3}$	dust
$\rho(t) \propto a(t)^{-4}$	radiation
$\rho(t) \propto$ constant	vacuum energy

The source may be a mixture of the components above. If they are non-interacting, then the energy-momentum tensor are separately conserved, and deductions from (5.3) hold for each component.

For any single components, we can substitute into (5.1) to get

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}\rho_0\left(\frac{a_0}{a}\right)^{3(1+w)} - \frac{k}{a^2},$$

and we can integrate this to find the scale factor $a(t)$.

5.6 Some simple solutions

It is convenient to use the *conformal time coordinate*

$$\eta = \int^t \frac{dt'}{a(t')}$$

which ensures that $d\eta = \frac{dt}{a(t)}$,

The FLRW metric is

$$ds^2 = a(\eta)^2(d\eta^2 + d\Sigma^2).$$

The Friedmann equation for a single component of a perfect fluid becomes

$$\left(\frac{da}{d\eta}\right)^2 + ka^2 = C^2 a^{1-3w},$$

where $C^2 = \frac{8\pi\rho_0}{3}a^{3(1+w)}$ is a constant.

We now find solutions for various values of w and k.

(i) **Radiation** $\left(w = \frac{1}{3}\right)$

We have

$$a(\eta) = \begin{cases} C\sin\eta, & k = 1 \\ C\eta, & k = 0 \\ C\sinh\eta, & k = -1 \end{cases}$$

with the choice of integration constant such that $a(0) = 0$, since for each solution, there is a point in the past such that $a = 0$.

Note:

- $\rho \propto a^{-4} \to \infty$ as $a \to 0$, so this model predicts a singularity at the Big Bang.
- For $k = 0, -1$, the universe expands indefinitely.
- For $k = 1$, the universe eventually contracts at the Big Crunch at $\eta = \pi$.

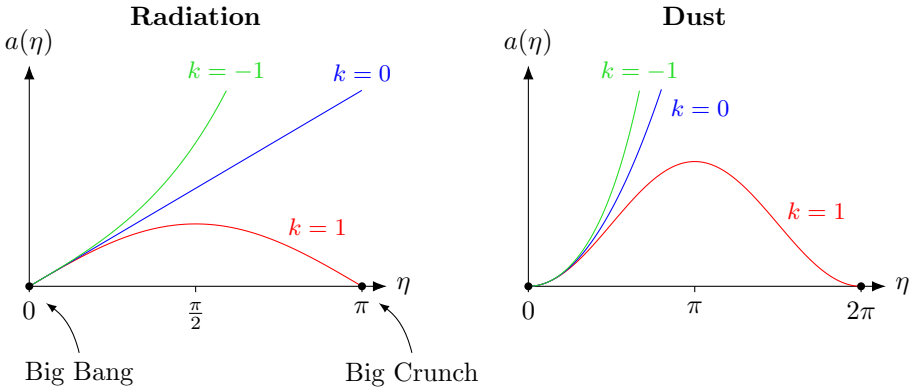
(ii) **Dust** $(w = 0)$

We have

$$a(\eta) = \begin{cases} \frac{C^2}{2}(1 - \cos\eta), & k = 1 \\ \frac{C^2}{4}\eta^2, & k = 0 \\ \frac{C^2}{2}(1 - \cosh\eta), & k = -1 \end{cases}$$

with η chosen such that $a(0) = 0$. Again, we find a Big Bang at the point $\eta = 0$.

As in (i), we have indefinite expansion for $k = 0, -1$, and a Big Crunch for $k = 1$.



(iii) **Vacuum/dark energy** $(w=-1)$

In this case, the equation of state is given by

$$\rho = -p = \frac{\Lambda}{8\pi} > 0\,.$$

The Friedmann equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{\Lambda}{3} = \lambda^2,$$

which has solutions

$$a(\eta) = \begin{cases} \frac{1}{\lambda}\cosh(\lambda t), & k=1 \\ \propto e^{\pm\lambda t}, & k=0 \\ \frac{1}{\lambda}\sinh(\lambda t), & k=-1 \end{cases}$$

In fact, these represent different slices of a maximally symmetric solution called *de Sitter spacetime*.

(iv) **Einstein static universe**

This is a combination of matter/dust with density ρ_m, and vacuum energy $(\Lambda > 0)$, so we have

$$\rho = \rho_m + \frac{\Lambda}{8\pi}\,, \qquad p = -\frac{\Lambda}{8\pi}\,.$$

So, from the Friedmann equations,

(5.1) :
$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3}\left(\rho_m + \frac{\Lambda}{8\pi}\right)\,,$$

(5.2) :
$$\begin{aligned} \frac{\ddot{a}}{a} &= -\frac{4\pi}{3}(\rho + 3p)\,, \\ &= -\frac{4\pi}{3}\left(\rho_m - \frac{\Lambda}{4\pi}\right)\,. \end{aligned}$$

We have a solution with $\dot{a} = \ddot{a} = 0$ provided $k = 1$:

$$\rho_m = \frac{\Lambda}{4\pi}\,, \qquad a^2 = \frac{1}{\Lambda}\,.$$

This is the *Einstein static universe*, but upon closer inspection, it is unstable.

5.6.1 Summary & comments

(i) Observations support our assumptions of homogeneity and isotropy on scales above several hundred Mpc.

$\Longrightarrow$ We have the FLRW metric and we live in a universe which is close to being spatially flat $k = 0$.

(ii) The universe is expanding, with Hubble constant

$$H_0 = \left(\frac{\dot{a}}{a}\right)_0 \approx 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}\,.$$

(iii) From the acceleration equation (5.2),

- matter produces *deceleration* $\ddot{a} < 0$.
- vacuum energy produces *acceleration* $\ddot{a} > 0$.

Observations suggest that Λ is very small but positive.

(iv) Contributions to energy density (at the present time) are from

- $\sim 70\%$ dark energy,
- $\sim 30\%$ matter,

but only 5% of this is standard matter (e.g. from stars, galaxies, ...).

The rest is *dark matter* (and there are negligible contributions from radiation).

(v) Simple solutions had shown an initial singularity (Big Bang) where $\rho \to \infty$.

The *Penrose-Hawking singularity theorems* show that such a singularity must be present, with reasonable assumptions on matter/energy sources.

6 Linearised Einstein equations

6.1 Reduction to wave equation

Consider a perturbation about Minkowski space

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}\,,$$

and work to first order in $|h_{\mu\nu}| \ll 1$.

Then, the inverse metric

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}\,,$$

where $h^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}h_{\alpha\beta}$ (to first order).

We will make use of coordinates changes

$$x^\alpha \longrightarrow \tilde{x}^\alpha = x^\alpha - \xi^\alpha(x)\,.$$

For $|\xi^\alpha| \ll 1$, this produces a change in the metric, or *gauge transformation*,

$$h_{\mu\nu} \longrightarrow \tilde{h}_{\mu\nu} = h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu\,,$$

which follows from the coordinate transformation rule

$$(\eta_{\mu\nu} + h_{\mu\nu})\frac{\partial x^\mu}{\partial\tilde{x}^\alpha}\frac{\partial x^\nu}{\partial\tilde{x}^\beta} = \eta_{\alpha\beta} + \tilde{h}_{\alpha\beta}\,,$$

where

$$\frac{\partial\tilde{x}^\alpha}{\partial x^\beta} = \eta^\alpha{}_\beta - \partial_\beta\xi^\alpha\,, \qquad \frac{\partial x^\alpha}{\partial\tilde{x}^\beta} = \eta^\alpha{}_\beta + \partial_\beta\xi^\alpha\,,$$

all to first order in $h_{\alpha\beta}$ and ξ^α.

To get to Einstein's equations, observe that

$$\begin{aligned}\Gamma_\alpha{}^\mu{}_\beta &= \frac{1}{2}\eta^{\mu\gamma}(h_{\alpha\gamma,\beta} + h_{\beta\gamma,\alpha} - h_{\alpha\beta,\gamma})\\ R_{\mu\nu\alpha\beta} &= \eta_{\mu\sigma}(\partial_\alpha\Gamma_\beta{}^\sigma{}_\nu - \partial_\beta\Gamma_\alpha{}^\sigma{}_\nu)\\ &= \frac{1}{2}(h_{\mu\beta,\nu\alpha} + h_{\nu\alpha,\mu\beta} - h_{\mu\alpha,\nu\beta} - h_{\nu\beta,\mu\alpha})\end{aligned}$$

all to first order in $h_{\alpha\beta}$.

Hence, the Ricci tensor and Ricci scalar are given by

$$\begin{aligned}R_{\nu\beta} &= \frac{1}{2}(-\partial_\mu\partial^\mu h_{\nu\beta} + \partial_\mu\partial_\beta h_\nu{}^\mu + \partial_\mu\partial_\nu h_\beta{}^\mu - \partial_\beta\partial_\nu h_\mu{}^\mu)\,,\\ R &= \eta^{\nu\beta}R_{\nu\beta} = -\partial_\mu\partial^\mu h + \partial_\mu\partial_\nu h^{\mu\nu}\,,\end{aligned}$$

where $h = h_\alpha{}^\alpha = \eta^{\alpha\beta}h_{\alpha\beta}$.

Finally, the Einstein tensor is given by

$$G_{\alpha\beta} = \frac{1}{2}(-\partial_\mu\partial^\mu h_{\alpha\beta} - \partial_\alpha\partial_\beta h + \partial_\mu\partial_\alpha h_\beta{}^\mu + \partial_\mu\partial_\beta h_\alpha{}^\mu - \eta_{\alpha\beta}\partial_\mu\partial_\nu h^{\mu\nu} + \eta_{\alpha\beta}\partial_\mu\partial^\mu h).$$

To simplify this, we use the following:

(i) Let the *trace-reversed metric perturbation* $\bar{h}_{\mu\nu}$ be given by

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}h\eta_{\mu\nu} \qquad\Longleftrightarrow\qquad h_{\mu\nu} = \bar{h}_{\mu\nu} - \frac{1}{2}\bar{h}\eta_{\mu\nu}\,,$$

where $\bar{h} = \bar{h}_\mu{}^\mu = -h$.

(ii) Change coordinates/make a gauge transformation

$$\bar{h}_{\mu\nu} \longrightarrow \bar{h}_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \eta_{\mu\nu}\partial_\alpha\xi^\alpha\,.$$

Using (i), we find that

$$G_{\alpha\beta} = \frac{1}{2}(-\partial_\mu\partial^\mu\bar{h}_{\alpha\beta} - \eta_{\alpha\beta}\partial_\mu\partial_\nu\bar{h}^{\mu\nu} + \partial_\mu\partial_\alpha\bar{h}_\beta{}^\mu + \partial_\mu\partial_\beta\bar{h}_\alpha{}^\mu).$$

The first term involves the *wave operator*

$$\partial_\mu\partial^\mu = -\left(\frac{\partial}{\partial t}\right)^2 + \frac{\partial}{\partial x^i}\frac{\partial}{\partial x^i}\,.$$

Other terms involve

$$\partial_\mu\bar{h}_\alpha{}^\mu \longrightarrow \partial_\mu\bar{h}_\alpha{}^\mu + \partial_\mu\partial^\mu\xi_\alpha\,.$$

We can choose ξ^α such that the RHS vanishes by solving the wave equation (with $-\partial_\mu\bar{h}_\alpha{}^\mu$ as a source).

With this choice, we have the *de Donder* or *Lorentz gauge*

$$\partial_\mu\bar{h}_\alpha{}^\mu = 0\,.$$

Then, in this choice of gauge, the Einstein tensor simplifies to

$$G_{\alpha\beta} = -\frac{1}{2}\partial_\mu\partial^\mu\bar{h}_{\alpha\beta}\,.$$

Using Einstein's equations $G_{\alpha\beta} = 8\pi T_{\alpha\beta}$, the *linearised Einstein equations* are (in units $c = G = 1$)

$$\partial_\mu\partial^\mu\bar{h}_{\alpha\beta} = -16\pi T_{\alpha\beta}\,. \tag{6.1}$$

Note:

We can make a further gauge transformation and maintain the choice of Lorentz/de Donder gauge, provided that

$$\partial_\mu\partial^\mu\xi^\alpha = 0\,,$$

so we have residual gauge transformations.

6.2 Gravitational waves

6.2.1 Plane wave solutions

We look for wave solutions of the linearised Einstein equations with $T_{\mu\nu} = 0$ of the form

$$\bar{h}_{\alpha\beta} = \mathrm{Re}\left[H_{\alpha\beta}e^{ik_\rho x^\rho}\right],$$

where k_ρ is a *real wavevector*, and $H_{\alpha\beta} = H_{\beta\alpha}$ is a complex matrix describing the *polarisation*.

To satisfy the wave equation, and de Donder gauge condition, we have

$$\begin{aligned}\partial_\mu\partial^\mu\bar{h}_{\alpha\beta} = 0 \qquad &\Longrightarrow \qquad k_\mu k^\mu = 0\,,\\ \partial^\mu\bar{h}_{\mu\alpha} = 0 \qquad &\Longrightarrow \qquad k_\mu H_{\mu\alpha} = 0\,.\end{aligned}$$

Example:

Consider a propagating wave with wavevector

$$k^\mu = k(1,0,0,1) \qquad k_\mu = k(-1,0,0,1)$$
$$\exp(ik_\rho x^\rho) = \exp(ik(t-x^3))$$

Here, waves propagate in the x^3-direction with speed $c = 1$, and the gauge condition is

$$k^\mu H_{\mu\alpha} = 0 \quad \Longleftrightarrow \quad H_{0\alpha} + H_{3\alpha} = 0\,.$$

Consider residual gauge transformations

$$\bar{h}_{\alpha\beta} \longrightarrow \bar{h}_{\alpha\beta} + \partial_\alpha \xi_\beta + \partial_\beta \xi_\alpha - \eta_{\alpha\beta}\partial_\gamma \xi^\gamma \,,$$

with $\partial_\mu \partial^\mu \xi_\alpha = 0$, applied to the plane wave solution

$$\bar{h}_{\alpha\beta} = \mathrm{Re}(H_{\alpha\beta} e^{ik_\rho x^\rho}) \,, \qquad k_\rho x^\rho = 0 \,.$$

Choosing

$$\xi_\alpha = \mathrm{Re}(-iX_\alpha e^{ik_\rho x^\rho})$$

for some X_α, we find

$$H_{\alpha\beta} \longrightarrow H_{\alpha\beta} + (k_\alpha X_\beta + k_\beta X_\alpha - \eta_{\alpha\beta} k_\gamma X^\gamma) \,.$$

Note

$$k^\beta H_{\beta\alpha} \longrightarrow k^\beta H_{\beta\alpha} \,,$$

confirming the preservation of the gauge condition.

We now find the degrees of freedom in the particular case (corresponding to a choice of coordinates), with wavevector

$$k^\mu = k(1,0,0,1), \qquad k_\mu = k(-1,0,0,1),$$

as in the above example, with $k = 1$, and with gauge condition

$$H_{0\alpha} + H_{3\alpha} = 0 \,.$$

- For $X_\alpha = (0, C, 0, 0)$, we find

$$\begin{aligned} H_{01} &\longrightarrow H_{01} + k_0 X_1 + k_1 X_0 - \eta_{01} k_\gamma X^\gamma \\ &= H_{01} - C \,. \end{aligned}$$

- Similarly, for $X_\alpha = (0, 0, C', 0)$, we find

$$H_{02} \longrightarrow H_{02} - C' \,.$$

We can use these gauge transformations to set $H_{01} = H_{02} = 0$, and from the gauge condition, $H_{31} = H_{32} = 0$.

- Next, for $X_\alpha = (A, 0, 0, B)$, noting that $k^\gamma X_\gamma = A + B$.

  With this choice,

$$\begin{aligned} H_{03} &\longrightarrow H_{03} + k_0 X_3 + k_3 X_0 \\ &= H_{03} - B + A \,. \\ H_{ij} &\longrightarrow H_{ij} - (A+B)\delta_{ij} \qquad (i,j = 1,2) \end{aligned}$$

  - Choose $A - B$ to make $H_{03} = 0 \quad \Longrightarrow \quad H_{00} = H_{30} = 0$.
  - Now, choose $A + B$ to make (H_{ij}) (for $i, j = 1, 2$) traceless.

In summary, we can make gauge transformations such that

$$H_{0\mu} = H_{3\mu} = 0 \,, \qquad H_{11} = -H_{22} \,,$$

where we can write the final form as

$$(H_{\mu\nu}) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & H_+ & H_\times & 0 \\ 0 & H_\times & -H_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

for constants H_+, $H_\times$.

This choice of gauge is called the *transverse-traceless gauge*.

Recall that the full solution is

$$h_{\alpha\beta} = \bar{h}_{\alpha\beta} = H_{\alpha\beta} e^{ik_\rho x^\rho} \,.$$

Note:

$h_{\alpha\beta}$ and $\bar{h}_{\alpha\beta}$ coincide due to this choice of gauge.

This describes waves propagating in the x^3-direction, with oscillations in the x^1-, x^2-directions, so gravitational waves are transverse with two polarisations.

6.2.2 Effects on test particles

Consider a gravitational wave in the z-direction (as above), with $\{x^i\} = \{x, y, z\}$ in the transverse-traceless gauge.

What is the effect on test particles in the (x, y)-plane?

Note:

$$\Gamma_0{}^\alpha{}_0 = \frac{1}{2}\eta^{\alpha\beta}(2\partial_0 h_{\beta 0} - \partial_\beta h_{00}) = 0$$

since we choose $H_{0\mu} = 0$.

If particles are initially at rest with 4-velocity $u^\alpha = (1, 0, 0, 0)$, then initially,

$$\frac{du^\alpha}{d\tau} = -\Gamma_0{}^\alpha{}_0 = 0\,.$$

Hence, particles remain at fixed positions, but their proper separation changes, as given by the metric

$$ds^2 = -dt^2 + (1 + h_+)^2 dx^2 + (1 - h_+)^2 dy^2 + 2h_\times dx\, dy + dz^2\,,$$

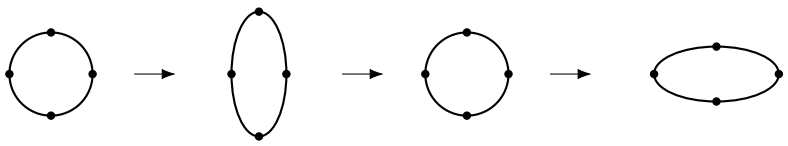
where $h_+ = \mathrm{Re}(H_+ e^{ik_\rho x^\rho})$, $h_\times = \mathrm{Re}(H_\times e^{ik_\rho x^\rho})$.

Examples:

Let (x, y) be fixed coordinates in the (x, y)-plane.

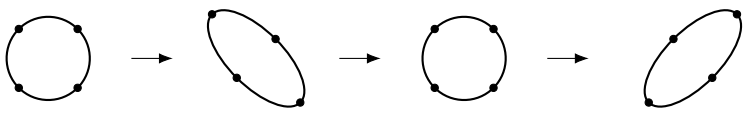
$H_+ \neq 0, \qquad H_\times = 0$

$$(\pm\delta, 0): \qquad \delta s^2 = 4(1 + h_+)\delta^2\,,$$
$$(0, \pm\delta): \qquad \delta s^2 = 4(1 - h_+)\delta^2\,.$$



$H_+ = 0, \qquad H_\times \neq 0$

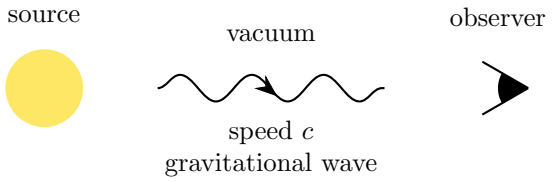
$$\frac{1}{\sqrt{2}}(\pm\delta, \pm\delta): \qquad \delta s^2 = 4(1 + h_\times)\delta^2\,,$$
$$\frac{1}{\sqrt{2}}(\pm\delta, \mp\delta): \qquad \delta s^2 = 4(1 - h_\times)\delta^2\,,$$



6.2.3 *Detection and quadrupole formula* *[Non-examinable]*

The first direct detection of gravitational waves was in 2015 by LIGO, where the accuracy required was 1 part in $\sim 10^{21}$.

Previously, there was indirect evidence related to the energy carried away by gravitational waves, in particular, from a binary star system (Hulse/Taylor, 1974).



Gravitational waves depend on

$$I_{ij}(t) = \int_{\text{source}} x_i x_j \rho(\mathbf{x}, t)\, d^3\mathbf{x}\,.$$

We assume that $d \ll \tau$ (in units c=1), where τ is the timescale on which the source varies.

The time variation of the metric for the observer is $h_{\alpha\beta}(t)$, which depends on

$$\ddot{I}_{ij}(t-r) \times \mathcal{O}\left(\frac{1}{r}\right),$$

using the far-field approximation

$$d \ll \tau \ll r\,.$$

Care is needed to define the energy carried away, but this results in the *quadrupole formula*

$$\langle P \rangle_t = \frac{1}{5} \langle \dddot{Q}_{ij}\, \ddot{Q}_{ij} \rangle_{t-r}\,.$$

This gives the energy flux across a spherical surface at radius r, where

$$Q_{ij} = I_{ij} - \frac{1}{3}\delta_{ij} I_{kk}\,.$$

The energy lost by this means of gravitational radiation matched the detailed observation of the rate of slowing of rotation of the binary pulsar system.

6.2.4 pp-wave metrics

There are exact solutions of the vacuum Einstein equations, corresponding to plane waves (parallel propagation)

$$ds^2 = H(u,x,y)du^2 - du\,dv + dx^2 + dy^2$$

with

$$u = t - z\,, \qquad v = t + z\,.$$

The only content of the vacuum Einstein equations are

$$R_{uu} = 0 \quad \Longleftrightarrow \quad (\partial_x^2 + \partial_y^2) H = 0\,.$$

Example:

Using complex notation,

$$H = H_0 e^{-iku}$$

which corresponds to a wave propogating in the z-direction.

6.2.5 Newtonian limit (revisited)

As discussed in previous sections, consider weak gravity with some Newtonian potential Φ, source/mass density ρ, and low velocities $v \ll 1$, for test particles/sources. (We will use units where $c = G = 1$.)

Introduce a small parameter ϵ to set scale and organise things

$$\Phi \sim \rho \sim v^2 \sim \epsilon \ll 1 \,,$$

and work in coordinates $\{x^0 = t, x^i\}$ with all quantities varying slowly, so

$$\frac{\partial}{\partial t} \ll \frac{\partial}{\partial x^i} \,.$$

The linearised Einstein equations become

$$\nabla^2 \bar{h}_{\alpha\beta} = -16\pi T_{\alpha\beta} \,,$$

with $\nabla^2 = \frac{\partial^2}{\partial x^i \partial x^i}$.

Previously, we found that $h_{00} = -2\Phi$ so that geodesic motion reproduces Newton's 2nd law in a gravitational potential Φ (see Section 2.3), and the (0,0)-component of Einstein's equations with $T_{00} = \rho$ gave $\nabla^2\Phi = 4\pi\rho$, with $\kappa = 8\pi$ in these units (see Section 5.1).

We now look at all components of the linearised Einstein equations and ask how the weak field metric is determined to $\mathcal{O}(\epsilon)$.

Assume that the orders match as follows (with $T_{\alpha\beta}$ components having the behaviour indicated):

$$\begin{aligned} \mathcal{O}(\epsilon) &: & \bar{h}_{00} && T_{00} &= \rho + \mathcal{O}(\epsilon^2) \\ \mathcal{O}(\epsilon^{\frac{3}{2}}) &: & \bar{h}_{0i} && T_{0i} &\sim \rho v_i \\ \mathcal{O}(\epsilon^2) &: & \bar{h}_{ij} && T_{ij} &\sim \rho v_i v_j \end{aligned}$$

[Compare this to the form for a perfect fluid.]

We find

$$\nabla^2 \bar{h}_{00} = -16\pi\rho \,,$$

which is consistent with the previous result, since

$$\begin{aligned} & \bar{h}_{00} = -4\Phi \\ \implies \qquad & \bar{h} = \eta^{\mu\nu} h_{\mu\nu} = 4\Phi \\ & h_{00} = \bar{h}_{00} - \frac{1}{2}\eta_{00}\bar{h} = -2\Phi \,, \end{aligned}$$

as required.

Then,

$$\begin{aligned} h_{0i} &= \bar{h}_{0i} \qquad (= \mathcal{O}(\epsilon^{\frac{3}{2}})) \\ h_{ij} &= \bar{h}_{ij} - \frac{1}{2}\eta_{ij}\bar{h} \\ &= \mathcal{O}(\epsilon^2) - \frac{1}{2}(4\Phi)\delta_{ij} \\ &= -2\Phi\delta_{ij} + \mathcal{O}(\epsilon^2). \end{aligned}$$

Hence, working to $\mathcal{O}(\epsilon)$, the general form of the weak field metric is

$$ds^2 = -(1+2\Phi)dt^2 + (1-2\Phi)\delta_{ij}dx^i dx^j$$

For a point mass M at $r = 0$,

$$\Phi = -\frac{M}{r},$$

where $\nabla^2\Phi = 0$ on $r > 0$.

This gives the weak field metric

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 + \frac{2M}{r}\right)(dr^2 + r^2\, d\Omega^2).$$

This is not quite what we get from the Schwarzschild metric for $\frac{M}{r} \ll 1$, but we can set

$$R = r\left(1 + \frac{2M}{r}\right)^{\frac{1}{2}} = r\left(1 + \frac{M}{r} + \mathcal{O}\left(\frac{M^2}{r^2}\right)\right)$$

$$\Longrightarrow \quad dR = dr\left(1 + \mathcal{O}\left(\frac{M^2}{r^2}\right)\right).$$

Hence, to first order in $\frac{M}{R} = \frac{M}{r} + \mathcal{O}\left(\frac{M^2}{r^2}\right)$, we find

$$ds^2 = -\left(1 - \frac{2M}{R}\right) dt^2 + \left(1 + \frac{2M}{R}\right) dR^2 + R^2\, d\Omega^2,$$

as expected for the Schwarzschild metric in coordinates $\{t, R, \theta, \varphi\}$.

7 Black holes

The Schwarzschild metric (in units $c = 1$)

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta \, d\varphi^2)$$

appears to have a singularity at $r = 0$ and $r = 2M$.

We look at the *Kretschmann scalar*

$$R^{\alpha\beta\gamma\delta} R_{\alpha\beta\gamma\delta} = \frac{48M^2}{r^6} \,,$$

which tends to 0 as $r \to \infty$, but is finite at $r = 2M$.

This implies

- $r = 0$ is a genuine singularity.
- $r = 2M$ may be a coordinate singularity, but is badly behaved.

7.1 Radial geodesics in $r > 2M$

Recall:

For radial geodesics in the Schwarzschild metric, in the region $r < 2M$, we have first integrals

$$\left(1 - \frac{2M}{r}\right) \frac{dt}{d\lambda} = E$$

$$\left(1 - \frac{2M}{r}\right) \dot{t}^2 - \left(1 - \frac{2M}{r}\right)^{-1} \dot{r}^2 = \kappa = \begin{cases} 1 \,, & \text{timelike geodesics,} \\ 0 \,, & \text{null geodesics,} \end{cases}$$

and we consider the case (which will only be relevant to timelike geodesics)

$$E = 1 \quad \iff \quad \text{the body is at rest } \dot{r} = 0 \text{ as } r \to \infty \,.$$

7.1.1 Timelike geodesics

The motion of an infalling body (using the above equations) is

$$\begin{aligned} 1 - \dot{r}^2 &= 1 - \frac{2M}{r} \\ \implies \qquad r^{\frac{1}{2}} \dot{r} &= -\sqrt{2M} \\ \implies \qquad \frac{2}{3}\left(r_1^{\frac{3}{2}} - r_0^{\frac{3}{2}}\right) &= -\sqrt{2M}(\tau_1 - \tau_0) \end{aligned}$$

where we start at $r_0 = r(\tau_0)$ and reach $r_1 = r(\tau_1) < r_0$ in finite proper time.

Nothing special happens for the observer travelling with the body as $r_1 \to 2M$.

In terms of coordinate time, however,

$$\begin{aligned} r^{\frac{1}{2}} \frac{dr}{dt} &= -\sqrt{2M}\left(1 - \frac{2M}{r}\right) \\ \implies \qquad -\sqrt{2M}(t_1 - t_0) &= \int_{r_0}^{r_1} \frac{r^{\frac{3}{2}}}{r - 2M} \, dr \end{aligned}$$

where the RHS is log divergent as $r_1 \to 2M$.

Since t is the proper time for a distant observer, stationary at large r, they never see the body reach $r = 2M$.

7.1.2 Null geodesics

For ingoing and outgoing geodesics,

$$dt = \pm\left(1 - \frac{2M}{r}\right)^{-1} dr\,.$$

Note:
$\dot{r} = 1$ so r is an affine parameter.

Set

$$dr_* = \left(1 - \frac{2M}{r}\right)^{-1} dr\,,$$

where

$$\frac{dr_*}{dr} = \left(1 - \frac{2M}{r}\right)^{-1} = 1 + \frac{1}{\left(\frac{r}{2M} - 1\right)}\,.$$

Choose constant of integration such that

$$r_* = r + 2M \log\left|\frac{r}{2M} - 1\right|.$$

The null geodesics in $r > 2M$ are given by

$$\begin{aligned} t - r_* &= \text{constant} \qquad \text{outgoing,} \\ t + r_* &= \text{constant} \qquad \text{ingoing.} \end{aligned}$$

7.2 Causal structure & horizon in Schwarzschild spacetime

7.2.1 Eddington-Finkelstein coordinates

Take $\{v, r, \theta, \varphi\}$ as spacetime coordinates, with *ingoing Eddington-Finkelstein coordinates*, where

$$\begin{aligned} v &= t + r_\star \\ dv &= dt + dr\left(1 - \frac{2M}{r}\right)^{-1} \end{aligned}$$

From Section 7.1, $v =$ constant on ingoing null geodesics in $r > 2M$.

In these coordinates,

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dv^2 + 2dv\,dr + r^2 d\Omega^2\,,$$

which has a singularity at $r = 0$, but is non-singular on $0 < r < \infty$, $-\infty < v < \infty$ (and the usual domains for θ, φ).

To check that the metric is non-singular, consider the 2×2 block of components for coordinates (v, r)

$$\det \begin{pmatrix} -\left(1 - \frac{2M}{r}\right) & 1 \\ 1 & 0 \end{pmatrix} \neq 0\,.$$

The change in coordinates $\{t, r\} \longrightarrow \{v, r\}$ produces the same new expression for ds^2 in both $r > 2M$ and $0 < r < 2M$.

7.2.2 Null geodesics using ingoing Eddington-Finkelstein coordinates

Look at the radial null geodesic equations in these new coordinates, with $0 < r < \infty$:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dv^2 + 2dv\, dr = 0\,.$$

We have the following solutions:

(i) $dv = 0 \quad \Longrightarrow \quad v = \text{constant for all } r > 0\,.$

(ii) $\left(1 - \frac{2M}{r}\right) dv = 2dr \quad \Longrightarrow \quad r = 2M \quad \text{or} \quad \frac{dv}{dr} = 2 + \frac{2}{\left(\frac{r}{2M} - 1\right)}$

$$\Longrightarrow \quad v = 2r + 4M \log\left|\frac{r}{2M} - 1\right| + \text{const.}$$

These solutions hold on $r > 0$, and to sketch them against r, introduce

$$t_* = v - r\,.$$

Note:

The ingoing Eddington-Finkelstein coordinate can be expressed as

$$t_* + r = v = t + r_*\,.$$

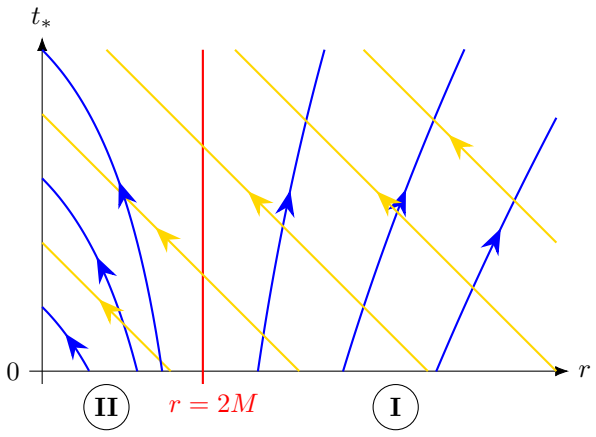
(i) $t_* = -r + \text{constant} \quad \Longrightarrow \quad \frac{dt_*}{dr} = -1$ are ingoing geodesics.

(ii) $t_* = r + 4M \log\left|\frac{r}{2M} - 1\right| + \text{constant}$

$$\Longrightarrow \quad \frac{dt_*}{dr} = \frac{\left(\frac{r}{2M} + 1\right)}{\left(\frac{r}{2M} - 1\right)} \begin{cases} > 0, & \text{outgoing for } r > 2M, \\ < 0, & \text{ingoing for } r < 2M. \end{cases}$$

From this, note also

$$\frac{dt_*}{dr} \to \begin{cases} 1\,, & \text{as } r \to \infty \\ -1\,, & \text{as } r \to 0 \end{cases}$$

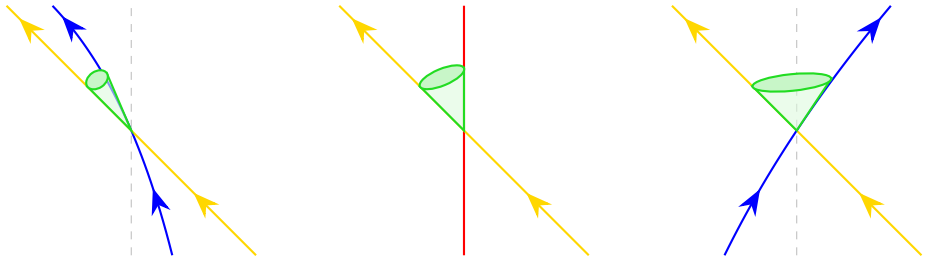


There are two regions:

- Region I $\quad (r > 2M)$
  - We have both ingoing and outgoing null geodesics, corresponding to the solutions seen earlier in Section 7.1.
- Region II $\quad (0 < r < 2M)$
  - All null geodesics are ingoing, and reach $r = 0$ in finite proper time.

7.2.3 Causal structure

From null geodesics, we can deduce the qualitative behaviour of any massive particle, since the tangent vector to any radial, timelike curve must lie within the forward light cones, as shown below.



$0 < r < 2M$ $\quad$ $r = 2M$ $\quad$ $r > 2M$

Any massive particle in region II reaches $r = 0$ in finite proper time. (See Example Sheet 4.)

Particles can pass through from region I to region II, but neither particle nor light can move from region II to region I. This is a *black hole*, and $r = 2M$ is the *event horizon*.

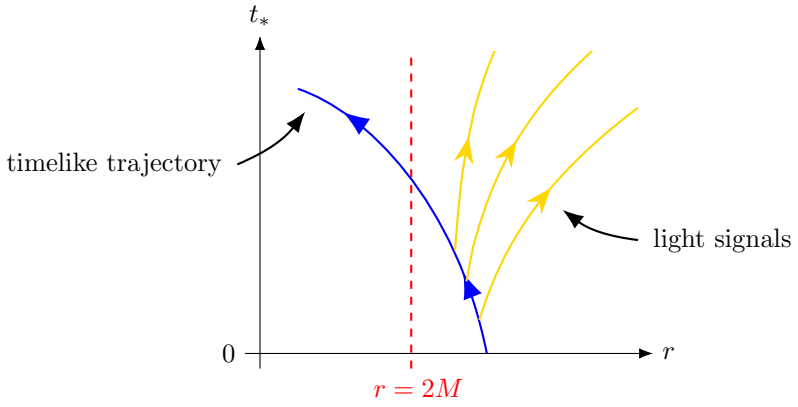
Note:

> Returning to our original coordinates, we see that r is a timelike coordinate in $0 < r < 2M$ (in the sense that the factors of $\left(1 - \frac{2M}{r}\right)$ are negative for $0 < r < 2M$ in the Schwarzschild metric, so we can treat t as a 'spacelike coordinate', and r as a 'timelike coordinate'), so it may be less surprising that all trajectories terminate at the surface $r = 0$ at some time in the future.

7.3 Gravitational collapse – a qualitative discussion

A black hole forms if a star contracts under its own gravity, until its mass is contained within the Schwarzschild radius $r_s = 2M$.

Consider an idealised spherical collapse. An infalling observer on the surface will reach $r = 2M$ in finite proper time. Light signals sent to a distant observer will become increasingly redshifted, and a collapsing star will appear to freeze and go dark in a very short period of time.



The process of gravitational collapse starts when nuclear fuel is used up, and stars begin to cool (thermal pressure decreases), contracting under gravity, until it is opposed by non-thermal (degeneracy) pressure, which leads to white dwarves or neutron stars. However, these have an upper limit of a few solar masses, called the *Chandrasekhar limit*. For more massive collapsing stars, they can contract indefinitely and form a black hole.

Realistic situations will not have exact spherical symmetry, but the formation of black holes is still ensured by *singularity theorems*. (See later.) Black holes are stable objects, and small perturbations will not change their formation under gravitational collapse.

Observations which confirm the existence of black holes:

- Cygnus X-1 (20 solar masses), which is an X-ray source
- Supermassive black holes (which are of $10^6 - 10^9$ solar masses) are expected at the centre of most galaxies.
- In 2015, gravitational waves were detected by LIGO, which originated from the merger of two black holes.
- The Event Horizon Telescope (2017–2019) produced the first direct (radio) image of a supermassive black hole (M87*) in 2019, and later an image of the black hole at the centre of the Milky Way (Sagittarius A*) in 2022.

7.4 Kruskal extension

In Section 7.2, we used ingoing Eddington-Finkelstein coordinates $\{v, r, \theta, \varphi\}$ with $v = t + r_*$ to extend the metric from region I ($r > 2M$) to region II ($0 < r < 2M$).

Similarly, we can consider *outgoing Eddington-Finkelstein coordinates* $\{u, r, \theta, \varphi\}$, with

$$u = t - r_* \,,$$

which allows the extension from region I to region III with $0 < r < 2M$.

However, region III is *distinct* from region II: any timelike/null geodesic *starts* at $r = 0$, crosses $r = 2M$ in finite proper time and cannot re-enter, which is the time-reversed situation of a black hole. This is a *white hole*.

The *maximal, Kruskal extension* of region I can be described using Kruskal coordinates $\{U, V, \theta, \varphi\}$, where in region I,

$$U = -e^{-u/4M}, \quad V = e^{v/4M} \,,$$

and note that

$$UV = -e^{r_*/2M} = -e^{r/2M}\left(\frac{r}{2M} - 1\right) .$$

From this, we can view $r = r(U, V)$, and

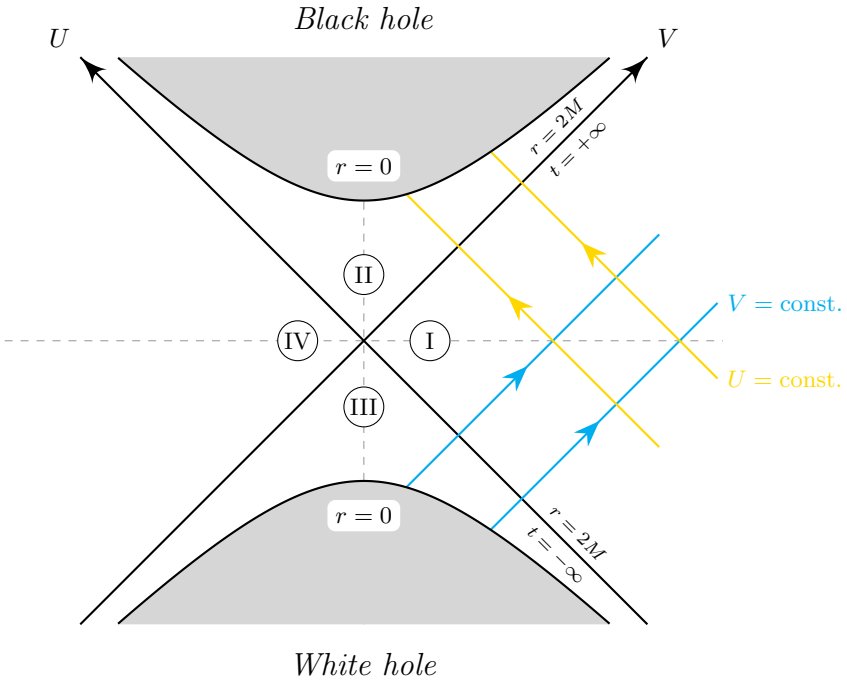
$$\begin{aligned} ds^2 &= -\left(1 - \frac{2M}{r}\right) du\, dv + r^2\, d\Omega^2 \\ &= -\frac{32M^3}{r} e^{-r/2M} dU\, dV + r^2\, d\Omega^2 \,. \end{aligned}$$

We can extend the metric from region I to other regions:

(I) $U < 0, \quad V > 0$ (III) $U < 0, \quad V < 0$

(II) $U > 0, \quad V > 0$ (IV) $U > 0, \quad V < 0$

- Null geodesics are parallel to the U, V axes, and correspond to $r = 2M$.
- $r = \text{const.}$ and $r \neq 2M$ are hyperbolas.
- Only regions I and II are relevant for physical processes (e.g. stellar collapse).
- Region IV is isometric to region I, but is causally disconnected from it.
- White holes are unstable, while black holes are stable.



7.5 Final remarks

7.5.1 Singularities & geodesics

- There are numerous powerful results which characterise black hole solutions. For example:

  Theorem 7.1: Israel's theorem

  If a (vacuum) spacetime is static, asymptotically flat, and has a regular horizon, then it is Schwarzschild.

- **Signal of a singularity:**

  Geodesic incompleteness occurs when some geodesic cannot be extended to arbitrary values of its parameter (into the past or the future). Then, there is a point where geodesics must terminate, and this point is a singularity.

  If this does not happen, then we say that the spacetime is *geodesically complete*.

  Examples of geodesics terminating at singularities:

  - $r = 0$ in Schwarzschild metric
  - $t = 0$ in some FLRW metrics

- The original singularity theorem was due to Penrose, where small departures from spherically symmetry do not affect the formation of a black hole with a singularity.

  Similar results (Hawking, Penrose, etc.) which apply to the expanding universe (with conditions on matter) imply that there must be an initial singularity or Big Bang.

7.5.2 Black hole thermodynamics

The horizon of a black hole is a null surface, but for $v = \text{const.}$ in Eddington-Finkelstein coordinates, we have the area of a 2-sphere

$$A = 4\pi(2M)^2\,,$$

or, restoring units,

$$A = 16\pi\left(\frac{GM}{c^2}\right)^2\,.$$

Remarkably, we can assign an entropy to a black hole, given by the *Bekenstein-Hawking formula*

$$S_{BH} = \frac{k_B c^3}{4G\hbar}A\,,$$

where k_B is *Boltzmann's constant*, and the *Hawking temperature* is

$$T = \frac{\hbar c^3}{8\pi GMk_B}.$$

Hawking showed that due to quantum effects in curved space, black holes radiate and emit a thermal (blackbody) spectrum of radiation, with temperature T ($\sim 6\times10^{-8}\,\text{K}$ for a black hole of solar mass).

Such effects provide important insights into the (as yet) incomplete understanding of quantum gravity.